

signus

Blind PSK/QAM/FSK/Chirp Demodulator — 필사용 코드북

실행에 필요한 코드 전량 · 23개 파일 · 3,925줄 · 총 79쪽 | 2026-08-24 · 868c402

한 줄이 100칸을 넘으면 왼쪽 줄번호 자리에 ↪ 표시가 붙고 아랫줄로 이어집니다 — 원래는 한 줄이니 붙여서 입력하세요. 세로 점선은 들여쓰기 4칸 눈금입니다.

0 · 프로젝트 설정

☐ pyproject.toml	패키지 정의 · 의존성 · ruff/pytest 설정	40줄	2쪽
☐ signus/__init__.py	패키지 진입점	3줄	3쪽

1 · 신호 입출력과 기준표

☐ signus/sigio.py	샘플 파일 I/O — 파일명이 fs·iq·real·샘플타임을 실어 나른다	180줄	4쪽
☐ signus/constellations.py	성상도 · 그레이 비트 매핑 · FSK 레벨 (공용 기준표)	99줄	7쪽

2 · 광대역 탐지와 채널 추출

☐ signus/spectrum.py	스펙트럼 · 위터폴 뷰 (표시 전용, 탐지엔 미사용)	36줄	9쪽
☐ signus/detect.py	광대역 탐지 — 캡처 안의 모든 방사체를 주파수/시간 상자로	107줄	10쪽
☐ signus/channelize.py	채널 추출 — 광대역에서 방사체 하나만 기저대역으로 끌어냄	32줄	12쪽
☐ signus/triage.py	채널 트리아지 — 복조 가능한 디지털 신호인지 판별	66줄	13쪽

3 · 수신 DSP 코어

☐ signus/dsp.py	수신 DSP 단계 (벡터화) — 이 프로젝트의 심장	390줄	15쪽
☐ signus/eq.py	블라인드 심볼간격 FIR 등화기 — CMA 획득 후 판정지향 LMS	44줄	22쪽
☐ signus/sync.py	블라인드 반복 프리앰블 동기 (정의 불요)	111줄	23쪽
☐ signus/classify.py	결정층 — 변조 분류 · 회전 정렬 · 락 품질 · SNR	123줄	25쪽
☐ signus/_accel.py	선택적 numba 가속 (없으면 동일한 순수 numpy로 풀백)	100줄	28쪽

4 · 변조별 수신 경로

☐ signus/fsk.py	FSK/CPM 경로 — 정보락선 게이트 + 주파수 판별기 복조	181줄	30쪽
☐ signus/chirp.py	선형 처프 / CSS(LoRa) 블라인드 탐지 · 특성화	97줄	34쪽

5 · 종단 파이프라인

☐ signus/pipeline.py	블라인드 복조 종단 조립 — 두 계열의 진입점	498줄	36쪽
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6 · 사용자 인터페이스

☐ signus/cli.py	CLI — analyze / survey / serve (합성·채점 명령은 개발기 전용)	207줄	45쪽
☐ signus/server.py	표준 라이브러리만 쓰는 웹 서버 + POST /api/analyze	89줄	49쪽

7 · 웹 프론트엔드

☐ signus/web/index.html	UI 구조	174줄	51쪽
☐ signus/web/style.css	UI 스타일	243줄	55쪽
☐ signus/web/app.js	업로드 · 분석 요청 · 성상도/위터폴 캔버스 렌더링	704줄	60쪽

8 · 장비 프로브 (signus/ 옆에 같은 이름으로 저장)

☐ probes/sigc.py	특징 4줄 + 단계 관찰 — 회신은 맥의 tools/sigc.py check	206줄	72쪽
☐ probes/sa.py	종합 관찰: 스펙트로그램+검출 띠+버스트별 $x^2 \cdot x^4 \cdot x^8$ -AM 판독	195줄	76쪽

```
1  [build-system]
2  requires = ["setuptools>=68"]
3  build-backend = "setuptools.build_meta"
4
5  [project]
6  name = "signus"
7  version = "2.0.0"
8  description = "Blind PSK/QAM demodulator: auto parameter detection + constellation UI"
9  requires-python = ">=3.11"
10 dependencies = ["numpy>=1.26", "scipy>=1.11"]
11
12 [project.optional-dependencies]
13 dev = ["pytest>=8.0", "ruff>=0.4"]
14 # 필사 인쇄물 발급기(tools/codebook.py) 전용. 분석기 본체는 여전히 numpy+scipy 둘뿐이다.
15 tools = ["pygments>=2.17"]
16 # Optional JIT for the sequential equalizer/carrier loops (16-34x on those loops).
17 # Falls back to identical pure-numpy when absent. NOTE: numba pins numpy<2.5.
18 fast = ["numba>=0.60"]
19
20 [project.scripts]
21 signus = "signus.cli:main"
22
23 [tool.setuptools.packages.find]
24 include = ["signus*"]
25
26 [tool.setuptools.package-data]
27 signus = ["web/*"] # the served frontend: without this a pip install 404s the whole UI
28
29 [tool.pytest.ini_options]
30 pythonpath = ["."]
31 testpaths = ["tests"]
32 addopts = "-q"
33 markers = ["stretch: slow/marginal cases, opt-in"]
34
35 [tool.ruff]
36 target-version = "py311"
37 line-length = 100
38
39 [tool.ruff.lint]
40 select = ["E", "F", "I", "UP", "B", "SIM", "NPY"]
```

```
1 """signus – blind signal demodulator with automatic parameter detection."""
2
3 __version__ = "2.0.0"
```

```

1  """Sample file I/O. Filename carries read-necessities ONLY (fs, cplx|real, sample
2  type, endian, bitrev); ground truth lives in a `<filename>.json` sidecar the
3  analyzer never reads. SigMF sidecars (`<stem>.sigmf-meta`) are also understood.
4  Endianness falls back to SIGNUS_ENDIAN when the filename carries no .be/.le token."""
5
6  import json
7  import os
8  import re
9  from dataclasses import dataclass
10
11 import numpy as np
12
13 # fs/rf 토막은 옛 밑줄 형식 전용이다. 점으로도 끊길 수 있어 경계에 둘 다 넣는다 -- 새 형식은
14 # 샘플레이트를 접두사 없이 cplx|real 바로 뒤에 놓으므로 이 정규식에 걸리지 않는다.
15 _FS_TOK = re.compile(r"(?:^[. _])fs(?:\d+(?:\.\d+)?(?:e[+-]?\d+)?(?::[. _]|$)", re.I)
16 _RF_TOK = re.compile(r"(?:^[. _])rf(?:\d+(?:\.\d+)?(?:e[+-]?\d+)?(?::[. _]|$)", re.I)
17
18 # token -> (numpy code, full-scale divisor, offset). u-types are offset binary
19 # (e.g. 80 / RTL-SDR u8: 0..255 with 128 = zero).
20 _DTYPES = {
21     "i8": ("i1", 128.0, 0.0), "u8": ("u1", 128.0, 128.0),
22     "i16": ("i2", 32768.0, 0.0), "u16": ("u2", 32768.0, 32768.0),
23     "f32": ("f4", 1.0, 0.0), "f64": ("f8", 1.0, 0.0),
24 }
25 # baudline-style aliases: <bits>t = two's complement, <bits>o = offset binary
26 _ALIAS = {"8t": "i8", "8o": "u8", "16t": "i16", "16o": "u16", "32f": "f32", "64f": "f64"}
27 _TOK = {v: k for k, v in _ALIAS.items()} # 저장할 땐 baudline 표기로 되돌린다
28 _FMT = {"cplx": "iq", "real": "real", "iq": "iq"} # iq = 옛 밑줄 형식의 이름
29 _BITREV = np.array([int(f"{i:08b}"[::-1]), 2) for i in range(256)], dtype=np.uint8)
30
31 # SigMF core:datatype -> (dtype token, fmt, endian)
32 _SIGMF = {"cf32": ("f32", "iq"), "ci16": ("i16", "iq"), "ci8": ("i8", "iq"),
33           "cu8": ("u8", "iq"), "rf32": ("f32", "real"), "ri16": ("i16", "real")}
34
35
36 @dataclass
37 class Meta:
38     fs: float | None = None
39     fmt: str | None = None # 'iq' | 'real'
40     dtype: str = "i16" # key of _DTYPES
41     endian: str = "le" # 'le' | 'be' (floats included)
42     bitrev: bool = False # bit order reversed within each byte
43     rf_center: float | None = None # RF centre freq (Hz); None = unknown (report baseband)
44
45     def ok(self) -> bool:
46         return self.fs is not None and self.fmt in ("iq", "real") and self.dtype in _DTYPES
47
48
49 def parse_name(name: str) -> Meta:
50     """파일명이 읽기 필수 정보를 실어 나른다 (정답은 사이드카에만 있고 절대 읽지 않는다).
51
52     새 형식 <이름>.cplx|real.<샘플레이트 정수>.<16t>[.be][.bitrev].pcm
53     옛 형식 <이름>_fs<샘플레이트>_iq|real_<i16>[.be][.bitrev].iq
54
55     샘플레이트는 새 형식에서 접두사 없이 온다 -- 그래서 '숫자처럼 생긴 토막'이 아니라
56     cplx|real **바로 다음 자리**로 찾는다. 이름 자체가 숫자여도(20260801.cplx...) 안 헛갈린다.
57     확장자를 떼지 않고 통째로 쪼갬다: 새 형식은 마지막 토막(.pcm)이 포맷을 뜻하지 않고,
58     옛 형식은 확장자(.iq)가 이미 다른 토막과 같은 말을 하므로 어느 쪽도 손해가 없다."""
59     base = os.path.basename(name).lower()
60     toks = re.split(r"[. _]", base)

```

```

61 # 진짜 포맷 토막 = "숫자 + 아는 제원 토막"을 데리고 다니는 **마지막** 후보. 세 오독을 막는다:
62 # · 라벨의 real/cplx/iq(rec_iq_20260801.cplx...)가 fs/fmt 를 강탈 -- 마지막 후보 규칙이 막고,
63 # · 점 긴 샘플레이트(cap.cplx.2.4e6... -> fs=2 Hz)와 라벨 숫자를 fs 로 발명(...iq_20260728.raw)
64 # -- 숫자 뒤가 제원 토막(16t/be/.../pcm)이어야 한다는 관문이 막는다. 관문에 걸리면 fs 없음으로
65 # 크게 죽는다: 조용한 오답 대신 시끄러운 실패가 이 저장소의 원칙이다.
66 cands = [k for k, t in enumerate(toks) if t in _FMT]
67 hits = [k for k in cands if k + 2 < len(toks) and toks[k + 1].isdigit()
68         and (toks[k + 2] in _DTYPES or toks[k + 2] in _ALIAS
69             or toks[k + 2] in ("be", "bitrev", "pcm") or toks[k + 2].startswith("rf"))]
70 i = hits[-1] if hits else (cands[0] if cands else None)
71 m = Meta()
72 if i is not None: # 'is not None' 이어야 한다 -- 0 번 토막(cplx...)도 유효하다
73     m.fmt = _FMT[toks[i]]
74     if i in hits:
75         m.fs = float(toks[i + 1])
76     if m.fs is None and (mt := _FS_TOK.search(base)):
77         m.fs = float(mt.group(1))
78     if rt := _RF_TOK.search(base):
79         m.rf_center = float(rt.group(1))
80     tail = toks[i + 1:] if i is not None else toks # 제원은 포맷 토막 **뒤**에만 산다 --
81     m.dtype = next((_ALIAS.get(t, t) for t in tail # 라벨의 u8/f32/be 가 제원을 오염 못 하게
82                   if t in _DTYPES or t in _ALIAS), "i16")
83     # 엔디안: 파일명 토큰(.be/.le)이 최우선, 없으면 SIGNUS_ENDIAN, 그것도 없으면 le.
84     # 실장비 녹음기는 BE 라 장비에서는 export SIGNUS_ENDIAN=be 한 줄로 전 파일이 읽힌다
85     # (2026-08-21: BE 캡처를 LE 로 읽어 균등분포 잡음으로 보이던 사고 뒤에 넣었다).
86     env = os.environ.get("SIGNUS_ENDIAN", "le").lower()
87     if env not in ("le", "be"):
88         raise ValueError(f"SIGNUS_ENDIAN 은 le 또는 be 여야 합니다: {env!r}")
89     m.endian = "be" if "be" in tail else ("le" if "le" in tail else env)
90     m.bitrev = "bitrev" in tail
91     return m
92
93
94 def parse_sigmf(path: str) -> Meta | None:
95     """Read `<stem>.sigmf-meta` (core:sample_rate + core:datatype) if present."""
96     try:
97         with open(os.path.splitext(path)[0] + ".sigmf-meta") as fh:
98             doc = json.load(fh)
99             g = doc["global"]
100             code = g["core:datatype"].replace("_le", "").replace("_be", "")
101             dtype, fmt = _SIGMF[code]
102             endian = "be" if g["core:datatype"].endswith("_be") else "le"
103             fs = float(g["core:sample_rate"])
104         except (OSError, KeyError, ValueError, TypeError, AttributeError):
105             return None # MANDATORY fields malformed: fall back to filename tokens
106     try:
107         # rf centre is OPTIONAL: a malformed value must cost only this field, never the whole
108         # sidecar -- discarding valid fs/fmt/dtype here silently re-read the file with filename
109         # tokens, which can lie about fs (a quiet garbage decode).
110         caps = doc.get("captures") or []
111         rf = caps[0].get("core:frequency") if caps else None
112         rf = None if rf is None else float(rf)
113     except (KeyError, ValueError, TypeError, AttributeError, IndexError):
114         rf = None
115     return Meta(fs, fmt, dtype, endian, rf_center=rf)
116
117
118 def make_name(label: str, m: Meta) -> str:
119     """저장은 새 형식만: <이름>.cplx|real.<샘플레이트>.<16t>[.be][.bitrev].pcm"""
120     extra = ("be" if m.endian == "be" else "") + (".bitrev" if m.bitrev else "")

```

```

121     kind = "real" if m.fmt == "real" else "cplx"
122     return f"{label}.{kind}.{m.fs:.0f}._TOK.get(m.dtype, m.dtype)}.{extra}.pcm"
123
124
125 def decode(raw: bytes, meta: Meta) -> np.ndarray:
126     """Raw bytes -> samples: complex128 for 'iq', float64 for 'real'.
127     Tolerates truncated files (drops trailing partial samples)."""
128     code, scale, off = _DTYPES[meta.dtype]
129     if meta.bitrev:
130         raw = _BITREV[np.frombuffer(raw, dtype=np.uint8)].tobytes()
131     step = np.dtype(code).itemsize
132     dt = ("<" if meta.endian == "le" else ">") + code
133     x = np.frombuffer(raw[: len(raw) - len(raw) % step], dtype=dt).astype(np.float64)
134     x = (x - off) / scale
135     if meta.fmt == "iq":
136         x = x[: x.size & ~1] # drop dangling half-sample
137         return x[0::2] + 1j * x[1::2]
138     return x
139
140
141 def read(path: str, meta: Meta | None = None) -> tuple[np.ndarray, Meta]:
142     meta = meta or parse_sigmf(path) or parse_name(path)
143     if not meta.ok():
144         raise ValueError(f"need fs + fmt (filename tokens, SigMF, or --fs/--fmt): {path}")
145     with open(path, "rb") as fh:
146         x = decode(fh.read(), meta)
147     if x.size < 256:
148         raise ValueError(f"신호가 너무 짧습니다 ({x.size} samples): {path}")
149     return x, meta
150
151
152 def write(path: str, x: np.ndarray, meta: Meta) -> None:
153     """Quantize (ints: scaled to 99.9th-percentile full scale) and write interleaved."""
154     code, scale, off = _DTYPES[meta.dtype]
155     flat = np.column_stack([x.real, x.imag]).ravel() if np.iscomplexobj(x) else x.real
156     if scale != 1.0: # integer types
157         peak = np.percentile(np.abs(flat), 99.9) or 1.0
158         flat = np.round(flat * (scale - 1) / peak) + off
159         info = np.iinfo(code)
160         flat = np.clip(flat, info.min, info.max)
161     out = flat.astype(("<" if meta.endian == "le" else ">") + code)
162     if meta.bitrev:
163         out = np.frombuffer(_BITREV[np.frombuffer(out.tobytes(), np.uint8)].tobytes(),
164                             dtype=out.dtype)
165     out.tofile(path)
166
167
168 def sidecar_write(path: str, meta: Meta, truth: dict, gen: dict) -> None:
169     doc = {"fs": meta.fs, "fmt": meta.fmt, "dtype": meta.dtype,
170           "endian": meta.endian, "bitrev": meta.bitrev, "truth": truth, "gen": gen}
171     with open(path + ".json", "w") as fh:
172         json.dump(doc, fh, indent=1)
173
174
175 def sidecar_read(path: str) -> dict | None:
176     try:
177         with open(path + ".json") as fh:
178             return json.load(fh)
179     except (OSError, ValueError):
180         return None

```

```

1  """Constellations, Gray labels, differential tables, FSK levels (shared reference)."""
2
3  import numpy as np
4
5  MODS = ("bpsk", "qpsk", "8psk", "16qam", "64qam")          # blind-classify targets
6  FSK_MODS = ("fsk2", "fsk4", "msk")
7  DIFF_MODS = ("dbpsk", "dqpsk", "pi4dqpsk")
8  GEN_MODS = MODS + ("32qam", "oqpsk") + DIFF_MODS + FSK_MODS # generator vocabulary
9
10 # (data-alphabet order, constellation rotational symmetry). pi4dqpsk: 4 transition
11 # values per symbol but an 8-point composite constellation.
12 _INFO = {"bpsk": (2, 2), "qpsk": (4, 4), "8psk": (8, 8), "16qam": (16, 4),
13          "64qam": (64, 4), "32qam": (32, 4), "oqpsk": (4, 4), "pi4dqpsk": (4, 8),
14          "dbpsk": (2, 2), "dqpsk": (4, 4), "fsk2": (2, 0), "fsk4": (4, 0), "msk": (2, 0)}
15
16 # differential phase increment per Gray symbol value
17 DIFF_PHASES = {
18     "dbpsk": np.array([0.0, np.pi]),
19     "dqpsk": np.array([0.0, np.pi / 2, np.pi, -np.pi / 2]),
20     "pi4dqpsk": np.array([np.pi / 4, 3 * np.pi / 4, -3 * np.pi / 4, -np.pi / 4]),
21 }
22
23
24 def family(mod: str) -> str:
25     return "fsk" if mod in FSK_MODS else "linear"
26
27
28 def mod_order(mod: str) -> int:
29     return _INFO[mod][0]
30
31
32 def mod_symmetry(mod: str) -> int:
33     """Lowest power p for which x**p strips the modulation (square QAM: 4)."""
34     return _INFO[mod][1]
35
36
37 def ideal_points(mod: str) -> np.ndarray:
38     if mod in ("bpsk", "dbpsk"):
39         return np.array([1.0 + 0j, -1.0 + 0j])
40     if mod in ("qpsk", "oqpsk", "dqpsk"):
41         return np.exp(1j * (np.pi / 4 + np.arange(4) * np.pi / 2))
42     if mod in ("8psk", "pi4dqpsk"):
43         return np.exp(2j * np.pi * np.arange(8) / 8)
44     if mod == "32qam": # 6x6 cross: grid minus the four corners
45         lv = np.arange(6) * 2.0 - 5
46         i, q = np.meshgrid(lv, lv)
47         pts = (i + 1j * q).ravel()
48         pts = pts[np.abs(pts.real * pts.imag) != 25]
49         return pts / np.sqrt(np.mean(np.abs(pts) ** 2))
50     n = 4 if mod == "16qam" else 8
51     lv = np.arange(n) * 2.0 - (n - 1)
52     i, q = np.meshgrid(lv, lv)
53     pts = (i + 1j * q).ravel()
54     return pts / np.sqrt(np.mean(np.abs(pts) ** 2))
55
56
57 def _gray(k: np.ndarray) -> np.ndarray:
58     return k ^ (k >> 1)
59
60

```

```

61 def bit_labels(mod: str) -> np.ndarray:
62     """Gray label per ideal_points index (PSK: phase Gray; square QAM: per-axis).
63     32qam uses plain index labels (cross grid has no clean per-axis Gray)."""
64     m = mod_order(mod)
65     k = np.arange(m)
66     if mod in ("16qam", "64qam"):
67         n = 4 if mod == "16qam" else 8
68         return _gray(k // n) * n + _gray(k % n)
69     if mod == "32qam":
70         return k
71     return _gray(k)
72
73
74 def fsk_levels(mod: str) -> tuple[np.ndarray, np.ndarray]:
75     """(frequency levels, Gray label per level) - e.g. fsk4: [-3,-1,1,3], [0,1,3,2]."""
76     m = mod_order(mod)
77     return np.arange(m) * 2.0 - (m - 1), _gray(np.arange(m))
78
79
80 def to_bits(labels: np.ndarray, mod: str) -> np.ndarray:
81     """Integer labels -> flat 0/1 array, MSB first, log2(M) bits per label."""
82     k = mod_order(mod).bit_length() - 1
83     return (labels[:, None] >> np.arange(k - 1, -1, -1) & 1).ravel().astype(np.uint8)
84
85
86 def demap_bits(symbols: np.ndarray, mod: str) -> np.ndarray:
87     """Hard-decide to nearest ideal point, return Gray bits."""
88     pts = ideal_points(mod)
89     idx = np.argmin(np.abs(symbols[:, None] - pts[None, :]), axis=1)
90     return to_bits(bit_labels(mod)[idx], mod)
91
92
93 def demap_diff_bits(symbols: np.ndarray, mod: str) -> np.ndarray:
94     """Differential demap: quantize successive phase differences to the mod's
95     transition set. Rotation-ambiguity-free (no absolute phase reference)."""
96     ph = DIFF_PHASES[mod]
97     d = np.angle(symbols[1:] * np.conj(symbols[:-1]))
98     err = np.angle(np.exp(1j * (d[:, None] - ph[None, :]))) # wrapped distance
99     return to_bits(np.argmin(np.abs(err), axis=1), mod)

```



```

1  """Spectrum + waterfall views of the burst (display only, never used for detection)."""
2
3  import numpy as np
4  from scipy.signal import welch
5
6
7  def _db(p: np.ndarray) -> np.ndarray:
8      return 10 * np.log10(p + 1e-20)
9
10
11 def spectrum(x: np.ndarray, fs: float, bins: int = 256) -> dict:
12     """Averaged PSD of the burst, decimated to `bins` points. Frequencies in kHz."""
13     nper = int(min(4096, max(64, x.size // 8)))
14     f, p = welch(x, fs=fs, nperseg=nper, return_onesided=False, detrend=False)
15     f, p = np.fft.fftshift(f), _db(np.fft.fftshift(p))
16     if f.size > bins: # max-pool so narrow tones survive decimation
17         k = f.size // bins
18         f, p = f[:k * bins].reshape(bins, k).mean(1), p[:k * bins].reshape(bins, k).max(1)
19     return {"f": np.round(f / 1e3, 3).tolist(), "db": np.round(p, 2).tolist()}
20
21
22 def waterfall(x: np.ndarray, fs: float, rows: int = 72, bins: int = 144) -> dict:
23     """Time-frequency magnitude (dB), row-major, robustly scaled by the caller."""
24     nfft = 1 << int(np.ceil(np.log2(max(bins * 2, 64))))
25     step = max(1, (x.size - nfft) // max(rows - 1, 1))
26     idx = np.arange(rows) * step
27     idx = idx[idx + nfft <= x.size]
28     if idx.size == 0:
29         return {"rows": 0, "bins": bins, "db": [], "dt": 0.0}
30     win = np.hanning(nfft)
31     seg = np.stack([x[i:i + nfft] * win for i in idx])
32     s = np.abs(np.fft.fftshift(np.fft.fft(seg, axis=1), axes=1)) ** 2
33     k = nfft // bins
34     s = s[:, :k * bins].reshape(idx.size, bins, k).mean(2)
35     return {"rows": int(idx.size), "bins": bins, "dt": float(step / fs),
36             "db": np.round(_db(s).ravel(), 2).tolist()}

```

```

1  """Wideband signal detection: find every emitter in a capture as a frequency/time
2  box. Turns signus from a single-signal demodulator into a survey tool. Frequency
3  detection runs on the time-averaged spectrogram (robust to signal width); the time
4  extent per band comes from the spectrogram. numpy + scipy only."""
5
6  from dataclasses import dataclass
7
8  import numpy as np
9  from scipy import ndimage
10 from scipy.signal import get_window, stft
11
12
13 @dataclass
14 class Detection:
15     fc: float          # centre frequency (Hz, baseband offset)
16     bw: float          # 99%-power bandwidth (Hz)
17     t0: int            # burst start / end (sample indices into the capture)
18     t1: int
19     snr_db: float      # peak-to-floor of the band (dB)
20
21     @property
22     def baud_hint(self) -> float:
23         """Rough symbol-rate prior: 99%-power BW is ~0.86 of occupied BW  $R_s(1+a)$ ."""
24         return self.bw / (0.86 * 1.25)
25
26
27 def _floor(s: np.ndarray, frac: float = 0.5, pct: int = 20) -> np.ndarray:
28     """Per-bin noise floor via a wide low-percentile filter. A 20th percentile over
29     a half-band window tracks the front-end tilt yet stays on the noise even when a
30     signal fills up to ~half the window -- so it survives both a narrowband emitter
31     and a single signal occupying 10-50% of the band."""
32     w = max(11, int(frac * s.size) | 1)
33     return ndimage.percentile_filter(s, pct, size=w, mode="nearest")
34
35
36 def detect(x: np.ndarray, fs: float, *, nfft: int = 4096, overlap: float = 0.5,
37           thr_db: float = 6.0, min_bw_bins: int = 3) -> list[Detection]:
38     """Detect emitters as (fc, bw, t0, t1) boxes, time/frequency ordered by fc.
39     Empty capture -> []; a single signal filling the band -> one whole-band box."""
40     if x.size == 0:
41         return []
42     # nfft never exceeds the record (else stft rejects an over-long window)
43     nfft = int(min(nfft, x.size, 1 << max(8, int(np.log2(max(x.size // 16, 256)))))
44     win = get_window("blackmanharris", nfft)
45     f, t, z = stft(x, fs=fs, window=win, nperseg=nfft, noverlap=int(nfft * overlap),
46                   return_onesided=False, boundary=None, padded=False)
47     f = np.fft.fftshift(f)
48     p = np.fft.fftshift(np.abs(z) ** 2, axes=0)          # (freq, time) linear power
49     s = p.mean(axis=1)                                  # time-averaged PSD
50     floor = _floor(s)
51     mask = s > floor * 10 ** (thr_db / 10)
52     close = max(3, int(0.012 * s.size) | 1)             # bridge intra-signal nulls
53     mask = ndimage.binary_opening(mask, structure=np.ones(2))
54     mask = ndimage.binary_closing(mask, structure=np.ones(close))
55
56     lab, n = ndimage.label(mask)
57     regions = [np.where(lab == i)[0] for i in range(1, n + 1)]
58     regions = [r for r in regions if r.size >= min_bw_bins]
59     merged: list[np.ndarray] = []
60     for r in sorted(regions, key=lambda r: r[0]):

```

```

61         # heal an intra-signal split (gap small vs the signal's OWN width) without
62         # swallowing a genuinely separate neighbour -- cap the merge gap at half the
63         # narrower blob, so adjacent channels (e.g. the 25 kHz marine raster) survive
64         if merged and r[0] - merged[-1][-1] <= max(close, min(merged[-1].size, r.size) // 2):
65             merged[-1] = np.arange(merged[-1][0], r[-1] + 1)
66         else:
67             merged.append(r)
68     if len(merged) >= 2: # frequency is circular: heal a signal straddling +-fs/2
69         wrap_gap = (s.size - 1 - merged[-1][-1]) + merged[0][0]
70         if wrap_gap <= max(close, min(merged[0].size, merged[-1].size) // 2):
71             merged[0] = np.concatenate([merged.pop(), merged[0]]) # wraps the edge
72
73     dets = [_measure(r, s, floor, p, f, t, fs, x.size) for r in merged]
74     if not dets and s.max() > 10 ** (thr_db / 10) * np.percentile(s, 20):
75         # floor defeated by a signal filling most of the band: one whole-band box,
76         # so survey() falls back to analysing the entire capture (legacy behaviour)
77         dets = [Detection(0.0, fs * 0.5, 0, x.size, 0.0)]
78     return dets
79
80
81 def _measure(r: np.ndarray, s: np.ndarray, floor: np.ndarray, p: np.ndarray,
82             f: np.ndarray, t: np.ndarray, fs: float, n: int) -> Detection:
83     """Estimate fc (floor-subtracted centroid), 99%-power bw, and time extent.
84     Handles a wrapped index set (a signal straddling +-fs/2, non-contiguous)."""
85     wrapped = r.size > 1 and not np.all(np.diff(r) == 1)
86     if wrapped:
87         idx = r # already spans the edge
88         fr = f[idx].astype(float).copy()
89         fr[idx < s.size // 2] += fs # lift the -fs/2 side onto a line
90     else:
91         idx = np.arange(max(0, r[0] - 3), min(s.size, r[-1] + 4)) # widen for skirts
92         fr = f[idx]
93     wgt = np.maximum(s[idx] - floor[idx], 0)
94     fc = float(np.sum(fr * wgt) / (np.sum(wgt) + 1e-30))
95     o = np.argsort(fr)
96     cs = np.cumsum(wgt[o]) / (np.sum(wgt) + 1e-30)
97     f_lo = fr[o][np.searchsorted(cs, 0.005)]
98     f_hi = fr[o][min(np.searchsorted(cs, 0.995), idx.size - 1)]
99     bw = float(abs(f_hi - f_lo))
100    if fc >= fs / 2: # fold a wrapped centroid back
101        fc -= fs
102    band = p[idx, :].sum(axis=0) # time profile of the band
103    on = np.where(band > max(np.median(band) * 2.0, band.max() * 0.1))[0]
104    t0 = int(t[on[0]] * fs) if on.size else 0
105    t1 = int(t[min(on[-1], t.size - 1)] * fs) if on.size else n
106    snr = float(10 * np.log10(s[idx].max() / (np.median(floor) + 1e-30)))
107    return Detection(fc, bw, t0, t1, snr)

```

```

1  """Channel extraction: pull one detected emitter out of a wideband capture as a
2  baseband stream, sized so the signal occupies ~1/4-1/3 of the reduced rate -- the
3  empirically-validated sweet spot for the downstream blind estimators (fill the band
4  and the matched filter starves; leave it too wide and noise dominates)."""
5
6  import numpy as np
7  from scipy.signal import firwin, resample_poly
8
9  from . import dsp
10 from .detect import Detection
11
12
13 def extract(x: np.ndarray, fs: float, det: Detection, *, target_frac: float = 0.28,
14            pad: float = 1.5, ntaps: int = 129) -> tuple[np.ndarray, float]:
15     """Mix det.fc to DC, low-pass to the channel, decimate; trim to the burst.
16     Returns (baseband IQ, channel sample rate)."""
17     bb = dsp.mix(x, fs, det.fc)
18     want = max(det.bw * pad, fs * 1e-4) # passband to keep
19     d = max(1, int(fs / (want / target_frac))) # want ~= target_frac * fs/d
20     fs_ch = fs / d
21     # Low-pass to the box ALWAYS, even when d==1 (a wide box, bw > ~0.28*fs): otherwise the
22     # "channel" is the whole capture merely mixed, and analyze locks onto a DIFFERENT, stronger
23     # emitter (reported as a confident duplicate). Decimate only when there is room to.
24     cutoff = min(0.9 * fs_ch / 2, want / 2) / (fs / 2)
25     bb = np.convolve(bb, firwin(ntaps, cutoff), "same")
26     if d > 1:
27         bb = resample_poly(bb, 1, d)
28     a, b = int(det.t0 // d), int(det.t1 // d)
29     if b - a >= 256: # trim empty time, keep guard
30         g = (b - a) // 20
31         bb = bb[max(0, a - g):b + g]
32     return bb.astype(np.complex128), fs_ch

```

```

1  """Per-channel triage: decide whether an extracted channel is a demodulable digital
2  signal or something the demodulator must NOT force onto a constellation (analog FM
3  voice, a CW tone/spur). Envelope CV is the SNR-robust separator: RRC-shaped PSK/QAM
4  always sit at CV>=0.25, while FSK/FM/CW are constant-envelope (CV<=0.05)."""
5
6  import numpy as np
7
8  from .chirp import is_chirp, sweeps_band
9  from .fsk import fsk_gate
10
11  _CV_CE = 0.15      # constant-envelope ceiling: below it a non-FSK signal is analog/tone
12  _TONE_PAR = 500.0  # M-th-power peak-to-mean above which a constant-envelope signal is CW
13
14
15  def _carrier_par(x: np.ndarray, powers: tuple[int, ...] = (1, 2, 4, 8)) -> float:
16      """Best peak-to-mean of the M-th-power magnitude spectrum. A CW tone spikes at
17      p=1; analog FM and noise stay flat at every power."""
18      x = x - x.mean()
19      nfft = 1 << int(np.ceil(np.log2(x.size)))
20      win = np.hanning(x.size)
21      best = 0.0
22      for p in powers:
23          spec = np.abs(np.fft.fft(x ** p * win, nfft))
24          best = max(best, float(spec.max() / (spec.mean() + 1e-30)))
25      return best
26
27
28  def _active(x: np.ndarray) -> np.ndarray:
29      """Trim leading/trailing dead air (envelope below half the top-quartile median). A bursty
30      channel's dead air corrupts BOTH triage discriminators -- it inflates the envelope CV past
31      fsk_gate's ceiling AND pollutes sweeps_band's IF histogram -- so every gate below must see
32      the emitter, not the silence around it. A continuous channel is returned unchanged."""
33      env = np.abs(x)
34      top = env[env > np.percentile(env, 75)]
35      if top.size == 0:
36          return x
37      on = np.flatnonzero(env > 0.5 * np.median(top))
38      if on.size < 256:
39          return x
40      return x[on[0]:on[-1] + 1]
41
42
43  def family(x: np.ndarray, fs: float) -> str:
44      """One of 'fsk' | 'chirp' | 'linear' | 'analog' | 'tone'. 'linear'/'fsk' go to the
45      demod; 'chirp'/'analog'/'tone' are reported as-is (never force-fit to a constellation)."""
46      x = _active(x)
47      if fsk_gate(x, fs):
48          # a linear FMCW chirp / CSS (LoRa) trips fsk_gate too -- its swept IF reads bimodal to
49          # the gate -- and would then be force-demodulated into confident garbage FSK symbols.
50          # is_chirp fires on both a chirp and (at some h/ baud) real FSK, so the IF-SWEEP test
51          # breaks the tie: a genuine band-sweep is characterized as chirp, real FSK stays FSK.
52          if is_chirp(x, fs) and sweeps_band(x, fs):
53              return "chirp"
54          return "fsk"
55      if is_chirp(x, fs) and sweeps_band(x, fs):
56          # linear chirp / CSS (LoRa) -- constant-envelope, would otherwise fall through to
57          # 'analog' (no M-power tone). The tie-break is REQUIRED here too: a bursty integer-h
58          # FSK channel (dead air inflates envelope CV past fsk_gate) also trips is_chirp via
59          # its CPFSK beat tones -- without sweeps_band it was labeled 'chirp' and never
60          # demodulated. A non-sweeping channel falls through, and the demod's burst isolation

```

```
61         # recovers the FSK.
62         return "chirp"
63     a = np.abs(x)
64     if a.std() / (a.mean() + 1e-12) < _CV_CE:          # constant envelope, not FSK
65         return "tone" if _carrier_par(x) >= _TONE_PAR else "analog"
66     return "linear"
```

```

1  """Receiver DSP stages for blind PSK/QAM demodulation. Vectorized; per the house
2  anti-shared-bug rule this imports only constellation reference data, never gen.py."""
3
4  from fractions import Fraction
5
6  import numpy as np
7  from scipy.ndimage import uniform_filter1d
8  from scipy.signal import get_window, hilbert, resample_poly, stft, welch
9
10 from ._accel import dd_carrier
11 from .constellations import ideal_points
12
13
14 def analytic(x: np.ndarray) -> np.ndarray:
15     """Complex passthrough as complex128; real input -> Hilbert analytic signal. Non-finite
16     samples are zeroed first: a single NaN/Inf otherwise spreads through every FFT (and Inf
17     overflows on the  $|x|^2$  the estimators take)."""
18     x = np.nan_to_num(x, posinf=0.0, neginf=0.0)
19     x = np.where(np.abs(x) > 1e150, 0.0, x) # a finite spike that overflows  $|x|^2$  is corrupt too
20     if np.iscomplexobj(x):
21         return x.astype(np.complex128)
22     return hilbert(np.asarray(x, dtype=np.float64)).astype(np.complex128)
23
24
25 def _parab(ydb: np.ndarray, k: int) -> float:
26     """Sub-bin peak offset from a 3-point parabola fit (inputs already in dB)."""
27     if k <= 0 or k >= ydb.size - 1:
28         return 0.0
29     a, b, c = ydb[k - 1], ydb[k], ydb[k + 1]
30     d = a - 2 * b + c
31     # clamp to +-half a bin: a true peak's sub-bin offset cannot exceed that, but a near-flat
32     # ( $d \sim 0$ ) or non-max triple can explode the ratio and drive baud negative -> a resample crash.
33     return float(np.clip(0.5 * (a - c) / d, -0.5, 0.5)) if d != 0 else 0.0
34
35
36 def _kmeans1d(a: np.ndarray, k: int, iters: int = 50) -> tuple[np.ndarray, np.ndarray, float]:
37     """1-D k-means on quantile-seeded centers; return (centers, labels, rms spread).
38     Assignment is a running min over the k centers (no (N, k) broadcast temporary);
39     strict '<' keeps the lowest-index center on ties, so labels are identical to a
40     broadcast argmin. Shared by the classify amplitude-ring test and the FSK level demod."""
41     c = np.quantile(a, (np.arange(k) + 0.5) / k)
42     lab = np.zeros(a.size, dtype=int)
43     for _ in range(iters):
44         d = np.abs(a - c[0])
45         lab = np.zeros(a.size, dtype=int)
46         for j in range(1, k):
47             dj = np.abs(a - c[j])
48             closer = dj < d
49             lab[closer] = j
50             d[closer] = dj[closer]
51         nc = np.array([a[lab == j].mean() if np.any(lab == j) else c[j] for j in range(k)])
52         if np.allclose(nc, c):
53             break
54         c = nc
55     return c, lab, float(np.sqrt(np.mean((a - c[lab]) ** 2)))
56
57
58 # --- burst detection -----
59
60 _SPIKE_PAR = 60.0 # raw peak-to-mean power above which a candidate run is an impulse smear, not a

```

```

61 #          burst. Modulated bursts sit low: constant-envelope FSK ~1, RRC-shaped PSK/QAM
62 #          peaks ~6-12, high-order QAM tails to ~20; a single-sample impulse smeared over
63 #          a ~win-wide run scores ~win (hundreds+). 60 clears every real mod with margin.
64
65
66 def _otsu(v: np.ndarray, bins: int = 128) -> float:
67     """Otsu split threshold of a 1-D value distribution (log powers, column scores)."""
68     hist, edges = np.histogram(v, bins=bins)
69     centers = (edges[:-1] + edges[1:]) / 2
70     w = np.cumsum(hist).astype(float)
71     m = np.cumsum(hist * centers)
72     mb = m / np.where(w > 0, w, 1)
73     mf = (m[-1] - m) / np.where(w[-1] - w > 0, w[-1] - w, 1)
74     return float(centers[int(np.argmax(w * (w[-1] - w) * (mb - mf) ** 2))])
75
76
77 def _cell_power(x: np.ndarray, fs: float, nperseg: int = 256) -> tuple[np.ndarray, int, int]:
78     """Waterfall cell power for detection: (P[fbin, col], hop, nperseg). nperseg
79     auto-shrinks on short records; shared core for find_bursts (and, later, survey)."""
80     nperseg = int(min(nperseg, max(64, 1 << int(np.log2(max(x.size // 2, 64)))))
81     hop = nperseg // 2
82     win = get_window("blackmanharris", nperseg)
83     _, _, z = stft(x, fs=fs, window=win, nperseg=nperseg, noverlap=nperseg - hop,
84     ..., return_onesided=False, boundary=None, padded=False)
85     return np.abs(z) ** 2, hop, nperseg
86
87
88 def find_bursts(x: np.ndarray, fs: float) -> list[tuple[int, int]]:
89     """Waterfall (cell-level) burst detector. Per-bin noise floors give the same
90     narrowband processing gain the operator's eye gets on a spectrogram (a burst at
91     wideband 0 dB is +12 dB per cell when it occupies 1/16 of the band); column
92     scores then drive the run/merge/guard machinery on the time axis.
93     Returns bursts in time order, or [(0, size)] when nothing stands out."""
94     n = x.size
95     pw = np.abs(x) ** 2
96     # A burst is a TIME-ENERGY event. A continuous signal that merely shuffles its spectrum
97     # (multi-tone FSK) lights different cells per column and once shattered into 78 phantom
98     # bursts -- but its wideband envelope is FLAT. Envelope Otsu separation measured 0.014-
99     # 0.052 decades on continuous FSK vs >=0.30 on every real burst regime: nothing turns
100    # on or off -> the whole record is the burst.
101    lp = np.log10(uniform_filter1d(pw, max(64, n // 1000)) + 1e-20)
102    e_thr = _otsu(lp)
103    e_lo, e_hi = lp[lp < e_thr], lp[lp >= e_thr]
104    if not e_lo.size or not e_hi.size or float(e_hi.mean() - e_lo.mean()) < 0.12:
105        ... return [(0, n)]
106    # nperseg 128: bin 7.8 kHz keeps the narrowband gain for real signal widths while the
107    # 64-sample hop resolves inter-burst gaps down to ~200 samples (256 could not split a
108    # 300-sample gap -- no column fits inside it).
109    P, hop, nperseg = _cell_power(x, fs, nperseg=128)
110    if P.shape[1] < 4:
111        ... return [(0, n)]
112    # Per-bin floor at a LOW percentile: tracks coloured noise bin by bin and stays on the
113    # noise even when a bin is signal-occupied up to ~90% of the time (high-duty trains).
114    # A record-filling signal owns its bins' floors entirely -> flat score -> [(0, n)].
115    floor_f = np.percentile(P, 10, axis=1)[:, None]
116    r = P / (floor_f + 1e-30)
117    k = max(3, P.shape[0] // 16)
118    sc = np.log10(np.sort(r, axis=0)[-k:].mean(axis=0) + 1e-30) # column score (decades)
119    # A single noise column can spike +0.53 decades over base -- OVERLAPPING the weakest
120    # real bursts (+0.57). Like the eye, the discriminator is PERSISTENCE, not height: a

```



```

121 # 3-column smooth drops the noise worst-case to 0.31-0.33 while a real burst (>=3
122 # columns) keeps its level (0.56+) -- the gap the fixed bars below live in.
123 sc = uniform_filter1d(sc, 3)
124 # The noise base is the LOWEST MODE of the score histogram (Otsu, as the wideband
125 # version proved out): fixed quantile anchors cannot serve every duty regime, and
126 # spread ESTIMATES broke three ways (median blind >50% duty, two-sided MAD inflated by
127 # skirts, one-sided by the score's left tail). The margins are FIXED instead: a noise
128 # column's score is distribution-pinned at  $C = \log_{10}((\ln(\text{nbins}/k)+1)/0.105)$  whatever
129 # the noise level, so its fluctuation is a constant of (nbins, k) -- measured worst
130 # 0.33 over 24 noise records vs 0.56 for the weakest keeper burst.
131 thr = _otsu(sc, bins=64)
132 quiet = sc[sc < thr]
133 if not quiet.size:
134     return [(0, n)]
135 base = float(np.median(quiet))
136 # No noise-quiet column at all (base far above the pinned C): a continuous signal
137 # shuffling its spectrum (a 4-tone FSK read as 78 phantom bursts before this guard).
138 c_noise = float(np.log10((np.log(P.shape[0] / k) + 1) / 0.105))
139 if base > c_noise + 0.25:
140     return [(0, n)]
141 hi = base + 0.45
142 lo = base + 0.25
143 above_lo = sc >= lo
144 dif = np.diff(above_lo.astype(np.int8))
145 starts = list(np.where(dif == 1)[0] + 1)
146 ends = list(np.where(dif == -1)[0] + 1)
147 if above_lo[0]:
148     starts.insert(0, 0)
149 if above_lo[-1]:
150     ends.append(sc.size)
151 runs = [(s, e) for s, e in zip(starts, ends, strict=True) if (sc[s:e] >= hi).any()]
152 if not runs:
153     return [(0, n)]
154 merged = [list(runs[0])]
155 for s, e in runs[1:]:
156     # A real inter-burst gap returns to the score base; a marginal-level burst dips just
157     # below lo without reaching it and would otherwise shatter into fragments. Merge
158     # across any valley whose median stays off the base.
159     valley = sc[merged[-1][1]:s]
160     if s - merged[-1][1] < 2 or float(np.median(valley)) > base + 0.12:
161         merged[-1][1] = e
162     else:
163         merged.append([s, e])
164 # Shape gate: a modulated burst has a near-flat raw envelope (peak/mean ~ a few); an
165 # impulse smear is one spike over noise (ratio ~span). Reject spike-dominated candidates.
166 out, spiked = [], 0
167 for c0, c1 in merged:
168     if c1 - c0 < 2: # single-column flicker (noise / impulse smear)
169         continue
170     s, e = max(0, c0 * hop), min(n, (c1 - 1) * hop + nperseg)
171     if c1 - c0 >= 6:
172         # long bursts: trim one hop of smear/quantization slack per edge. The noise
173         # tail otherwise fed to the demod sits on a knife edge for dense QAM (58 extra
174         # leading samples flipped a 16qam@24dB slice to 64qam lock 18). Short runs keep
175         # their full span -- trimming there would eat into the burst itself.
176         s, e = s + hop, e - hop
177     seg = pw[s:e]
178     if float(seg.max() / (seg.mean() + 1e-30)) > _SPIKE_PAR:
179         spiked += e - s
180     continue

```

```

181         out.append((int(s), int(e)))
182         # A detection must EXPLAIN the candidate mass the SPIKE gate threw away. If an impulse
183         # inside the strongest burst killed its candidate, returning the survivors would hand
184         # analyze() the wrong emitter -- fall back instead. Only spike kills count: the
185         # single-column drops above are routine flicker filtering.
186         covered = sum(e - s for s, e in out)
187         if out and covered < 0.6 * (covered + spiked):
188             return [(0, n)]
189         # A continuous signal OWNS its bins' floors and is invisible to the cell path -- but its
190         # edge transients are not, and once returned [(0, 192)] for a 64k channel whose 16qam
191         # ran the whole record. If the detections cover almost none of the envelope's high-power
192         # mass, the cell path missed the main event: return the ENVELOPE's high span instead
193         # (not the raw record -- a dead tail fed to the demod flipped that 16qam to 64qam).
194         # 0.1 keeps partial detections (a strong burst next to a lo-only sibling covers ~30%).
195         if out and covered < 0.1 * int((lp >= e_thr).sum()):
196             idx = np.where(lp >= e_thr)[0]
197             return [(int(idx[0]), int(idx[-1]) + 1)]
198         return out or [(0, n)]
199
200
201 def find_burst(x: np.ndarray, fs: float) -> tuple[int, int]:
202     """The most energetic burst of find_bursts."""
203     return max(find_bursts(x, fs), key=lambda b: float(np.sum(np.abs(x[b[0]:b[1]]) ** 2)))
204
205
206 # --- carrier estimation -----
207
208 def est_carrier(
209     x: np.ndarray, fs: float, powers: tuple[int, ...] = (2, 4, 8),
210     nfft: int = 65536, blocks: int = 4,
211 ) -> tuple[float, int, bool]:
212     """Blind M-th-power carrier estimate; returns (fc, symmetry, ambiguous). No DC
213     nulling: the front-end is DC-blocked, so a peak at DC is a genuine fc=0."""
214     freqs = np.fft.fftshift(np.fft.fftfreq(nfft, 1 / fs))
215     best_par, best_s, sym = -1.0, None, powers[0]
216     for p in powers:
217         xp = x ** p
218         blk = xp.size // blocks
219         segs = [xp[i * blk:(i + 1) * blk] for i in range(blocks)] if blk >= 8 else [xp]
220         acc = np.zeros(nfft)
221         for s in segs:
222             acc += np.abs(np.fft.fftshift(np.fft.fft(s * np.hanning(s.size), nfft)))
223         spec = acc / len(segs)
224         par = spec.max() / spec.mean()
225         if par > best_par:
226             best_par, best_s, sym = par, spec, p
227
228     if best_s is None: # degenerate input (all-zero / no energy): no M-th-power tone exists
229         return 0.0, int(powers[0]), True
230     ydb = 20 * np.log10(best_s + 1e-20)
231     k = int(np.argmax(best_s))
232     f_peak = freqs[k] + _parab(ydb, k) * (freqs[1] - freqs[0])
233     fc = f_peak / sym
234     ambiguous = abs(sym * fc) > 0.4 * fs
235     return float(fc), int(sym), bool(ambiguous)
236
237
238 def resolve_alias(x: np.ndarray, fs: float, fc: float, sym: int) -> float:
239     """The M-th-power tone wraps mod fs, so fc is only known mod fs/sym. Candidates tile the
240     whole band at spacing fs/sym; pick the one whose alias CELL holds the most signal power

```

```

241     (wrap-aware). Cell-energy beats a spectral centroid at the band edge, where a signal
242     straddling +/-fs/2 corrupts the two-sided centroid (bpsk fc=0.495*fs picked fc=-5000)."""
243     f, pxx = welch(x, fs=fs, nperseg=min(4096, x.size), return_onesided=False)
244     p = np.maximum(pxx - np.median(pxx), 0)
245     # range must reach +/-fs/2 for EVERY sym: |k| up to sym//2 (an old fixed range(-2,3) left
246     # 8psk beyond 0.3125*fs with no candidate).
247     cands = [fc + k * fs / sym for k in range(-(sym // 2) - 1, sym // 2 + 2)]
248     """
249     ... if abs(fc + k * fs / sym) < 0.5 * fs]
250     half = fs / (2 * sym)
251     # each candidate owns a fs/sym-wide cell
252     return max(cands, key=lambda c: float(p[np.abs(((f - c + fs / 2) % fs) - fs / 2) < half].sum()))
253
254 def mix(x: np.ndarray, fs: float, fc: float) -> np.ndarray:
255     """Downconvert by fc (complex heterodyne)."""
256     return x * np.exp(-2j * np.pi * fc * np.arange(x.size) / fs)
257
258 # --- symbol rate + rolloff -----
259
260 def est_baud(
261     x: np.ndarray, fs: float, lo: float | None = None, hi: float | None = None,
262     blocks: int = 4,
263 ) -> tuple[float, float]:
264     """Cyclostationary line at the symbol rate in |x|^2; returns (baud, confidence).
265     Low confidence flags a weak line (near-zero rolloff); caller decides. Fewer blocks =
266     a longer, stronger line (worth trying on a short burst where the default 4 is too weak)."""
267     u = np.abs(x) ** 2
268     u = u - u.mean()
269     blk = u.size // blocks
270     segs = [u[i * blk:(i + 1) * blk] for i in range(blocks)] if blk >= 8 else [u]
271     nfft = 1 << max(16, int(np.ceil(np.log2(max(s.size for s in segs))))
272     acc = np.zeros(nfft // 2 + 1)
273     for s in segs:
274         acc += np.abs(np.fft.rfft(s * np.hanning(s.size), nfft))
275     spec = acc / len(segs)
276     freqs = np.fft.rfftfreq(nfft, 1 / fs)
277
278     lo = lo if lo is not None else 0.005 * fs
279     hi = hi if hi is not None else 0.45 * fs
280     band = (freqs >= lo) & (freqs <= hi)
281     if not band.any(): # degenerate window (e.g. noise-derived prior): default band
282         band = (freqs >= 0.005 * fs) & (freqs <= 0.45 * fs)
283     idx = np.where(band)[0]
284     k = idx[int(np.argmax(spec[idx]))]
285     ydb = 20 * np.log10(spec + 1e-20)
286     baud = freqs[k] + _parab(ydb, k) * (freqs[1] - freqs[0])
287     conf = spec[k] / (np.median(spec[band]) + 1e-20)
288     return float(baud), float(conf)
289
290
291 def occupied_bw(x: np.ndarray, fs: float) -> float:
292     """Two-sided occupied bandwidth without a baud prior: threshold 10% of the way
293     from the 25th-percentile floor to the 90th-percentile in-band level, walk
294     outward from DC, stop at the first >20-bin below-threshold gap (notch tolerant).
295     Returns 0.0 when no band hugs DC (caller must not trust the prior then)."""
296     f, pxx = welch(x, fs=fs, nperseg=min(8192, x.size), return_onesided=False)
297     floor = np.percentile(pxx, 25)
298     order = np.argsort(np.abs(f))
299     idx = np.flatnonzero(pxx[order] > floor + 0.10 * (np.percentile(pxx, 90) - floor))
300     if idx.size == 0:

```

```

301         return 0.0
302     gaps = np.flatnonzero(np.diff(idx) > 20)    # 틈 정지가 없으면 대역 밖 스퍼(믹싱돼
303     last = idx[gaps[0]] if gaps.size else idx[-1]    # -fc 에 떨어진 LO 누설)가 bw 를 부풀린다
304     return float(2 * np.abs(f[order][last]))
305
306
307 def est_rolloff(x: np.ndarray, fs: float, baud: float) -> float:
308     """Occupied-bandwidth band-edge estimate mapped through the v1 rolloff calibration."""
309     n = x.size
310     nper = min(8192, max(256, n // 8))
311     f, pxx = welch(x, fs=fs, nperseg=nper, return_onesided=False)
312     f, pxx = np.fft.fftshift(f), np.fft.fftshift(pxx)
313     af = np.abs(f)
314     inband = np.median(pxx[af < 0.25 * baud])
315     noise = np.median(pxx[af > 0.9 * baud])
316     thr = noise + 0.10 * (inband - noise)
317     band = (af > 0.3 * baud) & (af < 1.0 * baud) & (pxx > thr)
318     if not band.any():
319         return 0.35
320     raw = 2 * af[band].max() / baud - 1
321     return float(np.clip(1.5 * raw + 0.026, 0.05, 1.0))
322
323
324 # --- resampling + matched filter -----
325
326 def to_sps(x: np.ndarray, fs: float, baud: float, sps: int = 4) -> np.ndarray:
327     """Rational resample to exactly sps samples/symbol."""
328     r = Fraction(sps * baud / fs).limit_denominator(2000)
329     return resample_poly(x, r.numerator, r.denominator)
330
331
332 def _rx_rrc(alpha: float, span: int, sps: int) -> np.ndarray:
333     """RX matched RRC taps, unit energy, odd length; singularities at t=0 and
334     |t|=1/4a patched. Written independently of gen.py's TX shaper (anti-shared-bug)."""
335     m = (span * sps) // 2
336     t = np.arange(-m, m + 1) / sps
337     a = alpha
338     num = np.sin(np.pi * t * (1 - a)) + 4 * a * t * np.cos(np.pi * t * (1 + a))
339     den = np.pi * t * (1 - (4 * a * t) ** 2)
340     h = np.divide(num, den, out=np.zeros_like(t), where=den != 0)
341     h[t == 0] = 1 - a + 4 * a / np.pi
342     if a > 0:
343         s = np.isclose(np.abs(t), 1 / (4 * a))
344         h[s] = a / np.sqrt(2) * ((1 + 2 / np.pi) * np.sin(np.pi / (4 * a))
345                                + (1 - 2 / np.pi) * np.cos(np.pi / (4 * a)))
346     return h / np.sqrt(np.sum(h ** 2))
347
348
349 def matched(x: np.ndarray, sps: int, alpha: float, span: int = 10) -> np.ndarray:
350     """Apply the RX matched RRC filter."""
351     return np.convolve(x, _rx_rrc(alpha, span, sps), "same")
352
353
354 # --- timing recovery -----
355
356 def timing(x: np.ndarray, sps: int, block: int = 256, out: int = 1) -> np.ndarray:
357     """Block-wise Oerder-Meyr timing recovery; returns `out` samples/symbol
358     (1 = decisions, 2 = clock-tracked T/2 stream for the fractionally-spaced eq).
359     Per block the spectral-line phase gives the offset; unwrapped across blocks
360     to track clock drift, then interpolated per symbol."""

```

```

361     n = sps
362     nsym = x.size // n
363     if nsym < 1:
364         return x[:0].astype(np.complex128)
365     xt = x[:nsym * n]
366     p = np.abs(xt) ** 2
367     ph = np.exp(-2j * np.pi * np.arange(xt.size) / n)
368     bs = block * n
369     if xt.size <= bs: # single block: one global fractional offset
370         eps = np.full(nsym, -np.angle(np.sum(p * ph)) / (2 * np.pi))
371     else:
372         nblk = xt.size // bs
373         c = (p[:nblk * bs].reshape(nblk, bs) * ph[:nblk * bs].reshape(nblk, bs)).sum(1)
374         eps_b = np.unwrap(-np.angle(c)) / (2 * np.pi)
375         eps = np.interp(np.arange(nsym), (np.arange(nblk) + 0.5) * block, eps_b)
376     pos = (np.arange(nsym * out) // out + np.repeat(eps, out)) * n
377     idx = np.arange(xt.size)
378     return np.interp(pos, idx, xt.real) + 1j * np.interp(pos, idx, xt.imag)
379
380
381 # --- decision-directed carrier sync -----
382
383 def ddsync(symbols: np.ndarray, mod: str, alpha: float = 0.05, beta: float = 0.002) -> np.ndarray:
384     """Decision-directed PI carrier loop, general across PSK/QAM (never a PSK-only
385     Costas loop, which jitters QAM at the correct points). Sequential feedback (each
386     phase update depends on the prior decision) -> numba kernel when available."""
387     pts = ideal_points(mod)
388     z = symbols / np.sqrt(np.mean(np.abs(symbols) ** 2)) # AGC to unit power
389     return dd_carrier(np.ascontiguousarray(z, dtype=np.complex128),
390                      np.ascontiguousarray(pts, dtype=np.complex128), alpha, beta)

```

```

1  """Blind symbol-spaced FIR equalizer: CMA acquisition then decision-directed LMS,
2  to undo the multipath ISI (the generator's `taps=` channel) that a phase-only
3  carrier loop cannot remove."""
4
5  import numpy as np
6
7  from ._accel import cma_dd_equalize, cma_dd_equalize_fse
8  from .constellations import ideal_points
9
10
11 def equalize(symbols: np.ndarray, mod: str, n_taps: int = 11, mu_cma: float = 5e-3,
12             mu_dd: float = 2e-3, warmup: int = 1500) -> np.ndarray:
13     """Blind FIR equalizer for symbol-spaced samples; returns the cleaned symbols.
14
15     Phase 1 is constant-modulus (Godard) acquisition, phase 2 refines by decisions,
16     then the converged taps refilter the whole record so early symbols are clean too.
17     """
18     pts = ideal_points(mod)
19     r2 = np.mean(np.abs(pts) ** 4) / np.mean(np.abs(pts) ** 2) # Godard dispersion const
20     x = symbols / np.sqrt(np.mean(np.abs(symbols) ** 2)) # AGC to unit power
21     # The adaptive passes (every tap update depends on the previous output/decision) are
22     # the one part that cannot be vectorized; they run as a numba kernel when available.
23     return cma_dd_equalize(np.ascontiguousarray(x, dtype=np.complex128),
24                           np.ascontiguousarray(pts, dtype=np.complex128),
25                           float(r2), n_taps, mu_cma, mu_dd, warmup)
26
27
28 def equalize_fse(x2: np.ndarray, mod: str, n_taps: int = 75, mu_cma: float = 1e-3,
29                 mu_dd: float = 8e-4, warmup: int = 4000) -> np.ndarray:
30     """T/2 fractionally-spaced CMA->DD equalizer: 2 samples/symbol in, symbols out.
31
32     Sees the unaliased spectrum, so it inverts channels whose symbol-rate folding
33     puts zeros near the unit circle (e.g. a strong 1.5-symbol echo) where the
34     symbol-spaced `equalize` cannot. A strong echo's inverse is LONG (0.8 gain
35     needs ~70 T/2 taps for -20 dB); the clock must already be tracked, so feed it
36     `timing(ym, sps, out=2)` -- fixed taps cannot follow a drifting clock.
37     Calibrated on 2ray(0.8, 1.5 sym) x seeds 0-2: lock 77-80 (61 taps: 57-66)."""
38     pts = ideal_points(mod)
39     r2 = np.mean(np.abs(pts) ** 4) / np.mean(np.abs(pts) ** 2) # Godard dispersion const
40     x = x2 / np.sqrt(np.mean(np.abs(x2) ** 2)) # AGC to unit power
41     # Seven sequential passes (2 CMA + 4 DD + 1 tracked output); numba kernel when available.
42     return cma_dd_equalize_fse(np.ascontiguousarray(x, dtype=np.complex128),
43                               np.ascontiguousarray(pts, dtype=np.complex128),
44                               float(r2), n_taps, mu_cma, mu_dd, warmup)

```

```

1  """Blind repeated-preamble synchronization (definition-free).
2
3  A real packetized emission begins with a preamble: a short block repeated a few times so a
4  receiver can lock. The repetition is detectable and exploitable WITHOUT knowing its content --
5  the delay-and-correlate (Schmidl-Cox) metric  $M(d) = |\text{Sum conj}(x[d+k]) x[d+k+P]|^2 / (\text{Sum } |x|^2)^2$ 
6  rises to  $\sim 1$  over the span where two period- $P$  windows coincide (inside the preamble), and the PHASE
7  of that correlation is a low-variance carrier-frequency-offset (CFO) estimate that works from only a
8  few symbols -- where the blind  $M$ -th-power carrier estimator, needing many symbols to average, fails.
9
10 Used as a gated, keep-best rescue in pipeline.analyze(): a signal with no preamble yields no
11 plateau -> no rescue -> the existing blind chain is untouched (sweep stays byte-identical)."""
12
13 from dataclasses import dataclass
14
15 import numpy as np
16
17 _CONF_MIN = 0.80      # plateau strength floor (median M over the run); calibrated vs no-preamble
18 _RUN_MIN = 4.0        # plateau length >= this * period. RRC pulse memory spans a FIXED  $\sim 4.5$  symbols,
19 #                      so in periods it exceeds this only at a 1-2 symbol lag and dies by  $P \sim 3$  syms;
20 #                      a real preamble's plateau is  $(R-1)$  periods long -> this rejects the short
21 #                      memory (needs  $R \sim 5$  repeats; residual random false-alarms are caught by the
22 #                      pipeline keep-best-lock gate -- a spurious CFO lowers lock and is discarded).
23 _HARM_MIN = 0.50      # the SAME plateau must also correlate at lag  $2P$  (a real preamble,  $\geq 3$  repeats)
24 _VALLEY_MAX = 0.80    # ...AND DROP at lag  $1.5P$  (between harmonics): a real preamble is peaked at
25 #                      multiples of  $P$  (a valley in between), so  $\text{median}(M@1.5P) < 0.85 * \text{conf}$ . RRC pulse
26 #                      memory is a smooth decay ( $\sim$ equal at  $P$  and  $1.5P$ ) -> rejected. Peak/valley +
27 #                      run length is what lets the period be scanned directly, WITHOUT a reliable
28 #                      baud (est_baud is exactly what fails on the short bursts we target).
29 _PMIN = 12            # smallest period to scan (samples); below this is intra-symbol pulse memory
30 _PMAX = 512           # largest period to scan; a preamble block is short ( $L \leq \sim 12$  syms,  $\text{sps} \leq \sim 40$ )
31 _WMAX = 8192          # only the leading window is scanned -- a preamble sits at the burst START
32
33
34 @dataclass
35 class Preamble:
36     start: int         # sample index where the repeated region begins
37     end: int           # sample index where the preamble ends (data starts ~here)
38     period: int        # repetition period in samples (=  $L * \text{sps}$ )
39     cfo_hz: float      # carrier-frequency offset estimated from the preamble phase slope
40     conf: float        # 0..1 plateau confidence (median metric over the detected run)
41
42
43 def _metric(x: np.ndarray, P: int) -> tuple[np.ndarray, np.ndarray]:
44     """Normalized delay-correlation  $M(d)$  in  $[0,1]$  and the raw correlation  $P_{\text{met}}(d)$ , lag  $P$ , window
45      $W=P$ . Normalized by BOTH windows' energy (Cauchy-Schwarz) so  $M \leq 1$  -- a true correlation
46     coefficient,  $\sim 1$  only where the two period- $P$  windows are genuinely identical (a preamble)."""
47     prod = np.conj(x[:-P]) * x[P:]          #  $\text{conj}(x[n]) x[n+P]$ 
48     ea = np.abs(x[:-P]) ** 2                # first-window energy
49     eb = np.abs(x[P:]) ** 2                 # second-window energy
50     cp = np.concatenate([[0.0 + 0j], np.cumsum(prod)])
51     ca = np.concatenate([[0.0], np.cumsum(ea)])
52     cb = np.concatenate([[0.0], np.cumsum(eb)])
53     W = P
54     pm = cp[W:] - cp[:-W]                   # Sum  $\text{prod}[d:d+W]$ 
55     da = ca[W:] - ca[:-W]
56     db = cb[W:] - cb[:-W]
57     return np.abs(pm) ** 2 / (da * db + 1e-30), pm
58
59
60 def _seg_median(x: np.ndarray, lag: int, s: int, ln: int) -> float:

```

```

61     """Median of the delay-correlation metric at `lag` over the plateau [s, s+ln)."""
62     m = _metric(x, lag)[0]
63     e = min(s + ln, m.size)
64     return float(np.median(m[s:e])) if e > s else 0.0
65
66
67 def _longest_run(mask: np.ndarray) -> tuple[int, int]:
68     """(start, length) of the longest True run in a boolean array; (0, 0) if none."""
69     if not mask.any():
70         return 0, 0
71     d = np.diff(np.concatenate([[0], mask.view(np.int8), [0]]))
72     starts = np.where(d == 1)[0]
73     ends = np.where(d == -1)[0]
74     i = int(np.argmax(ends - starts))
75     return int(starts[i]), int(ends[i] - starts[i])
76
77
78 def find_preamble(x: np.ndarray, fs: float, baud_hint: float | None = None) -> Preamble | None:
79     """Detect a repeated preamble and estimate its (start, end, period, CF0, confidence).
80     Returns None when no repeated block stands out. The period is scanned DIRECTLY (not derived
81     from baud) so it works even when est_baud fails on the short burst -- exactly the target case.
82     baud_hint, if given, only tightens the scan's upper bound for speed."""
83     x = np.asarray(x, dtype=np.complex128)
84     x = x - x.mean()
85     if x.size < 256 or not np.all(np.isfinite(x)):
86         return None
87     xw = x[:_WMAX] # a preamble sits at the burst start
88     n = xw.size
89     pmax = _PMAX if not (baud_hint and baud_hint > 0) else min(_PMAX, int(12 * fs / baud_hint))
90     pmax = min(pmax, (n - 8) // 3) # need >=3 windows for the 2P harmonic check
91     best = None
92     for P in range(_PMIN, max(_PMIN + 1, pmax + 1)):
93         M, pm = _metric(xw, P)
94         run_start, run_len = _longest_run(M >= _CONF_MIN)
95         if run_len < _RUN_MIN * P:
96             continue
97         conf = float(np.median(M[run_start:run_start + run_len]))
98         # peak/valley gate: a genuinely periodic preamble correlates at 2P (harmonic) but DROPS at
99         # 1.5P (between harmonics); RRC pulse memory (which fakes a plateau at a 1-2 symbol lag) is
100         # a smooth decay -- high at both -> rejected, letting the period be scanned directly.
101         harm = _seg_median(xw, 2 * P, run_start, run_len)
102         valley = _seg_median(xw, int(round(1.5 * P)), run_start, run_len)
103         if harm < _HARM_MIN or valley > _VALLEY_MAX * conf:
104             continue
105         score = run_len * conf # long AND strong wins
106         if best is None or score > best[0]:
107             ang = float(np.angle(pm[run_start:run_start + run_len].sum())) # coherent, wrap-safe
108             cfo = ang * fs / (2 * np.pi * P)
109             end = run_start + run_len + 2 * P # past the last correlated repeat -> data
110             best = (score, Preamble(run_start, min(end, x.size), P, cfo, conf))
111     return best[1] if best else None

```



```

1  """Decision layer: modulation classification, rotation alignment, lock quality, SNR."""
2
3  from dataclasses import dataclass
4
5  import numpy as np
6
7  from .constellations import ideal_points, mod_symmetry
8  from .dsp import _kmeans1d
9
10 # Amplitude-ring separators, calibrated through the real front-end (seeds 0-3, snr 14-25):
11 # 16qam <=0.80 and 64qam <=0.99 vs 32qam >=1.16 on s3*s9/s5**2.
12 _D32 = 1.07
13 _C42_PSK = 0.80 # constant-modulus 4th cumulant: qpsk ~1.0, QAM <0.7
14 _CM_TOL = 0.2 # 1-ring spread guard for constant-modulus signals
15 _R39 = 2.7 # s3/s9: 16qam ~2.1, 64qam >=3.0
16
17
18 def _agc(z: np.ndarray) -> np.ndarray:
19     """Scale to unit average power."""
20     return z / np.sqrt(np.mean(np.abs(z) ** 2))
21
22
23 def _c42(z: np.ndarray) -> float:
24     """Magnitude of the normalized 4th-order cumulant C42 (rotation invariant)."""
25     z = _agc(z)
26     p = np.mean(np.abs(z) ** 2)
27     return float(abs(np.mean(np.abs(z) ** 4) - abs(np.mean(z**2)) ** 2 - 2 * p**2))
28
29
30 def _spread(a: np.ndarray, k: int) -> float:
31     """RMS within-cluster spread from the shared 1-D k-means (sorted-value ring test)."""
32     return _kmeans1d(a, k)[2]
33
34
35 def classify(z: np.ndarray, symmetry: int) -> str:
36     """Symmetry 2->bpsk, 8->8psk; 4-> qpsk/16qam/32qam/64qam via C42 then amplitude rings.
37
38     Not blind classes (documented): oqpsk collides with square-QAM on every ring/cumulant
39     feature; pi4dqpsk's 4th power carries a pi-per-symbol rotation, so the carrier estimate
40     absorbs baud/8 and it arrives here looking exactly like qpsk (its bits still decode
41     correctly via differential demapping).
42     """
43     if symmetry == 2:
44         return "bpsk"
45     if symmetry == 8:
46         return "8psk"
47     z = _agc(z)
48     if _c42(z) >= _C42_PSK:
49         return "qpsk"
50     a = np.sort(np.abs(z))
51     s1 = _spread(a, 1)
52     if s1 <= _CM_TOL: # constant-modulus guard (a PSK cloud that slipped the C42 gate)
53         return "qpsk"
54     s3, s5, s9 = _spread(a, 3), _spread(a, 5), _spread(a, 9)
55     if s3 * s9 / (s5 * s5 + 1e-18) > _D32: # only a 5-ring cross collapses at k=5
56         return "32qam"
57     # amplitude-ring ratio, NEVER best-fit EVM (a denser grid always fits closer)
58     return "16qam" if s3 / (s9 + 1e-12) < _R39 else "64qam"
59
60

```

```

61 _SNR_CEIL = 40.0 # above this the moments cannot resolve the noise term
62
63
64 def snr_m2m4(z: np.ndarray, mod: str) -> float:
65     """Blind symbol-SNR (dB) from the 2nd/4th moments, corrected for the
66     constellation kurtosis. Saturates at _SNR_CEIL (noise below the moments'
67     resolution); NaN when the moments leave the valid region entirely."""
68     ka = np.mean(np.abs(ideal_points(mod)) ** 4) # unit-power points => already /M2**2
69     m2, m4 = np.mean(np.abs(z) ** 2), np.mean(np.abs(z) ** 4)
70     disc = 2 * m2**2 - m4
71     if disc <= 0:
72         return float("nan")
73     s = np.sqrt(disc / (2 - ka))
74     n = m2 - s
75     return min(float(10 * np.log10(s / n)), _SNR_CEIL) if n > 0 else _SNR_CEIL
76
77
78 def _nearest_dist(w: np.ndarray, pts: np.ndarray) -> np.ndarray:
79     """Distance to nearest ideal point. Running min over the M points instead of a
80     (... , M) broadcast temporary: identical values, far less memory, ~1.5-5x faster."""
81     d = np.abs(w - pts[0])
82     for p in pts[1:]:
83         d = np.minimum(d, np.abs(w - p))
84     return d
85
86
87 def align(z: np.ndarray, mod: str) -> np.ndarray:
88     """AGC + resolve the M-fold rotation ambiguity by minimizing nearest-point distance."""
89     z = _agc(z)
90     pts = ideal_points(mod)
91     sub = z[:1000]
92     ang = np.linspace(0, 2 * np.pi / mod_symmetry(mod), 64, endpoint=False)
93     rot = sub[None, :] * np.exp(1j * ang)[: , None]
94     cost = _nearest_dist(rot, pts).mean(axis=1)
95     i = int(np.argmin(cost))
96     c0, c1, c2 = cost[(i - 1) % 64], cost[i], cost[(i + 1) % 64] # parabolic sub-grid refine
97     denom = c0 - 2 * c1 + c2
98     # clip: on a flat cost surface (e.g. pure noise) the fit can explode
99     delta = float(np.clip(0.5 * (c0 - c2) / denom, -1, 1)) if denom else 0.0
100    return z * np.exp(1j * (ang[i] + delta * (ang[1] - ang[0])))
101
102
103 @dataclass
104 class Quality:
105     evm: float
106     mer_db: float
107     lock: float
108     occupied: int
109
110
111 def quality(z: np.ndarray, mod: str) -> Quality:
112     """Alignment + EVM/MER, plus an occupancy-gated lock score in [0, 100]."""
113     z = align(z, mod)
114     pts = ideal_points(mod)
115     m = pts.size
116     idx = np.argmin(np.abs(z[:, None] - pts[None, :]), axis=1)
117     d = pts[idx]
118     evm = float(np.sqrt(np.mean(np.abs(z - d) ** 2) / np.mean(np.abs(d) ** 2)))
119     mer_db = float(-20 * np.log10(evm + 1e-12))
120     thr = max(3, 0.6 * len(z) / m)

```

```
121 |     occupied = int(np.sum(np.bincount(idx, minlength=m) >= thr))
122 |     lock = float(np.clip((mer_db - 6) / 19 * 100, 0, 100) * (occupied / m))
123 |     return Quality(evm, mer_db, lock, occupied)
```

```

1  """Optional numba acceleration for the sequential feedback loops (adaptive equalizer
2  taps, decision-directed carrier) that CANNOT be vectorized -- each update depends on
3  the previous output, so they are the only python-loop hot spots in an otherwise
4  vectorized pipeline. Numba is an OPTIONAL extra ('pip install signus[fast]'): when it
5  is absent 'njit' degrades to a no-op and the identical bodies run as plain numpy, so
6  results are unchanged and the numpy+scipy-only baseline still works. The kernels use
7  the same expressions as the pure-numpy originals, so the numba path stays numerically
8  equivalent (verified: byte-identical 'w @ win', argmin/abs/angle exact)."""
9
10 import os
11
12 import numpy as np
13
14 HAVE_NUMBA = False
15 if not os.environ.get("SIGNUS_NO_NUMBA"): # escape hatch: force the pure-numpy path
16     try:
17         import numba
18         njit = numba.njit(cache=True, fastmath=False) # fastmath off: keep IEEE results
19         HAVE_NUMBA = True
20     except ImportError: # numba absent: same bodies run as plain numpy (slower, identical)
21         pass
22 if not HAVE_NUMBA:
23     def njit(fn):
24         return fn
25
26
27 @njit
28 def cma_dd_equalize(x, pts, r2, n_taps, mu_cma, mu_dd, warmup):
29     """Symbol-spaced CMA acquisition -> decision-directed LMS -> converged refilter."""
30     n = x.size
31     mid = n_taps // 2
32     xp = np.concatenate((np.zeros(mid, dtype=np.complex128), x,
33                          np.zeros(mid, dtype=np.complex128)))
34     w = np.zeros(n_taps, dtype=np.complex128)
35     w[mid] = 1.0
36     out = np.empty(n, dtype=np.complex128)
37     warm = min(warmup, n)
38     for i in range(warm): # Phase 1: CMA (blind, modulus-only)
39         win = xp[i:i + n_taps]
40         y = w @ win
41         w = w - mu_cma * (y * (abs(y) ** 2 - r2)) * np.conj(win)
42     for i in range(warm, n): # Phase 2: decision-directed LMS
43         win = xp[i:i + n_taps]
44         y = w @ win
45         d = pts[np.argmin(np.abs(y - pts))]
46         w = w - mu_dd * (y - d) * np.conj(win)
47     for i in range(n): # refilter whole record with final taps
48         win = xp[i:i + n_taps]
49         y = w @ win
50         d = pts[np.argmin(np.abs(y - pts))]
51         w = w - mu_dd * (y - d) * np.conj(win)
52         out[i] = y
53     return out
54
55
56 @njit
57 def cma_dd_equalize_fse(x, pts, r2, n_taps, mu_cma, mu_dd, warmup):
58     """T/2 fractionally-spaced: 2 CMA passes -> 4 DD passes -> tracked output pass."""
59     nsym = x.size // 2
60     mid = n_taps // 2

```

```

61     xp = np.concatenate((np.zeros(mid, dtype=np.complex128), x,
62                             np.zeros(n_taps, dtype=np.complex128)))
63     w = np.zeros(n_taps, dtype=np.complex128)
64     w[mid] = 1.0
65     out = np.empty(nsym, dtype=np.complex128)
66     warm = min(warmup, nsym)
67     for _ in range(2):                                     # CMA acquisition, 2 data-reuse passes
68         for i in range(warm):
69             win = xp[2 * i:2 * i + n_taps]
70             y = w @ win
71             w = w - mu_cma * (y * (abs(y) ** 2 - r2)) * np.conj(win)
72     for _ in range(4):                                     # decision-directed refinement
73         for i in range(nsym):
74             win = xp[2 * i:2 * i + n_taps]
75             y = w @ win
76             d = pts[np.argmin(np.abs(y - pts))]
77             w = w - mu_dd * (y - d) * np.conj(win)
78     for i in range(nsym):                                   # final tracked output pass
79         win = xp[2 * i:2 * i + n_taps]
80         y = w @ win
81         d = pts[np.argmin(np.abs(y - pts))]
82         w = w - mu_dd * (y - d) * np.conj(win)
83         out[i] = y
84     return out
85
86
87 @njit
88 def dd_carrier(z, pts, alpha, beta):
89     """Per-symbol decision-directed PI carrier loop (general PSK/QAM, not PSK Costas)."""
90     out = np.empty(z.size, dtype=np.complex128)
91     phase = 0.0
92     freq = 0.0
93     for i in range(z.size):
94         y = z[i] * np.exp(-1j * phase)
95         d = pts[np.argmin(np.abs(y - pts))]
96         e = np.angle(y * np.conj(d))
97         out[i] = y
98         phase += freq + alpha * e
99         freq += beta * e
100    return out

```

```

1  """FSK/CPM receive path: constant-envelope gate + frequency-discriminator demod.
2  Runs on the DC-blocked analytic burst, before any carrier work (linear front-end
3  does not apply). Imports only constellation reference data (anti-shared-bug rule)."""
4
5  import numpy as np
6  from scipy.ndimage import uniform_filter1d
7  from scipy.signal import firwin, resample_poly, welch
8
9  from .constellations import fsk_levels, to_bits
10 from .dsp import _kmeans1d, _parab, analytic, mix
11
12 _CV_MAX = 0.24      # envelope coeff-of-variation ceiling. Recalibrated on a broad grid: real
13 # FSK (all h x SNR>=10 x high baud) tops out at cv 0.224, but constant-modulus PSK at very
14 # high baud (fs/baud~3) or near +-fs/2 dips to cv>=0.250 and used to slip the old 0.30 gate
15 # (false FSK). 0.24 sits in the 0.026 gap -> keeps every real FSK, rejects the PSK misfires.
16 _SEP_MIN = 3.1      # instantaneous-freq 2-means separation floor (FSK>=3.39, rest<=2.85)
17 _K4_RATIO = 2.6      # sp(k=2)/sp(k=4) collapse above which 4 levels beat 2
18 _MIN_SYMS = 8        # fewer symbols than this cannot cluster 2/4 levels (empty-quantile crash)
19 _PROM_MIN = 10.0     # a symbol-rate line peak/median below this is unreliable -- BUT only confident-
20 #                    wrong when paired with too few symbols (a long low-index burst has a weak line
21 #                    yet a correct baud). The sweep sits at prom 23-73.
22 _NSYM_MIN = 200     # ...so reject a weak line only below this symbol count. A 100 floor was tried
23 #                    (to recover more short bursts) but an independent grid found weak-prominence
24 #                    half-rate folds in the 100-199 band that still cluster tight enough to lock >=60
25 #                    (msk h=0.5 baud 24-32% off, lock 61-65) -- confident-wrong. 200 keeps them out;
26 #                    precision over recall (the short-burst confident-wrong is the cardinal sin).
27 _DECIM_FRAC = 0.30  # decimate a heavily-oversampled burst so the tones fill ~this of Nyquist
28 _DECIM_MIN = 3       # ...but only when that needs a factor >= this, so the sps~10 demod grid
29 #                    (fsk2/fsk4/msk baud=fs/10) is untouched -> the sweep stays byte-identical
30
31
32 def _bw99(x: np.ndarray, fs: float) -> float:
33     """Two-sided 99%-power occupied bandwidth (Hz), noise-pedestal subtracted."""
34     f, p = welch(x, fs=fs, nperseg=min(4096, x.size), return_onesided=False, detrend=False)
35     f, p = np.fft.fftshift(f), np.fft.fftshift(p)
36     p = np.maximum(p - np.median(p), 0.0)
37     cs = np.cumsum(p) / (p.sum() + 1e-30)
38     lo = f[min(np.searchsorted(cs, 0.005), f.size - 1)] # clamp BOTH edges: a degenerate PSD
39     hi = f[min(np.searchsorted(cs, 0.995), f.size - 1)] # (all-zero after floor subtraction)
40     return float(abs(hi - lo))                          # otherwise indexes one past the end
41
42
43 def _dcblock(x: np.ndarray) -> np.ndarray:
44     """Complex128 analytic burst, DC-blocked (front-end convention)."""
45     x = analytic(x)
46     return x - x.mean()
47
48
49 def _ifreq(x: np.ndarray, fs: float) -> np.ndarray:
50     """Instantaneous frequency (Hz) from sample-to-sample phase differences."""
51     return np.angle(x[1:] * np.conj(x[:-1])) * fs / (2 * np.pi)
52
53
54 def _sep(a: np.ndarray) -> float:
55     """2-means between/within separation ratio of a 1-D sample set."""
56     c, _, sp = _kmeans1d(a, 2)
57     return abs(c[1] - c[0]) / (sp + 1e-9)
58
59
60 def _est_baud(f: np.ndarray, fs: float) -> tuple[float, float]:

```

```

61 """Symbol rate from the spectral line of |diff(smoothed f)| (transition impulses),
62 harmonic-folded to the fundamental (2-level lines alias to 2*baud). Returns
63 (baud, prominence) where prominence = peak/median of the in-band line: a weak line
64 (few symbols, no clean symbol clock) means the baud is UNRELIABLE -> caller rejects."""
65 d = np.abs(np.diff(uniform_filter1d(f, 2)))
66 d = d - d.mean()
67 n = 1 << int(np.ceil(np.log2(d.size)))
68 spec = np.abs(np.fft.rfft(d * np.hanning(d.size), n))
69 freqs = np.fft.rfftfreq(n, 1 / fs)
70 df = freqs[1] - freqs[0]
71 lo, hi = 0.002 * fs, 0.45 * fs
72 idx = np.where((freqs >= lo) & (freqs <= hi))[0]
73 prom = float(spec[idx].max() / (np.median(spec[idx]) + 1e-20)) if idx.size else 0.0
74 k = idx[int(np.argmax(spec[idx]))]
75
76 def _hp(kk: int) -> float:                                     # harmonic-comb energy: the TRUE fundamental
77     """return float(sum(spec[m * kk] for m in range(1, 6) if m * kk < spec.size)) # captures most
78     # in-range harmonics ONLY: clamping out-of-range ones to the last bin multi-counted the
79     # Nyquist bin, inflating _hp at a 2*baud peak enough to veto the legitimate fold -> 2x baud.
80
81 for div in (3, 2):
82     ks = int(round(freqs[k] / div / df))
83     if freqs[ks] >= lo:
84         w = spec[max(0, ks - 2):ks + 3]
85         kk = max(0, ks - 2) + int(np.argmax(w)) if w.size else k
86         # fold to the sub-harmonic ONLY if it is genuinely the fundamental -- its harmonic comb
87         # must beat the current peak's. A real baud/2 fundamental captures MORE harmonic lines
88         # than the 2*baud peak; spurious low-h/low-SNR energy at baud/2 does not (it once
89         # slipped the 0.34 gate and folded baud -> baud/2: a confident WRONG decode). The comb
90         # guard vetoes that. On the sps~10 grid the fold never fired (0.34 unmet) -> sweep same.
91         if w.size and w.max() > 0.34 * spec[k] and _hp(kk) >= _hp(k):
92             k = kk
93     return float(freqs[k] + _parab(20 * np.log10(spec + 1e-20), k) * df), prom
94
95
96 def fsk_gate(x: np.ndarray, fs: float) -> bool:
97     """True when the burst looks like FSK/CPM: near-constant envelope AND a bimodal
98     instantaneous frequency. Calibrated on generate() over all GEN_MODS x seeds 0..3
99     x snr {25,15}: every linear mod fails one test with margin -- bpsk/dbpsk have
100     cv~0.40 (>_CV_MAX), the rest have freq-sep<=2.85 (<_SEP_MIN); fsk2/fsk4/msk keep
101     cv<=0.22 and freq-sep>=3.39 down to snr 10 (margins ~0.08 and ~0.29).
102     Recenter by the mean instantaneous frequency first: at a large carrier offset a
103     linear mod's phase-transition spikes wrap past +/-fs/2 and fake a second mode."""
104     x = _dcblock(x)
105     a = np.abs(x)
106     if a.std() / (a.mean() + 1e-12) >= _CV_MAX:
107         return False
108     f = _ifreq(x, fs)
109     x = x * np.exp(-2j * np.pi * np.mean(f) / fs * np.arange(x.size))
110     return bool(_sep(uniform_filter1d(_ifreq(x, fs), 4)) > _SEP_MIN)
111
112
113 def analyze_fsk(x: np.ndarray, fs: float) -> dict:
114     """Full FSK/MSK demod by frequency discrimination. Returns mod, fc, baud, h,
115     lock (0-100), level_freqs (Hz, sorted, centered), symbols (normalized f per
116     symbol) and Gray-demapped bits."""
117     x = _dcblock(x)
118     f = _ifreq(x, fs)
119     # A carrier OFFSET (tones sit at fc +/- dev, not around DC) corrupts the symbol-rate estimate two
120     # ways -- off-centre it trips _est_baud's harmonic fold, and the tones fall outside the decim

```

```

121 # LPF passband -- either way a confident WRONG baud. Recentre the discriminator on the carrier
122 # (mean instantaneous freq) first; only for a significant offset, so a centred burst
123 # (fc~0, the whole demod/sweep grid) is byte-identical. Levels are already centred downstream.
124 fc0 = float(np.mean(f))
125 if abs(fc0) > 0.005 * fs:
126     x = mix(x, fs, fc0)
127     f = _ifreq(x, fs)
128 else:
129     fc0 = 0.0
130 baud, prom = _est_baud(f, fs)
131 # heavy oversampling (fs/baud >> 10) buries the symbol-rate line under broadband transition
132 # artifacts -> _est_baud locks a spurious peak (baud ~25% off) while the clean tones still
133 # cluster (lock ~94): a confident WRONG decode. Re-estimate baud on a decimated copy (tones ->
134 # ~_DECIM_FRAC of Nyquist, as the survey channelizer does). Gated at _DECIM_MIN so the sps~10
135 # grid is untouched (fsk2/fsk4/msk baud=fs/10 -> d<3 -> no re-estimate -> sweep byte-identical).
136 bw = _bw99(x, fs)
137 d = int(_DECIM_FRAC * fs / bw) if bw > 0 else 1
138 if d >= _DECIM_MIN:
139     xd = np.convolve(x, firwin(129, min(0.9 * (fs / d) / 2, bw) / (fs / 2)), "same")
140     baud, prom = _est_baud(_ifreq(resample_poly(xd, 1, d), fs / d), fs / d)
141 if baud <= 0:
142     raise ValueError("FSK 버스트에서 심볼율을 추정할 수 없습니다 (너무 짧음)")
143 sps = fs / baud
144 f = uniform_filter1d(f, max(1, int(round(sps / 2))))
145 resid = float(f.mean())
146 fc = fc0 + resid # absolute carrier = recentre offset + in-band residual
147 fcen = f - resid
148 nsym = int(fcen.size / sps) - 1
149 # a WEAK symbol-rate line (prom < _PROM_MIN) is unreliable ONLY when symbols are also too few to
150 # average (nsym < _NSYM_MIN): a short burst's spurious peak still clusters tight (few points ->
151 # high lock) = a confident WRONG baud. A long low-index (h~0.35) burst also has a weak line but
152 # MANY symbols -> its baud is correct -> NOT rejected. Sweep: nsym~3000/prom 23-73 -> untouched.
153 if nsym < _MIN_SYMS or (prom < _PROM_MIN and nsym < _NSYM_MIN):
154     raise ValueError(f"FSK 버스트가 너무 짧거나 심볼율이 불확실합니다 ({nsym} 심볼)")
155 ctr = (np.arange(nsym) + 0.5) * sps
156
157 # symbol sampling phase: the offset whose sampled levels separate best
158 v, best = fcen[:nsym], -1.0
159 for o in np.linspace(0, sps, 8, endpoint=False):
160     s = fcen[np.clip((ctr + o).astype(int), 0, fcen.size - 1)]
161     sep = _sep(s)
162     if sep > best:
163         best, v = sep, s
164
165 # level count via spread-collapse (k=2 vs k=4), then cluster
166 k = 4 if _kmeans1d(v, 2)[2] / (_kmeans1d(v, 4)[2] + 1e-9) > _K4_RATIO else 2
167 c, lab, spread = _kmeans1d(v, k)
168 order = np.argsort(c)
169 ranks = np.argsort(order)[lab] # per-symbol level index, 0 = lowest freq
170 csort = c[order]
171 mid = float(csort.mean()) # debias center: symmetric levels sum ~0
172 fc, csort, v = fc + mid, csort - mid, v - mid
173 spacing = float(np.mean(np.diff(csort)))
174 h = spacing / baud
175
176 mod = "msk" if (k == 2 and abs(h - 0.5) < 0.15) else f"fsk{k}"
177 _, glabels = fsk_levels(mod)
178 bits = to_bits(glabels[ranks], mod).astype(np.uint8)
179 lock = float(np.clip((spacing / (spread + 1e-9) - 1.5) / 6.5 * 100, 0, 100))
180 return dict(mod=mod, fc=fc, baud=baud, h=h, lock=lock, level_freqs=csort,

```


181 |

symbols=v / (spacing / 2), bits=bits)

```

1  """Blind linear-chirp / CSS (LoRa) detection + characterization. A constant-envelope
2  signal whose instantaneous frequency ramps linearly de-chirps to a tone: the delay-
3  conjugate  $x[t] \cdot \text{conj}(x[t-\tau])$  is a beat tone at  $\mu \cdot \tau$  (NONZERO), unlike FSK/CW (beat at
4  DC), PSK/QAM (not constant-envelope), or FM-voice/noise (broadband). Characterize-only --
5  reported like analog/tone, NEVER demodulated: blind CSS symbol decode needs preamble
6  CF0/ST0 sync plus LoRa's reverse-engineered Gray/interleave/Hamming/whitening framing.
7  Calibrated on gen chirps vs every other family: chirp beat-PAR $\geq$ 936, non-chirp $\leq$ 264."""
8
9  import numpy as np
10 from scipy.ndimage import uniform_filter1d
11 from scipy.signal import welch
12
13 _PAR_MIN = 400.0    # delay-conjugate beat-tone peak-to-mean floor (chirp $\geq$ 936, non-chirp $\leq$ 264)
14
15
16 def _bw99(x: np.ndarray, fs: float) -> float:
17     """99%-power occupied bandwidth (a chirp fills its band ~flat). Floor-subtracted so the
18     noise pedestal in the channel's LPF passband does not inflate the width."""
19     f, p = welch(x, fs=fs, nperseg=min(4096, x.size), return_onesided=False, detrend=False)
20     f, p = np.fft.fftshift(f), np.fft.fftshift(p)
21     p = np.maximum(p - np.median(p), 0.0)    # remove the noise pedestal
22     cs = np.cumsum(p) / (p.sum() + 1e-30)
23     lo = f[min(np.searchsorted(cs, 0.005), f.size - 1)] # clamp BOTH edges: a degenerate PSD
24     hi = f[min(np.searchsorted(cs, 0.995), f.size - 1)] # (all-zero after floor subtraction)
25     return float(abs(hi - lo))                # otherwise indexes one past the end
26
27
28 def _beat(x: np.ndarray, fs: float) -> tuple[float, float]:
29     """(peak-to-mean,  $\mu$ [Hz/s]) of the DC-nulled delay-conjugate spectrum."""
30     x = x - x.mean()
31     tau = max(1, x.size // 200)
32     y = x[tau:] * np.conj(x[:-tau])
33     nf = 1 << int(np.ceil(np.log2(y.size)))
34     spec = np.abs(np.fft.fftshift(np.fft.fft((y - y.mean()) * np.hanning(y.size), nf))) ** 2
35     fr = np.fft.fftshift(np.fft.fftfreq(nf, 1 / fs))
36     spec[np.abs(fr) < 0.002 * fs] = 0.0    # null DC band: FSK/CW/tone beat here, chirps do not
37     k = int(np.argmax(spec))
38     return float(spec[k] / (spec.mean() + 1e-30)), float(fr[k] * fs / tau)
39
40
41 def is_chirp(x: np.ndarray, fs: float) -> bool:
42     """True when a channel is a linear chirp / CSS (LoRa/FMCW). The beat-tone PAR is the
43     discriminator: a huge margin (chirp $\geq$ 936 vs PSK/QAM $\leq$ 94, FSK $\leq$ 91, FM $\leq$ 264, noise $\leq$ 13)
44     makes an envelope-CV pre-gate redundant AND harmful (it rejects noisy-but-clear chirps)."""
45     if x.size < 256:
46         return False
47     return _beat(x, fs)[0] >= _PAR_MIN
48
49
50 _SWEEP_PM_MAX = 1.45    # instantaneous-freq histogram peak-to-mean: a band-sweep stays ~1
51 _SWEEP_OCC_MIN = 0.6    # ...and fills its band; FSK sits on 2-4 discrete tones (peaky, sparse)
52
53
54 def sweeps_band(x: np.ndarray, fs: float, bins: int = 40) -> bool:
55     """True when the instantaneous frequency SWEEPS the channel (linear chirp / CSS) rather
56     than hopping between a few discrete FSK tones. A linear FMCW ramp reads bimodal to
57     fsk_gate AND trips is_chirp's beat test, so triage needs this to tell a genuine band-sweep
58     from FSK: a sweep's IF histogram is flat and fully occupied; FSK's spikes at its tones.
59     Calibrated on extracted channels: 0/222 FSK/MSK (snr 8-40) pass, 214/216 FMCW + 60/60 LoRa
60     pass (misses only the narrowest, gentlest ramps -- which stay 'fsk', never a regression).

```

```

61     BOTH the raw and a lightly-smoothed IF must look like a sweep: at moderate SNR (~14) noise
62     flattens integer-h FSK's raw histogram to sweep-like pm ~1.44 (knife-edge), but smoothing
63     restores its tone spikes (pm 2.5+) while a genuine ramp stays flat (pm ~1.1) -- and at LOW
64     SNR smoothing can flatten FSK instead, which the raw view still catches. The conjunction
65     only ever narrows 'sweep', so every calibrated chirp above keeps its label (margin >=0.3)."""
66     if x.size < 256:
67         return False
68     x = x - x.mean()
69     f0 = np.diff(np.unwrap(np.angle(x))) * fs / (2 * np.pi)
70     for f in (f0, uniform_filter1d(f0, 4)):
71         lo, hi = np.percentile(f, 1), np.percentile(f, 99)
72         if hi - lo < 1:
73             return False
74         fv = f[(f >= lo) & (f <= hi)]
75         h = np.histogram(fv, bins=bins)[0] / fv.size
76         pm = h.max() / (h.mean() + 1e-30)          # discrete tones spike this; a sweep ~1
77         occ = float((h > 0.005).mean())           # a sweep fills the band; tones occupy few bins
78         if not (pm < _SWEEP_PM_MAX and occ >= _SWEEP_OCC_MIN):
79             return False
80     return True
81
82
83 def analyze_chirp(x: np.ndarray, fs: float) -> dict:
84     """Characterize a chirp: {mu[Hz/s], up, bw, par, sf, rs, tsym}. A LoRa hypothesis
85     (sf, symbol rate, symbol time) is filled when bw^2/|mu| snaps to 2^(7..12); otherwise
86     it stays a generic linear chirp / FMCW. bw is measured on the channel, not asserted."""
87     par, mu = _beat(x, fs)
88     bw = _bw99(x, fs)
89     sf = rs = tsym = None
90     if mu != 0 and bw > 0:
91         r = float(np.log2(bw * bw / abs(mu)))
92         if 7 <= round(r) <= 12 and abs(r - round(r)) < 0.3:    # power-of-two chip count -> LoRa
93             sf = int(round(r))
94             rs = abs(mu) / bw
95             tsym = 2 ** sf / bw
96     return {"mu": mu, "up": bool(mu > 0), "bw": float(bw),
97             "par": par, "sf": sf, "rs": rs, "tsym": tsym}

```

```

1  """End-to-end blind demodulation. Two families:
2      linear (PSK/QAM): front-end -> carrier -> baud -> matched filter -> timing ->
3          ..... classify -> fine sync -> [equalizer rescue] -> quality
4      fsk (CPFSK/MSK): frequency discriminator (gated by constant envelope)
5  Classification runs BEFORE fine sync so the decision-directed loop knows the
6  constellation. Differential demapping is an option, not a detection: dqpsk and
7  qpsk share a constellation."""
8
9  import json
10 from dataclasses import dataclass, field
11
12 import numpy as np
13
14 from . import classify as cl
15 from . import dsp, triage
16 from .channelize import extract
17 from .chirp import analyze_chirp, is_chirp, sweeps_band
18 from .constellations import demap_bits, demap_diff_bits, mod_order
19 from .detect import Detection, detect
20 from .eq import equalize, equalize_fse
21 from .fsk import analyze_fsk, fsk_gate
22 from .sigio import Meta, parse_name, read
23 from .spectrum import spectrum, waterfall
24 from .sync import find_preamble
25
26 _BITS_CAP = 65536
27 _EQ_LOCK = 60.0    # below this a linear signal is worth an equalizer attempt
28 _EQ_GAIN = 3.0     # ...keep it only if lock improves by this much
29 _EQ_FLOOR = 50.0   # ...and clears this floor, so a wrong-mod fit is never locked in
30 _EQ_QAM_FLOOR = 60.0 # ...but a DENSE-QAM (32/64) eq fit must clear a higher bar: the DD loop can
31 #                   drag a lower-order cloud onto a 32/64 lattice at a marginal lock (~51), a
32 #                   confident WRONG order. Genuine multipath recoveries clear ~72; the sweep's
33 #                   eq cases are qpsk/16qam (never 32/64 via eq) so this bar is byte-identical.
34 _BAUD_GAIN = 5.0   # an in-band baud hypothesis must beat the global one by this lock
35 _SHORT_LOCK = 60.0 # below this, retry the baud line with fewer blocks (short-burst rescue)
36 _SYNC_LOCK = 60.0  # below this, try a repeated-preamble carrier/timing sync (packetized bursts)
37 _ALIAS_MARGIN = 10.0 # a preamble carrier alias must beat the next by this lock, else ambiguous.
38 #                   This is the LOAD-BEARING rotation guard: even when the coarse anchor itself
39 #                   aliases and validates a rotation at ~80 lock, that rotation wins by a thin
40 #                   margin (~8) -- so a 10-lock margin refuses it. Because the margin catches it,
41 #                   the lock floor can stay modest (78, not 82) -> ~5% more genuine recoveries.
42 _SYNC_ACCEPT = 78.0 # ...AND clear this absolute lock: below it correct/rotated aliases overlap. A
43 #                   65 floor was tried (to recover more moderate bursts) but an independent
44 #                   snr-12 grid found 8psk rotated aliases locking 71-72 in the opened 65-78 band
45 #                   (carrier off by one baud/L step -> confident garbage bits). Even 78 leaks a
46 #                   a few hard-corner rotations (a separate pre-existing gate limit): hold at 78.
47 _SYNC_ADIST = 1.5  # ...AND sit within this many alias steps of the coarse est_carrier fc -- a
48 #                   soft backstop to the margin: it catches the one rotation the margin misses,
49 #                   but tuned LOOSE (1.5, not 0.75) so it does not needlessly reject genuine
50 #                   large-offset recoveries. The three together give 0 confident-wrong across
51 #                   1920 hard-corner bursts at the max recall this gate reaches.
52 _DIFF_OF = {"bpsk": "dbpsk", "qpsk": "dqpsk"} # pi4dqpsk arrives as qpsk -> dqpsk
53
54
55 @dataclass
56 class Result:
57     meta: Meta
58     family: str          # 'linear' | 'fsk'
59     burst: tuple[int, int]
60     fc: float

```

```

61     baud: float
62     mod: str
63     lock: float
64     symbols: np.ndarray          # aligned symbols (linear) / normalized levels (fsk)
65     bits: np.ndarray
66     iq_corr: np.ndarray          # exportable corrected baseband
67     burst_x: np.ndarray = field(repr=False, default=None) # for spectrum views
68     symmetry: int = 0
69     carrier_ambiguous: bool = False
70     baud_conf: float = 0.0
71     rolloff: float | None = None
72     h: float | None = None       # FSK modulation index
73     evm: float = float("nan")
74     mer_db: float = float("nan")
75     occupied: int = 0
76     snr_est: float = float("nan")
77     eq_applied: bool = False
78     eq_mode: str | None = None   # 'sym' | 'fse'
79     alias_resolved: bool = False
80     baud_fallback: bool = False
81     bursts: list = field(default_factory=list)
82     burst_idx: int = 0
83     preamble: object = None     # sync.Preamble when a repeated preamble drove the decode
84
85     def to_json(self, max_points: int = 6000, views: bool = True) -> dict:
86         z = np.nan_to_num(self.symbols) # a dead/DC capture -> NaN symbols -> invalid JSON
87         if z.size > max_points: # keep TIME ORDER: the UI animates this sequence
88             z = z[np.linspace(0, z.size - 1, max_points).astype(int)]
89         det = {"fc": round(self.fc, 3), "baud": round(self.baud, 3), "mod": self.mod,
90              "symmetry": self.symmetry, "carrier_ambiguous": self.carrier_ambiguous,
91              "baud_conf": round(self.baud_conf, 1),
92              "alias_resolved": self.alias_resolved,
93              "baud_fallback": self.baud_fallback}
94         det["rolloff"] = None if self.rolloff is None else round(self.rolloff, 3)
95         rf = self.meta.rf_center # real RF = capture centre + baseband fc
96         det["rf_hz"] = None if rf is None else round(rf + self.fc, 3)
97         if self.h is not None:
98             det["h"] = round(self.h, 3)
99         if self.preamble is not None: # a repeated preamble drove the sync
100             p = self.preamble
101             det["preamble"] = {"period": p.period, "cfo_hz": round(p.cfo_hz, 1),
102                               "conf": round(p.conf, 2), "start": p.start, "end": p.end}
103         doc = {
104             "fs": self.meta.fs, "fmt": self.meta.fmt, "dtype": self.meta.dtype, # dtype 은 한 줄
105             "rf_center": self.meta.rf_center, # 보고에 필수: lock 0 의 1순위 용의자다
106
107             "family": self.family, "n_samples": int(self.iq_corr.size),
108             "burst": {"start": self.burst[0], "end": self.burst[1]},
109             "bursts": [{"start": s, "end": e} for s, e in self.bursts],
110             "burst_idx": self.burst_idx,
111             "detected": det,
112             "quality": {"lock": _r(self.lock, 1) or 0.0, "mer_db": _r(self.mer_db),
113                        "evm": _r(self.evm, 4), "occupied": int(self.occupied)},
114             "snr_est_db": _r(self.snr_est),
115             "eq": {"applied": self.eq_applied, "mode": self.eq_mode},
116             "constellation": {"i": np.round(z.real, 4).tolist(),
117                              "q": np.round(z.imag, 4).tolist()},
118             "bits": "".join(map(str, self.bits[:_BITS_CAP])),
119         }
120         if views and self.burst_x is not None:

```

```

121         doc["spectrum"] = spectrum(self.burst_x, self.meta.fs)
122         doc["waterfall"] = waterfall(self.burst_x, self.meta.fs)
123     return doc
124
125     def save_iq(self, path: str) -> None:
126         np.column_stack([self.iq_corr.real, self.iq_corr.imag]).astype("<f4").tofile(path)
127
128     def save_symbols(self, path: str) -> None:
129         np.save(path, self.symbols.astype(np.complex64))
130
131     def save_bits(self, path: str, packed: bool = False) -> None:
132         if packed:
133             np.packbits(self.bits).tofile(path)
134         else:
135             with open(path, "w") as fh:
136                 fh.write("".join(map(str, self.bits)))
137
138     def save_report(self, path: str) -> None:
139         with open(path, "w") as fh:
140             json.dump(self.to_json(), fh, indent=1)
141
142
143     def _r(v: float, nd: int = 2) -> float | None:
144         return None if v is None or not np.isfinite(v) else round(float(v), nd)
145
146
147     def _fine(z: np.ndarray, mod: str) -> np.ndarray:
148         """Bulk-phase removal, decision-directed carrier loop, final rotation lock."""
149         return cl.align(dsp.ddsync(cl.align(z, mod), mod), mod)
150
151
152     def _blocks(n: int) -> int:
153         """M-th-power spectra are averaged INCOHERENTLY, so a weak tone (high symmetry,
154         low SNR) wants fewer, longer segments; only long records need more for
155         stationarity. Worth ~2 dB at the 8PSK edge over a fixed blocks=4."""
156         return int(np.clip(n // 30000, 1, 8))
157
158
159     def _demod(xd: np.ndarray, fs: float, baud: float, symmetry: int) -> tuple:
160         """Matched filter + timing + classify + fine sync at one baud hypothesis."""
161         alpha = dsp.est_rolloff(xd, fs, baud)
162         ym = dsp.matched(dsp.to_sps(xd, fs, baud), 4, alpha)
163         raw = dsp.timing(ym, 4)
164         mod = cl.classify(raw, symmetry)
165         # align BEFORE ddsync: the coarse carrier leaves only micro-Hz residual, so removing
166         # the bulk phase first kills the loop transient that would corrupt the first symbols
167         syms = _fine(raw, mod)
168         return alpha, ym, raw, mod, syms, cl.quality(syms, mod)
169
170
171     def _rescue(eq_fn, z: np.ndarray, seed_mod: str, symmetry: int) -> tuple:
172         """Equalize with a seed constellation, then let the OPENED eye pick the mod:
173         heavy ISI corrupts the ring test and even the 2/8 symmetry vote, so re-classify
174         with the estimated symmetry AND the neutral order-4 gate, keep the best lock."""
175         z1 = eq_fn(z, seed_mod)
176         cand_s = []
177         for m in {cl.classify(z1, symmetry), cl.classify(z1, 4)}: # symmetry too, per the docstring
178             s = _fine(z1 if m == seed_mod else eq_fn(z, m), m)
179             cand_s.append((cl.quality(s, m), s, m))
180         best = max(cand_s, key=lambda c: c[0].lock)

```

```

181     # Occam de-fold for the PSK point-doubling: CMA is modulus-only, so a bpsk signal under a
182     # multipath echo ALSO fits a qpsk ring at a marginally higher lock (phantom 2->4 points).
183     # A bpsk candidate that clears the accept floor cannot be an over-fit of qpsk, so prefer it.
184     # Only fires when symmetry==2 put bpsk in the set; a genuine qpsk reads symmetry 4 -> the
185     # candidate set is the qpsk singleton -> this branch is unreachable and it stays untouched.
186     lo = [c for c in cands if c[2] == "bpsk" and c[0].lock >= _EQ_FLOOR]
187     if lo and best[2] == "qpsk":
188         return lo[0]
189     return best
190
191
192 def analyze(x: np.ndarray, meta: Meta, diff: bool = False,
193            burst: int | None = None) -> Result:
194     """Run the blind chain. `diff` demaps phase transitions (D-BPSK/D-QPSK);
195     `burst` selects one detected burst (default: the most energetic)."""
196     if x.size:
197         # pathological UNIFORM scale must be fixed BEFORE analytic(): its magnitude sanitizer
198         # zeroes samples above 1e150 sample-WISE, so a whole capture near/over that ceiling was
199         # wiped wholesale (garbage mods, lock=nan at 1e154) before the rms-normalize below could
200         # ever help. The 99th percentile is robust to spike outliers (a single corrupt 1e300
201         # sample leaves it at signal scale -> untouched here, still zeroed sample-wise later);
202         # normal captures sit far inside (1e-100, 1e100) -> untouched -> sweep byte-identical.
203         scale = float(np.percentile(np.abs(x[:16]), 99)) # strided: scale detection needs the
204         # BULK magnitude, not every sample -- 16x cheaper on a large direct capture
205         if np.isfinite(scale) and scale > 0 and not (1e-100 < scale < 1e100):
206             x = x / scale
207     x = dsp.analytic(x)
208     x = x - x.mean() # DC block: LO leakage otherwise corrupts every estimator
209     if x.size < 64: # empty / trivially short: nothing to estimate, and it crashes downstream
210         raise ValueError(f"신호가 너무 짧습니다 ({x.size} 샘플)")
211     rms = float(np.sqrt(np.mean(np.abs(x) ** 2)))
212     if rms and (rms < 1e-6 or rms > 1e6): # only PATHOLOGICAL scale: normalize so the scale-variant
213         x = x / rms # spots (fsk_gate's CV epsilon, est_carrier's x*p over/
214         # underflow) stay invariant. Normal captures rms~0(1) ->
215         # untouched -> the acceptance sweep is byte-identical.
216     bursts = dsp.find_bursts(x, meta.fs)
217     if burst is None or not 0 <= burst < len(bursts):
218         burst = int(np.argmax([np.sum(np.abs(x[s:e]) ** 2) for s, e in bursts]))
219     s, e = bursts[burst]
220     xb = x[s:e] if e - s >= 64 else x
221
222     # chirp gate -- same conjunctive tie-break as triage.family, so direct analyze / survey_web
223     # single mode agree with survey's label: an FMCW/CSS sweep trips fsk_gate (bimodal IF) and
224     # would force-demodulate into confident garbage FSK. sweeps_band keeps real FSK (0/222 pass)
225     # out of this branch, so only a genuine band-sweep is refused -- never demodulated.
226     if is_chirp(xb, meta.fs) and sweeps_band(xb, meta.fs):
227         info = analyze_chirp(xb, meta.fs)
228         kind = f"LoRa 추정 SF{info['sf']}" if info['sf'] else f"μ={info['mu']:.3g} Hz/s"
229         raise ValueError(f"처프(FMCW/CSS) 신호입니다 ({kind}, bw≈{info['bw']:.3g} Hz) - "
230             f"성상도 복조 대상이 아닙니다 (서베이 모드에서 특성 보고)")
231
232     if fsk_gate(xb, meta.fs):
233         r = analyze_fsk(xb, meta.fs)
234         sym = np.asarray(r["symbols"], dtype=np.complex128)
235         return Result(meta, "fsk", (s, e), r["fc"], r["baud"], r["mod"], r["lock"],
236             sym, r["bits"], xb, burst_x=xb, h=r["h"],
237             bursts=bursts, burst_idx=burst)
238
239     fc0, symmetry, _ = dsp.est_carrier(xb, meta.fs, blocks=_blocks(xb.size))
240     fc = dsp.resolve_alias(xb, meta.fs, fc0, symmetry)

```

```

241     amb = abs(symmetry * fc) > 0.4 * meta.fs
242     xd = dsp.mix(xb, meta.fs, fc)
243
244     baud, conf = dsp.est_baud(xd, meta.fs)
245     fell = False
246     alpha, ym, raw, mod, syms, q = _demod(xd, meta.fs, baud, symmetry)
247     bw = dsp.occupied_bw(xd, meta.fs)
248     if bw and not (0.72 * bw <= baud <= 1.05 * bw):
249         # the global |x|^2 peak disagrees with the occupied band: below alpha~0.1
250         # (and under harsh channels) data junk outruns the true line. Demod the
251         # in-band hypotheses too and let lock decide, with a clear margin.
252         for lo in (0.80, 0.67):
253             b2, c2 = dsp.est_baud(xd, meta.fs, lo=lo * bw, hi=1.02 * bw)
254             if abs(b2 - baud) <= 0.01 * b2:
255                 continue
256             t = _demod(xd, meta.fs, b2, symmetry)
257             if t[5].lock > q.lock + _BAUD_GAIN:
258                 alpha, ym, raw, mod, syms, q = t
259                 baud, conf, fell = b2, c2, True
260
261     if q.lock < _SHORT_LOCK:
262         # short / low-SNR burst rescue: the default 4-block |x|^2 line splits the record
263         # too finely and locks a junk peak. Fewer, longer blocks give a stronger line (the
264         # carrier_blocks logic, applied to baud). Run both and keep only a clearly-better
265         # lock -- so a long, already-locked signal (CORE) never enters here and is untouched.
266         for nb in (2, 1):
267             b2, c2 = dsp.est_baud(xd, meta.fs, blocks=nb)
268             if abs(b2 - baud) <= 0.01 * b2:
269                 continue
270             t = _demod(xd, meta.fs, b2, symmetry)
271             if t[5].lock > q.lock + _BAUD_GAIN:
272                 alpha, ym, raw, mod, syms, q = t
273                 baud, conf, fell = b2, c2, True
274
275     eq_applied, eq_mode, pre = False, None, None
276     if q.lock < _EQ_LOCK: # ISI (multipath) is what a phase-only loop cannot fix
277         # CMA is modulus-only, so it opens the eye without knowing the constellation.
278         qe, se, me = _rescue(lambda z, m: equalize(z, m), raw, mod, symmetry)
279         mode = "sym"
280         if qe.lock < max(_EQ_FLOOR, q.lock + _EQ_GAIN):
281             # symbol-spaced FIR could not invert the channel; retry T/2-spaced on
282             # the clock-tracked 2 sps stream (unaliasd spectrum, longer inverse)
283             q2, s2, m2 = _rescue(lambda z, m: equalize_fse(z, m),
284                                   dsp.timing(ym, 4, out=2), mod, symmetry)
285             if q2.lock > qe.lock:
286                 qe, se, me, mode = q2, s2, m2, "fse"
287         floor = _EQ_QAM_FLOOR if me in ("32qam", "64qam") else _EQ_FLOOR
288         if qe.lock > q.lock + _EQ_GAIN and qe.lock >= floor:
289             syms, mod, q, eq_applied, eq_mode = se, me, qe, True, mode
290     elif mod in ("16qam", "32qam", "64qam"):
291         # confident square-QAM at high lock: a benign post-echo can fold a LOWER-order signal
292         # onto a QAM lattice with a clean-looking eye, so the low-lock rescue above never fires.
293         # Equalize once and accept ONLY if a strictly lower order emerges -- verified never to
294         # demote a genuine QAM (0/36 across 16/32/64qam x snr x seeds), so real QAM is untouched.
295         qe, se, me = _rescue(lambda z, m: equalize(z, m), raw, mod, symmetry)
296         mode = "sym"
297         if mod_order(me) < mod_order(mod) and qe.lock < _EQ_FLOOR:
298             # a lower order emerged but the symbol-spaced eye stayed shut (echo > ~2 symbols);
299             # retry T/2 fractionally-spaced. Only runs once a de-fold is INDICATED, so a genuine
300             # QAM (no lower order) never pays for the FSE.

```



```

301         qe, se, me = _rescue(lambda z, m: equalize_fse(z, m),
302                               dsp.timing(ym, 4, out=2), mod, symmetry)
303         mode = "fse"
304         if mod_order(me) < mod_order(mod) and qe.lock >= _EQ_FLOOR:
305             syms, mod, q, eq_applied, eq_mode = se, me, qe, True, mode
306
307     if q.lock < _SYNC_LOCK:
308         # packetized-burst rescue -- LAST, after the equalizer: a repeated preamble pins the
309         # carrier/ baud/ timing from only a few symbols, where the blind M-th-power estimator (needing
310         # many symbols to average) fails. Runs only if the eq rescue above did NOT lift the lock, so
311         # a multipath signal (which the eq handles, and whose ISI can fake a short period) never
312         # reaches here. Detect the preamble, mix by its CFO, demod the DATA past it, keep-best -- a
313         # random / no-preamble signal yields no plateau or a CFO that does not improve lock and is
314         # dropped (the sweep stays byte-identical). The preamble period P = L*sps supplies the baud
315         # (fs*L/P per block length L), the phase slope gives the carrier (+fs/P resolves the L-fold
316         # alias); every PSK/QAM symmetry is tried -- the right (carrier, baud, sym) locks clean.
317         ps = find_preamble(xb, meta.fs, baud_hint=baud)
318         if ps is not None:
319             # The Schmidl-Cox CFO is ambiguous modulo fs/P = baud/L: an M-PSK carrier off by baud/L
320             # rotates a whole constellation position per symbol, so the eye still looks locked but
321             # the bits shift -> a confident WRONG decode if the wrong alias is trusted. The coarse
322             # M-th-power fc lands within ~1 alias step of the truth even on these short bursts, so
323             # CENTRE the alias scan on it (k0) and sweep +-3 steps -- this reaches the correct alias
324             # for any offset (not just |fc|<baud/2); keep-best over them picks the cleanest eye.
325             step = meta.fs / ps.period
326             k0 = round((fc - ps.cfo_hz) / step)
327             cands = []
328             for b2 in sorted({round(meta.fs * L / ps.period) for L in (3, 4, 6, 8)}):
329                 for k in range(k0 - 3, k0 + 4):
330                     cfo = ps.cfo_hz + k * step
331                     data = dsp.mix(xb, meta.fs, cfo)[ps.end:]
332                     if data.size < 256:
333                         continue
334                     for sm in (2, 4, 8):
335                         t = _demod(data, meta.fs, float(b2), sm)
336                         cands.append((t[5].lock, cfo, float(b2), sm, t))
337             cands.sort(key=lambda c: -c[0])
338             # Adjacent aliases both look like clean M-PSK, so lock alone can pick the WRONG one by a
339             # hair. Require the winner to beat the best DIFFERENT-carrier candidate by a clear gap;
340             # otherwise the alias is genuinely ambiguous -> leave the honest low-lock result rather
341             # than emit a confident wrong decode. (Same-carrier baud/sym variants do not compete.)
342             # PRECISION GATE: the alias ambiguity (carrier off by baud/L) is blind-ambiguous -- a
343             # rotated alias still locks AS HIGH AS the truth (both are clean M-PSK), so lock alone
344             # cannot separate them (a rotated 8psk alias was measured locking 78 with ber 0.38). Two
345             # conjunctive conditions favour precision over recall: (1) a strong absolute lock, and
346             # (2) the winner sits within _SYNC_ADIST alias steps of the coarse est_carrier fc -- a
347             # baud/L rotation is >=1 step off the true carrier, so a far-from-anchor winner is a
348             # rotation. Genuine recoveries have a near-anchor carrier (adist<=0.5, lock 88+); on the
349             # hardest bursts the anchor itself is unreliable -> both conditions fail -> refuse.
350             if cands and cands[0][0] >= _SYNC_ACCEPT and cands[0][0] > q.lock + _EQ_GAIN:
351                 top = cands[0]
352                 other = next((c for c in cands if abs(c[1] - top[1]) > step / 2), None)
353                 anchored = abs(top[1] - fc) <= _SYNC_ADIST * step
354                 if anchored and (other is None or top[0] - other[0] >= _ALIAS_MARGIN):
355                     alpha, ym, raw, mod, syms, q = top[4]
356                     fc, baud, symmetry = top[1], top[2], top[3]
357                     eq_applied, eq_mode, pre = False, None, ps
358                     # diagnostics must describe the ACCEPTED decode, not the abandoned blind
359                     # attempt: the carrier came from the preamble (not resolve_alias -> fc0=fc
360                     # keeps alias_resolved False), the baud from the preamble period (no

```

```

361         # spectral line -> conf 0, no fallback), and ambiguity follows the new fc.
362         fc0, conf, fell = fc, 0.0, False
363         amb = abs(symmetry * fc) > 0.4 * meta.fs
364
365     bits = demap_diff_bits(syms, _DIFF_OF[mod]) if diff and mod in _DIFF_OF \
366     else demap_bits(syms, mod)
367     return Result(meta, "linear", (s, e), fc, baud, mod, q.lock, syms, bits, ym,
368         burst_x=xb, symmetry=symmetry, carrier_ambiguous=amb, baud_conf=conf,
369         rolloff=alpha, evm=q.evm, mer_db=q.mer_db, occupied=q.occupied,
370         snr_est=cl.snr_m2m4(syms, mod), eq_applied=eq_applied, eq_mode=eq_mode,
371         alias_resolved=abs(fc - fc0) > 1.0, baud_fallback=fell,
372         bursts=bursts, burst_idx=burst, preamble=pre)
373
374
375 def analyze_file(path: str, fs: float | None = None, fmt: str | None = None,
376     dtype: str | None = None, endian: str | None = None,
377     bitrev: bool | None = None, diff: bool = False,
378     burst: int | None = None, rf: float | None = None) -> Result:
379     """Analyze a file; explicit args override SigMF/filename tokens."""
380     from .sigio import parse_sigmf
381     m = parse_sigmf(path) or parse_name(path)
382     meta = Meta(fs or m.fs, fmt or m.fmt, dtype or m.dtype,
383         endian or m.endian, m.bitrev if bitrev is None else bitrev,
384         rf_center=rf if rf is not None else m.rf_center)
385     x, meta = read(path, meta)
386     return analyze(x, meta, diff, burst)
387
388
389 # --- wideband survey: many emitters in one capture -----
390
391 @dataclass
392 class Emitter:
393     detection: Detection
394     kind: str # linear|fsk|chirp|analog|tone|tooshort|error
395     abs_fc: float # emitter carrier vs capture centre (Hz)
396     result: Result | None = None # demod result for digital kinds
397     info: dict | None = None # characterization for non-digital kinds (e.g. chirp params)
398
399     def to_json(self) -> dict:
400         d = self.detection
401         doc = {"kind": self.kind, "abs_fc": round(self.abs_fc, 3),
402             "det": {"fc": round(d.fc, 3), "bw": round(d.bw, 3),
403                 "t0": d.t0, "t1": d.t1, "snr_db": _r(d.snr_db),
404                 "baud_hint": round(d.baud_hint, 1)}}
405         if self.info is not None:
406             doc["info"] = self.info
407         if self.result is not None:
408             r = self.result
409             doc.update(mod=r.mod, baud=round(r.baud, 3), lock=round(r.lock, 1),
410                 mer_db=_r(r.mer_db), evm=_r(r.evm, 4), family=r.family)
411         return doc
412
413     def to_detail(self, rf_center: float | None = None) -> dict:
414         """Web drill-down payload: the box summary plus, for a digital emitter, the
415         full analyze result so the UI reuses its single-signal detail view verbatim.
416         The channel is demodulated at baseband (no rf_center), and r.fc is only the
417         residual within-channel offset -- so the emitter's absolute RF is the capture
418         centre plus abs_fc (the full offset from capture centre), injected here."""
419         doc = self.to_json()
420         if self.result is not None:

```

```

421         doc["result"] = self.result.to_json()
422         if rf_center is not None:
423             doc["result"]["detected"]["rf_hz"] = round(rf_center + self.abs_fc, 3)
424         return doc
425
426
427 @dataclass
428 class Survey:
429     meta: Meta
430     emitters: list[Emitter] = field(default_factory=list)
431
432     def to_json(self) -> dict:
433         return {"fs": self.meta.fs, "fmt": self.meta.fmt, "dtype": self.meta.dtype,
434                 "n_emitters": len(self.emitters),
435                 "emitters": [e.to_json() for e in self.emitters]}
436
437
438 def survey(x: np.ndarray, meta: Meta, *, diff: bool = False, **detect_kw) -> Survey:
439     """Detect every emitter in a wideband capture, extract each to its own baseband
440     channel, and demodulate the digital ones with the unchanged single-signal chain.
441     A capture holding one signal that fills the band collapses to a single emitter."""
442     xa = dsp.analytic(x)
443     xa = xa - xa.mean()
444     emitters = []
445     for d in detect(xa, meta.fs, **detect_kw):
446         ch, fs_ch = extract(xa, meta.fs, d)
447         if ch.size < 256:
448             emitters.append(Emitter(d, "tooshort", d.fc))
449             continue
450         kind = triage.family(ch, fs_ch)
451         if kind in ("linear", "fsk"):
452             # triage only gates digital vs not; analyze's own family is authoritative.
453             # Isolate each channel: a degenerate one must not abort the whole survey.
454             try:
455                 r = analyze(ch, Meta(fs_ch, "iq", "f32", "le", False), diff=diff)
456                 emitters.append(Emitter(d, r.family, d.fc + r.fc, r))
457             except Exception:
458                 emitters.append(Emitter(d, "error", d.fc))
459         else:
460             info = analyze_chirp(ch, fs_ch) if kind == "chirp" else None
461             emitters.append(Emitter(d, kind, d.fc, info=info))
462     return Survey(meta, emitters)
463
464
465 def survey_web(x: np.ndarray, meta: Meta, *, diff: bool = False) -> dict:
466     """Web survey payload for /api/survey. When detect sees <=1 emitter the capture
467     is effectively one signal -> 'single' mode returns the UNCHANGED direct analyze()
468     result (correct fc, matches /api/analyze). Otherwise 'survey' mode adds a whole-
469     capture overview waterfall and every emitter's drill-down detail. Additive: does
470     not alter survey()/Survey/Emitter, so CLI + existing tests are unaffected."""
471     xa = dsp.analytic(x)
472     xa = xa - xa.mean()
473     if len(detect(xa, meta.fs)) <= 1:
474         r = analyze(xa, meta, diff=diff)
475         out = {"mode": "single", "result": r.to_json()}
476         if len(r.bursts) > 1: # multi-burst signal -> whole-record map for burst selection
477             out["overview"] = {"n": int(xa.size), "fs": meta.fs,
478                                "waterfall": waterfall(xa, meta.fs)}
479         return out
480     sv = survey(x, meta, diff=diff)

```

```

481     return {"mode": "survey", "fs": meta.fs, "fmt": meta.fmt, "rf_center": meta.rf_center,
482            "overview": {"fs": meta.fs, "n": int(xa.size), "spectrum": spectrum(xa, meta.fs),
483                       "waterfall": waterfall(xa, meta.fs)},
484            "emitters": [e.to_detail(meta.rf_center) for e in sv.emitters]}
485
486
487 def survey_file(path: str, fs: float | None = None, fmt: str | None = None,
488                dtype: str | None = None, endian: str | None = None,
489                bitrev: bool | None = None, diff: bool = False,
490                rf: float | None = None) -> Survey:
491     """Survey a capture file; explicit args override SigMF/filename tokens."""
492     from .sigio import parse_sigmf
493     m = parse_sigmf(path) or parse_name(path)
494     meta = Meta(fs or m.fs, fmt or m.fmt, dtype or m.dtype,
495               endian or m.endian, m.bitrev if bitrev is None else bitrev,
496               rf_center=rf if rf is not None else m.rf_center)
497     x, meta = read(path, meta)
498     return survey(x, meta, diff=diff)

```

```

1  """CLI: analyze FILE / survey / serve. 합성·채점(gen/dataset/sweep)은 개발기 전용 lab.py 가
2  단다 -- 격리망 장비에는 그 파일이 없어 명령 자체가 없다."""
3
4  import argparse
5  import importlib.util
6  import re
7  import sys
8  from zlib import crc32
9
10 from .pipeline import analyze_file, survey_file
11 from .sigio import sidecar_read
12
13 _DTYPE_CHOICES = ("i8", "u8", "i16", "u16", "f32", "f64")
14
15
16 # --- 손으로 옮기는 한 줄 -----
17 # 실신호 장비는 인터넷도 클립보드도 없다. 결과는 사람이 화면을 보고 받아쳐서 나온다. 그래서
18 # 이 블록은 반드시 여기(필사 대상)에 있어야 한다 -- tools/ 에 두면 그 도구부터 필사해야 하는
19 # 본말전도가 된다. 줄 끝의 검출 코드가 "옮기다 한 글자 틀림"을 잡는다.
20
21 _ALPHA = "0123456789abcdefghijklmnopqrstuvwxyz" # 0/o, 1/l/i 처럼 화면에서 헛갈리는 글자를 뺀 32자
22 _BRIEF_FLAGS = [("eq", lambda d: d["eq"]["applied"]),
23                 ("al", lambda d: d["detected"]["alias_resolved"]),
24                 ("fb", lambda d: d["detected"]["baud_fallback"]),
25                 ("amb", lambda d: d["detected"]["carrier_ambiguous"]),
26                 ("pre", lambda d: "preamble" in d["detected"])]
27
28
29 def check_code(text: str) -> str:
30     """받아친 줄의 오타 검출 코드 (4글자 = 20비트). 공백은 '한 칸으로 줄이되 없애지는'
31     않는다 -- 전부 지우면 'fs1000000 16qam' 과 'fs10000001 6qam' (샘플레이트 10배!) 이 같은
32     코드가 되어, 값이 바뀌는 오타 32종이 조용히 통과했다. 대소문자와 줄 간격은 계속 무시한다."""
33     body = re.sub(r"\s+", " ", re.sub(r"#[0-9a-z]{4}\b", "", text.lower())).strip()
34     n = crc32(body.encode())
35     return "".join(_ALPHA[(n >> (5 * i)) & 31] for i in (3, 2, 1, 0))
36
37
38 def brief(doc: dict, mode: str) -> str:
39     """손으로 옮기는 한 줄 + 검출 코드. Result.to_json() 디셔너리에서 만든다 -- 객체
40     속성을 직접 포맷하면 to_json 이 이미 한 반올림과 두 번 겹쳐 장비와 맥이 갈린다."""
41     head = f"sig2 {mode} fs{doc['fs']:.0f} {doc['fmt']}--{doc['dtype']}"
42     if mode == "sv":
43         lines = [f"{head} n{doc['n_emitters']}"]
44         for i, e in enumerate(doc["emitters"][:12]):
45             baud = f" bd{e['baud']:.0f}" if e.get("baud") else ""
46             lock = f" lk{e['lock']:.0f}" if e.get("lock") is not None else ""
47             lines.append(f"{i} fc{e['abs_fc']:.0f}{baud}{lock} {e.get('mod') or e['kind']}")
48         if len(doc["emitters"]) > 12:
49             lines.append(f"...{len(doc['emitters']) - 12}개 생략")
50     else:
51         d, q = doc["detected"], doc["quality"]
52         p = [head, d["mod"], f"fc{d['fc']:.0f}", f"bd{d['baud']:.0f}", f"lk{q['lock']:.0f}"]
53         if q["mer_db"] is not None:
54             p.append(f"mer{q['mer_db']:.1f}")
55         if d.get("h") is not None: # FSK 는 롤오프 대신 변조지수
56             p.append(f"h{d['h']:.2f}")
57         elif d["rolloff"] is not None:
58             p.append(f"rl{d['rolloff']:.2f}")
59         if len(doc["bursts"]) > 1:
60             p.append(f"b{doc['burst_idx'] + 1}/{len(doc['bursts'])}")

```

```

61         p += [f for f, get in _BRIEF_FLAGS if get(doc)]
62         lines = [" ".join(p)]
63         body = "\n".join(lines)
64         return f"{body} #{check_code(body)}"
65
66
67 # --- analyze / survey / serve -----
68
69 def _analyze(args: argparse.Namespace) -> int:
70     r = analyze_file(args.file, args.fs, args.fmt, args.dtype,
71                     "be" if args.be else None, args.bitrev or None, args.diff,
72                     None if args.burst is None else args.burst - 1, rf=args.rf)
73     j = r.to_json(views=False) # 한 줄 요약도 여기서 만든다 (반올림이 한 번만 걸리게)
74     d = j["detected"]
75     truth = (sidecar_read(args.file) or {}).get("truth") # display only, never detection
76     if len(r.bursts) > 1:
77         print(f"시간상 버스트 {len(r.bursts)}개 - 그중 {r.burst_idx + 1}번을 분석"
78               " (--burst N 으로 선택)")
79     print(f"변조      {d['mod']}") + (f"      (정답 {truth['mod']})" if truth else "")
80     print(f"중심주파수 {d['fc']:.1f} Hz" + (f"      (정답 {truth['fc']:.1f})" if truth else ""))
81     if d["rf_hz"] is not None:
82         print(f"실제 RF      {d['rf_hz'] / 1e6:.6f} MHz")
83     print(f"baud      {d['baud']:.1f} Hz" + (f"      (정답 {truth['baud']:.1f})" if truth else ""))
84     fsk = r.family == "fsk"
85     tail = f"변조지수 h {d['h']:.2f}" if fsk else f"롤오프      {d['rolloff']:.2f}"
86     mer = "" if fsk else f"MER {r.mer_db:.1f} dB"
87     print(f"{tail}    lock {r.lock:.1f}{mer}")
88     if r.eq_applied:
89         print("다중경로 보정(등화기) 적용"
90               " + (" (T/2 분수간격)" if r.eq_mode == "fse" else " (심볼간격)"))
91     if r.alias_resolved:
92         print("중심주파수 접힘(앨리어스) 보정: 후보 중 스펙트럼 무게중심에 맞는 것을 선택")
93     if r.baud_fallback:
94         print("baud 폴백: 스펙트럼에서 baud 선을 못 찾아 점유대역폭으로 추정 - baud 신뢰 낮음")
95     if r.carrier_ambiguous:
96         print("경고: 중심주파수가 접혀 있을 수 있음 - fs 에 비해 너무 높다"
97               " (fc 를 그대로 믿지 말 것)")
98     if args.save_iq:
99         r.save_iq(args.save_iq)
100    if args.save_symbols:
101        r.save_symbols(args.save_symbols)
102    if args.save_bits:
103        r.save_bits(args.save_bits, args.packed)
104    if args.report:
105        r.save_report(args.report)
106    if args.brief: # 사람용 출력을 대체하지 않고 맨 끝에 한 줄 더 --
107        print(brief(j, "an")) # 조기 return 이면 위 저장들이 조용히 무시된다
108    return 0
109
110
111 def _fhz(v: float) -> str:
112     a = abs(v)
113     if a >= 1e6:
114         return f"{v / 1e6:+.3f} MHz"
115     if a >= 1e3:
116         return f"{v / 1e3:+.2f} kHz"
117     return f"{v:+.0f} Hz"
118
119
120 def _survey(args: argparse.Namespace) -> int:

```

```

121     """Wideband survey: detect every emitter, demodulate the digital ones."""
122     s = survey_file(args.file, args.fs, args.fmt, args.dtype,
123                     "be" if args.be else None, args.bitrev or None, args.diff, rf=args.rf)
124     rf0 = s.meta.rf_center
125     note = f" · RF 중심 {rf0 / 1e6:.3f} MHz" if rf0 is not None else ""
126     print(f"주파수상 방사체 {len(s.emitters)}개 - 시간상 버스트 수가 아님"
127           f" (샘플레이트 {s.meta.fs:.0f} Hz{note})")
128     print(f"{'#':>2} {'실제 RF' if rf0 is not None else '중심주파수'}:>12} {'대역폭':>10}"
129           f" {'분류':>7} {'변조':>7} {'baud':>11} {'lock':>5}")
130     for i, e in enumerate(s.emitters):
131         r = e.result
132         mod = r.mod if r else "-"
133         baud = f"{r.baud:.0f}" if r else "-"
134         lock = f"{r.lock:.0f}" if r else "-"
135         kind = {"linear": "디지털", "fsk": "FSK", "analog": "아날로그",
136                "tone": "순수톤", "tooshort": "너무짧음", "error": "오류"}.get(e.kind, e.kind)
137         fc = e.abs_fc if rf0 is None else rf0 + e.abs_fc
138         print(f"{'i':>2} {_fhz(fc):>12} {_fhz(e.detection.bw):>10} {kind:>7}"
139               f" {mod:>7} {baud:>11} {lock:>5}")
140     if args.report:
141         import json
142         with open(args.report, "w") as fh:
143             json.dump(s.to_json(), fh, indent=1)
144     if args.brief:
145         print(brief(s.to_json(), "sv"))
146     return 0
147
148
149 def _read_args(p: argparse.ArgumentParser) -> None:
150     """Read-side sample-format overrides, shared by analyze/survey (sidecar/filename win)."""
151     p.add_argument("--fs", type=float)
152     p.add_argument("--fmt", choices=("iq", "real"))
153     p.add_argument("--dtype", choices=_DTYPE_CHOICES)
154     p.add_argument("--be", action="store_true", help="big-endian 샘플")
155     p.add_argument("--bitrev", action="store_true", help="바이트 내 비트 역순")
156     p.add_argument("--rf", type=float, help="RF 중심주파수 (Hz) - 실제 주파수로 보고")
157     p.add_argument("--brief", action="store_true",
158                     help="손으로 옮길 한 줄 + 오타 검출 코드를 끝에 덧붙임")
159
160
161 def main(argv: list[str] | None = None) -> int:
162     ap = argparse.ArgumentParser(prog="signus")
163     sub = ap.add_subparsers(dest="cmd", required=True)
164
165     a = sub.add_parser("analyze", help="블라인드 복조 + 제원 탐지")
166     a.add_argument("file")
167     _read_args(a)
168     a.add_argument("--save-iq")
169     a.add_argument("--save-symbols")
170     a.add_argument("--save-bits")
171     a.add_argument("--packed", action="store_true")
172     a.add_argument("--report")
173     a.add_argument("--diff", action="store_true",
174                     help="차동 디맵(D-BPSK/D-QPSK): 회전 모호성 없이 비트 복원")
175     a.add_argument("--burst", type=int, help="분석할 버스트 번호 (1부터; 기본 최강 버스트)")
176
177     sv = sub.add_parser("survey", help="광대역 캡처의 모든 신호 탐지 + 복조")
178     sv.add_argument("file")
179     _read_args(sv)
180     sv.add_argument("--diff", action="store_true", help="차동 디맵")

```

```

181     sv.add_argument("--report", help="JSON 리포트 저장 경로")
182
183     s = sub.add_parser("serve", help="웹 UI 서버")
184     s.add_argument("--host", default="127.0.0.1")
185     s.add_argument("--port", type=int, default=8000)
186
187     # 합성 생성·채점(gen/dataset/sweep)은 개발기 전용이다. 격리망 장비에는 lab.py 를
188     # 옮기지 않으므로 그 장비에서는 이 명령들이 아예 존재하지 않는다. try/except 로 삼키면
189     # 개발기에서 lab 내부의 진짜 임포트 오류까지 "명령 없음"으로 둔갑하니 파일 유무만 본다.
190     if importlib.util.find_spec("signus.lab"):
191         from . import lab
192         lab.add_commands(sub)
193
194     args = ap.parse_args(argv)
195     if args.cmd == "analyze":
196         return _analyze(args)
197     if args.cmd == "survey":
198         return _survey(args)
199     if args.cmd == "serve":
200         from .server import run
201         run(args.host, args.port)
202         return 0
203     return args.run(args)          # lab 이 단 명령 (gen / dataset / sweep)
204
205
206 if __name__ == "__main__":
207     sys.exit(main())

```



```

1  """Stdlib-only web server: static frontend + POST /api/analyze (raw body,
2  filename + optional fs/fmt/dtype overrides in the query string – no multipart)."""
3
4  import json
5  from http.server import BaseHTTPRequestHandler, ThreadingHTTPServer
6  from pathlib import Path
7  from urllib.parse import parse_qs, urlparse
8
9  from .pipeline import analyze, survey_web
10 from .sigio import Meta, decode, parse_name
11
12 _WEB = Path(__file__).resolve().parent / "web" # inside the package so wheels/sdists ship it
13 _MIME = {"html": "text/html", "css": "text/css", "js": "text/javascript",
14         ".json": "application/json", ".png": "image/png", ".svg": "image/svg+xml"}
15 _MAX_BODY = 256 * 1024 * 1024
16
17
18 class Handler(BaseHTTPRequestHandler):
19     def log_message(self, *_: object) -> None:
20         pass
21
22     def _json(self, code: int, obj: dict) -> None:
23         body = json.dumps(obj).encode()
24         self.send_response(code)
25         self.send_header("Content-Type", "application/json")
26         self.send_header("Content-Length", str(len(body)))
27         self.end_headers()
28         self.wfile.write(body)
29
30     def _meta(self, q: dict) -> Meta:
31         """Build Meta from query params with filename fallback; raise on unknown."""
32         name = q.get("name", [""])[0]
33         m = parse_name(name)
34         meta = Meta(float(q["fs"][0]) if "fs" in q else m.fs,
35                    q.get("fmt", [m.fmt])[0], q.get("dtype", [m.dtype])[0],
36                    q.get("endian", [m.endian])[0],
37                    q.get("bitrev", ["1" if m.bitrev else "0"])[0] == "1",
38                    rf_center=float(q["rf"][0]) if "rf" in q else m.rf_center)
39         if not meta.ok():
40             raise ValueError(f"샘플레이트/포맷을 알 수 없습니다: {name}")
41         return meta
42
43     def do_POST(self) -> None: # noqa: N802 (BaseHTTPRequestHandler API)
44         url = urlparse(self.path)
45         if url.path not in ("/api/analyze", "/api/survey"):
46             self._json(404, {"error": "not found"})
47             return
48         q = parse_qs(url.query)
49         n = int(self.headers.get("Content-Length") or 0)
50         if n <= 0:
51             self._json(411, {"error": "Content-Length 필요"})
52             return
53         if n > _MAX_BODY:
54             self._json(413, {"error": "파일이 너무 큼니다 (최대 256MB)"})
55             return
56         try:
57             meta = self._meta(q)
58             burst = int(q["burst"][0]) if "burst" in q else None
59             x = decode(self.rfile.read(n), meta)
60             if x.size < 256:

```

```

61         raise ValueError("신호가 너무 짧습니다")
62     except ValueError as exc:
63         self._json(400, {"error": str(exc)})
64     return
65     try:
66         out = survey_web(x, meta) if url.path == "/api/survey" \
67             else analyze(x, meta, burst=burst).to_json()
68         self._json(200, out)
69     except Exception as exc: # surface DSP failures to the UI
70         self._json(500, {"error": f"분석 실패: {exc}"})
71
72     def do_GET(self) -> None: # noqa: N802
73         rel = urlparse(self.path).path.lstrip("/") or "index.html"
74         target = (_WEB / rel).resolve()
75         if not (target.is_file() and target.is_relative_to(_WEB.resolve())):
76             self._json(404, {"error": "not found"})
77         return
78         body = target.read_bytes()
79         self.send_response(200)
80         self.send_header("Content-Type", _MIME.get(target.suffix, "application/octet-stream"))
81         self.send_header("Content-Length", str(len(body)))
82         self.end_headers()
83         self.wfile.write(body)
84
85
86     def run(host: str = "127.0.0.1", port: int = 8000) -> None:
87         srv = ThreadingHTTPServer((host, port), Handler)
88         print(f"signus UI -> http://{host}:{srv.server_address[1]}")
89         srv.serve_forever()

```

```

1  <!doctype html>
2  <html lang="ko">
3  <head>
4  <meta charset="utf-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1">
6  <title>Signus - 신호 복조 분석기</title>
7  <link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 1
4    6 16'%3E%3Crect width='16' height='16' rx='5' fill='%233182F6'/%3E%3C/svg%3E">
8  <link rel="stylesheet" href="/style.css">
9  <script defer src="/app.js"></script>
10 </head>
11 <body>
12 <header class="topbar">
13   <div class="brand">
14     <span class="logo" aria-hidden="true"></span>
15     <div>
16       <h1>Signus</h1>
17       <p class="tagline">블라인드 PSK/QAM/FSK 신호 복조 분석기</p>
18     </div>
19   </div>
20 </header>
21
22 <main class="wrap">
23   <section id="dropCard" class="card drop" tabindex="0" role="button"
24     aria-label="신호 파일 선택 또는 드래그">
25     <input id="fileInput" type="file" multiple hidden>
26     <div class="drop-icon" aria-hidden="true"></div>
27     <p class="drop-title">신호 파일을 여기에 놓으세요</p>
28     <p class="drop-sub">여러 개를 놓으면 일괄 분석 · <code>.json</code> 정답 · <code>.sigmf-meta</co
4       de> 지원</p>
29   </section>
30
31   <section id="metaForm" class="card form hidden">
32     <p class="form-title">파일 이름에서 정보를 찾지 못했어요. 직접 입력해주세요.</p>
33     <div class="form-row">
34       <label>샘플레이트 (Hz)</label>
35       <input id="mFs" type="number" min="1" step="any" placeholder="예: 1000000"></label>
36       <label>포맷</label>
37       <select id="mFmt">
38         <option value="">선택</option><option value="iq">IQ</option><option value="real">Real</opt
4         ion>
39       </select></label>
40       <label>데이터 타입</label>
41       <select id="mDtype">
42         <option value="i16">i16 (16t)</option><option value="i8">i8 (8t)</option>
43         <option value="u8">u8 (8o)</option><option value="u16">u16 (16o)</option>
44         <option value="f32">f32</option><option value="f64">f64</option>
45       </select></label>
46     </div>
47     <button id="metaGo" class="btn">분석 시작</button>
48   </section>
49
50   <section id="loading" class="card loading hidden" aria-live="polite">
51     <div class="spinner" aria-hidden="true"></div>
52     <p id="loadMsg">신호를 분석하고 있어요...</p>
53   </section>
54
55   <section id="errorCard" class="card error hidden" role="alert">
56     <div class="err-mark" aria-hidden="true">!</div>
57     <div>

```

```

58     <p class="err-title">분석에 실패했어요</p>
59     <p id="errMsg" class="err-msg"></p>
60 </div>
61 </section>
62
63 <!-- Batch results table -->
64 <section id="batchCard" class="card hidden">
65     <div class="card-head">
66         <h2 class="card-title">일괄 분석 결과</h2>
67         <button id="dlCsv" class="btn ghost small">CSV 내보내기</button>
68     </div>
69     <div class="table-wrap">
70         <table class="batch">
71             <thead><tr><th>파일</th><th>변조</th><th>중심주파수</th><th>심볼레이트</th><th>Lock</th></tr>
72             <tbody id="batchBody"></tbody>
73         </table>
74     </div>
75     <p class="hint">행을 클릭하면 상세 화면으로 이동합니다.</p>
76 </section>
77
78 <!-- Wideband survey: overview waterfall with detected-emitter boxes -->
79 <section id="surveyCard" class="card hidden">
80     <div class="card-head">
81         <h2 class="card-title">광대역 Survey</h2>
82         <span id="survCount" class="pill ok"></span>
83     </div>
84     <div class="wf-wrap">
85         <canvas id="survFall" class="fall"></canvas>
86         <div id="survBoxes" class="boxes"></div>
87     </div>
88     <p class="hint">상자 또는 아래 목록을 클릭하면 그 신호의 상세로 이동합니다. (세로=시간, 가로=주
89     파수)</p>
90     <div id="survInfo" class="survinfo hidden"></div>
91     <div class="table-wrap">
92         <table class="batch">
93             <thead><tr><th>중심주파수</th><th>종류</th><th>변조</th><th>심볼레이트</th><th>Lock</th></tr>
94             <tbody id="survBody"></tbody>
95         </table>
96     </div>
97 </section>
98
99 <section id="results" class="results hidden">
100     <button id="backBatch" class="btn ghost small hidden">← 목록으로</button>
101
102     <div class="card summary">
103         <div class="summary-head">
104             <span id="statusPill" class="pill"></span>
105             <p id="fileName" class="file-name"></p>
106         </div>
107         <div class="summary-body">
108             <div class="gauge">
109                 <svg viewBox="0 0 120 120" class="donut" aria-hidden="true">
110                     <circle class="donut-track" cx="60" cy="60" r="52"></circle>
111                     <circle id="donutArc" class="donut-arc" cx="60" cy="60" r="52"></circle>
112                 </svg>
113                 <div class="gauge-center"><span id="lockNum" class="gauge-num">0</span>
114                 <span class="gauge-cap tip" title="별자리가 얼마나 조밀·정렬됐는지 (0~100). 60 이상이
115                 면 복조 신뢰">LOCK</span></div>

```

```

114     </div>
115     <div class="metrics">
116         <div class="metric"><span class="metric-label tip" title="변조 오차비 - 높을수록 깨끗 (dB)"
117             ">MER</span>
118             <span id="merVal" class="metric-val"></span></div>
119         <div class="metric"><span class="metric-label tip" title="오차 벡터 크기 - 낮을수록 좋음 (
120             %)">EVM</span>
121             <span id="evmVal" class="metric-val"></span></div>
122         <div class="metric"><span class="metric-label tip" title="추정 심볼 신호 대 잡음비 (dB)">S
123             NR</span>
124             <span id="snrVal" class="metric-val"></span></div>
125     </div>
126 </div>
127
128 <div id="burstMap" class="card hidden">
129     <h2 class="card-title">버스트 지도</h2>
130     <div class="wf-wrap">
131         <canvas id="burstFall" class="fall"></canvas>
132         <div id="burstBoxes" class="boxes"></div>
133     </div>
134     <p class="hint">전체 녹음의 시간-주파수. 버스트 구간을 클릭하면 그 구간을 분석합니다.</p>
135 </div>
136
137 <div class="card">
138     <h2 class="card-title">스펙트럼 · 워터폴</h2>
139     <canvas id="specCanvas" class="spec"></canvas>
140     <canvas id="fallCanvas" class="fall"></canvas>
141 </div>
142
143 <div class="card const-card">
144     <div class="card-head">
145         <h2 class="card-title">성상도</h2>
146         <div class="play-row">
147             <button id="playBtn" class="btn ghost small" aria-label="재생/일시정지">⏮ 일시정지</button>
148             <select id="speedSel" class="sel small" aria-label="재생 속도">
149                 <option value="1">1x</option><option value="4" selected>4x</option>
150                 <option value="16">16x</option>
151             </select>
152         </div>
153     </div>
154     <canvas id="constCanvas" class="const"></canvas>
155     <input id="scrub" class="scrub" type="range" min="0" max="1000" value="0"
156         aria-label="재생 위치">
157     <p id="playInfo" class="hint"></p>
158 </div>
159
160 <div id="paramGrid" class="param-grid"></div>
161
162 <div class="card downloads">
163     <h2 class="card-title">다운로드</h2>
164     <div class="dl-row">
165         <button id="copyBits" class="btn ghost">비트 복사</button>
166         <button id="dlBits" class="btn ghost">비트 .txt</button>
167         <button id="dlSyms" class="btn ghost">심볼 .csv</button>
168         <button id="dlReport" class="btn ghost">리포트 .json</button>
169     </div>

```

```
170     </div>
171   </section>
172 </main>
173 </body>
174 </html>
```

```

1  :root {
2    --bg: #f4f6f8; --card: #fff; --ink: #191f28; --sub: #6b7684;
3    --blue: #3182F6; --blue-weak: #e8f1fe; --line: #eef1f4;
4    --green: #12b886; --amber: #f59f00; --red: #fa5252;
5    --field: #fff; --hover: #f7fafd; --border: #dfe4ea; /* theme-sensitive surfaces */
6    --ok-bg: #e6f7f1; --warn-bg: #fff4e0; --bad-bg: #ffecec;
7    --radius: 20px; --shadow: 0 6px 24px rgba(30, 42, 66, .07);
8  }
9  /* dark mode: follows the OS. The signal canvases are already dark, so they blend in. */
10 @media (prefers-color-scheme: dark) {
11   :root {
12     --bg: #0e131b; --card: #161d28; --ink: #e6ebf2; --sub: #94a1b2;
13     --blue-weak: #17273c; --line: #232c39; --field: #1b2330; --hover: #1d2632;
14     --border: #2b3543; --ok-bg: #102c24; --warn-bg: #302512; --bad-bg: #321b1e;
15     --shadow: 0 6px 24px rgba(0, 0, 0, .38);
16   }
17 }
18
19 * { box-sizing: border-box; }
20
21 body {
22   margin: 0; background: var(--bg); color: var(--ink);
23   font-family: -apple-system, 'Apple SD Gothic Neo', 'Segoe UI', 'Malgun Gothic', sans-serif;
24   line-height: 1.5; -webkit-font-smoothing: antialiased;
25 }
26
27 .wrap { max-width: 580px; margin: 0 auto; padding: 8px 20px 64px; }
28
29 .topbar { max-width: 580px; margin: 0 auto; padding: 28px 20px 12px; }
30 .brand { display: flex; align-items: center; gap: 12px; }
31 .logo {
32   width: 40px; height: 40px; border-radius: 12px; flex: none;
33   background: linear-gradient(135deg, var(--blue), #6aa8ff);
34 }
35 h1 { margin: 0; font-size: 22px; font-weight: 800; letter-spacing: -.4px; }
36 .tagline { margin: 2px 0 0; color: var(--sub); font-size: 13px; }
37
38 .card {
39   background: var(--card); border-radius: var(--radius); padding: 20px;
40   box-shadow: var(--shadow); margin-top: 16px; animation: rise .35s ease both;
41 }
42 .card-title { margin: 0 0 12px; font-size: 15px; font-weight: 700; }
43 .hidden { display: none !important; }
44
45 @keyframes rise { from { opacity: 0; transform: translateY(10px); } to { opacity: 1; transform: none
46   ; } }
47
48 /* Dropzone */
49 .drop {
50   text-align: center; cursor: pointer; border: 2px dashed var(--border);
51   transition: border-color .18s, background .18s, transform .18s;
52 }
53 .drop:hover, .drop:focus-visible { border-color: var(--blue); background: var(--hover); outline: none; }
54 .drop.drag { border-color: var(--blue); background: var(--blue-weak); transform: scale(1.01); }
55 .drop-icon {
56   width: 52px; height: 52px; margin: 6px auto 14px; border-radius: 16px;
57   background: var(--blue-weak); position: relative;
58 }
59 .drop-icon::after {

```

```

59   content: ''; position: absolute; inset: 16px 15px; border: 3px solid var(--blue);
60   border-radius: 3px; border-top: none; border-right: none;
61   transform: rotate(45deg) translate(2px, -2px);
62 }
63 .drop-title { margin: 0; font-size: 16px; font-weight: 700; }
64 .drop-sub { margin: 6px 0 0; color: var(--sub); font-size: 13px; }
65 code { background: var(--line); padding: 1px 6px; border-radius: 6px; font-size: 12px; }
66
67 /* Meta form */
68 .form-title { margin: 0 0 12px; font-size: 14px; font-weight: 600; }
69 .form-row { display: grid; grid-template-columns: 1fr 1fr 1fr; gap: 12px; }
70 .form-row label { display: flex; flex-direction: column; gap: 6px; font-size: 12px; color: var(--sub)
  4   ); }
71 input, select {
72   font: inherit; padding: 10px 12px; border: 1px solid var(--border); border-radius: 12px;
73   background: var(--field); color: var(--ink);
74 }
75 input:focus, select:focus { outline: 2px solid var(--blue); border-color: transparent; }
76
77 .btn {
78   margin-top: 16px; width: 100%; padding: 13px; border: none; border-radius: 14px;
79   background: var(--blue); color: #fff; font-size: 15px; font-weight: 700; cursor: pointer;
80   transition: filter .15s, transform .1s;
81 }
82 .btn:hover { filter: brightness(1.05); }
83 .btn:active { transform: scale(.98); }
84 .btn.ghost { background: var(--blue-weak); color: var(--blue); margin-top: 0; }
85
86 /* Loading */
87 .loading { display: flex; align-items: center; gap: 14px; color: var(--sub); }
88 .spinner {
89   width: 26px; height: 26px; border: 3px solid var(--line);
90   border-top-color: var(--blue); border-radius: 50%; animation: spin .8s linear infinite;
91 }
92 @keyframes spin { to { transform: rotate(360deg); } }
93
94 /* Error */
95 .error { display: flex; gap: 14px; align-items: center; border-left: 4px solid var(--red); }
96 .err-mark {
97   width: 34px; height: 34px; flex: none; border-radius: 50%; background: var(--bad-bg);
98   color: var(--red); font-weight: 800; display: grid; place-items: center;
99 }
100 .err-title { margin: 0; font-weight: 700; }
101 .err-msg { margin: 2px 0 0; color: var(--sub); font-size: 14px; }
102
103 /* Summary + gauge */
104 .summary-head { display: flex; align-items: center; gap: 12px; margin-bottom: 16px; }
105 .pill { padding: 6px 14px; border-radius: 999px; font-size: 13px; font-weight: 700; }
106 .pill.ok { background: var(--ok-bg); color: var(--green); }
107 .pill.warn { background: var(--warn-bg); color: var(--amber); }
108 .pill.bad { background: var(--bad-bg); color: var(--red); }
109 .file-name { margin: 0; font-size: 13px; color: var(--sub); overflow: hidden; text-overflow: ellipsi
  4   s; white-space: nowrap; }
110
111 .summary-body { display: flex; align-items: center; gap: 24px; }
112 .gauge { position: relative; width: 120px; height: 120px; flex: none; }
113 .donut { width: 120px; height: 120px; }
114 .donut-track { fill: none; stroke: var(--line); stroke-width: 12; }
115 .donut-arc {
116   fill: none; stroke: var(--blue); stroke-width: 12; stroke-linecap: round;

```



```

117   transform: rotate(-90deg); transform-origin: center;
118   transition: stroke-dashoffset .9s cubic-bezier(.22, .61, .36, 1);
119 }
120 .gauge-center { position: absolute; inset: 0; display: grid; place-items: center; }
121 .gauge-num { font-size: 34px; font-weight: 800; line-height: 1; font-variant-numeric: tabular-nums
122   ; }
123 .gauge-cap { font-size: 11px; color: var(--sub); letter-spacing: 1px; }
124
125 .metrics { display: flex; flex-direction: column; gap: 12px; }
126 .metric { display: flex; flex-direction: column; gap: 2px; }
127 .metric-label { font-size: 12px; color: var(--sub); }
128 .tip { cursor: help; text-decoration: underline dotted; text-underline-offset: 3px;
129   text-decoration-color: color-mix(in srgb, var(--sub) 45%, transparent); }
130 .metric-val { font-size: 20px; font-weight: 700; font-variant-numeric: tabular-nums; }
131
132 /* Card head with actions: spacing lives on the CONTAINER so the title and the
133   action stay vertically centered on the same line */
134 .card-head { display: flex; align-items: center; justify-content: space-between;
135   gap: 12px; margin-bottom: 12px; }
136 .card-head .card-title { margin-bottom: 0; }
137 .btn.small { width: auto; margin-top: 0; padding: 8px 14px; font-size: 13px; border-radius: 10px; }
138 .sel.small {
139   font: inherit; font-size: 13px; font-weight: 600; padding: 7px 10px; border-radius: 10px;
140   border: 1px solid var(--border); background: var(--field); color: var(--ink);
141 }
142 .play-row { display: flex; gap: 8px; align-items: center; }
143 .hint { margin: 8px 0 0; font-size: 12px; color: var(--sub); }
144
145 /* Spectrum + waterfall */
146 .spec { width: 100%; height: 120px; display: block; border-radius: 12px 12px 0 0; background: #0d132
147   0; }
148 .fall { width: 100%; height: 150px; display: block; border-radius: 0 0 12px 12px; background: #0d132
149   0; }
150
151 /* Constellation + playback */
152 .const { width: 100%; aspect-ratio: 1; border-radius: 14px; display: block; background: #0d1320; }
153 .scrub { width: 100%; margin-top: 12px; accent-color: var(--blue); }
154
155 /* Batch table */
156 .table-wrap { overflow-x: auto; }
157 table.batch { width: 100%; border-collapse: collapse; font-size: 13.5px; }
158 table.batch th {
159   text-align: left; font-size: 12px; color: var(--sub); font-weight: 600;
160   padding: 8px 10px; border-bottom: 1px solid var(--line);
161 }
162 table.batch td { padding: 11px 10px; border-bottom: 1px solid var(--line);
163   font-variant-numeric: tabular-nums; }
164 table.batch .pill { padding: 3px 10px; font-size: 12px; }
165 table.batch tbody tr { cursor: pointer; transition: background .12s; }
166 table.batch tbody tr:hover { background: var(--hover); }
167 table.batch .fn { font-weight: 600; max-width: 210px; overflow: hidden;
168   text-overflow: ellipsis; white-space: nowrap; }
169
170 /* Param grid */
171 .param-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 16px; margin-top: 16px; }
172 .param { padding: 16px; margin: 0; animation: rise .4s ease both; }
173 .param-label { font-size: 12px; color: var(--sub); }
174 .param-val { font-size: 22px; font-weight: 800; margin-top: 4px; letter-spacing: -.3px;
175   font-variant-numeric: tabular-nums; }
176 .param-note { font-size: 11px; color: var(--sub); margin-top: 2px; }

```

```

174 .chips { display: flex; flex-wrap: wrap; gap: 6px; margin-top: 10px; }
175 .chips:empty { display: none; } /* no flags -> no ghost margin */
176 .chip { font-size: 11px; font-weight: 700; padding: 4px 9px; border-radius: 999px; }
177 .chip.hit { background: var(--ok-bg); color: var(--green); }
178 .chip.miss { background: var(--bad-bg); color: var(--red); }
179 .chip.warn { background: var(--warn-bg); color: var(--amber); }
180
181 /* Burst selector (chips that act) */
182 .chip-btn {
183   font: inherit; font-size: 11px; font-weight: 700; padding: 4px 10px;
184   border-radius: 999px; border: 1px solid var(--border); background: var(--field);
185   color: var(--sub); cursor: pointer; transition: color .12s, border-color .12s;
186 }
187 .chip-btn:hover { border-color: var(--blue); color: var(--blue); }
188 .chip-btn.on { background: var(--blue-weak); color: var(--blue); border-color: transparent; }
189
190 /* Downloads */
191 .dl-row { display: flex; gap: 10px; flex-wrap: wrap; }
192 .dl-row .btn { flex: 1; min-width: 120px; }
193
194 /* Survey overview: waterfall + detected-emitter boxes */
195 .wf-wrap { position: relative; }
196 .wf-wrap .fall { height: 230px; border-radius: 12px; }
197 .boxes { position: absolute; inset: 0; pointer-events: none; }
198 .box {
199   position: absolute; box-sizing: border-box; border: 2px solid var(--blue);
200   border-radius: 5px; background: rgba(49, 130, 246, .10); cursor: pointer;
201   pointer-events: auto; min-width: 7px; min-height: 12px;
202   transition: background .12s, box-shadow .12s;
203 }
204 .box:hover, .box:focus-visible {
205   background: rgba(49, 130, 246, .24); box-shadow: 0 0 0 2px rgba(255, 255, 255, .55); outline: none
206   ;
207 }
208 .box-lbl { /* hidden by default (a spectrogram full of labels is unreadable); shown on hover */
209   position: absolute; top: -17px; left: -2px; white-space: nowrap; font-size: 10px;
210   font-weight: 700; color: #fff; background: rgba(13, 19, 32, .85); padding: 1px 6px;
211   border-radius: 6px; pointer-events: none; display: none; z-index: 2;
212 }
213 .box:hover .box-lbl, .box:focus-visible .box-lbl, .box.sel .box-lbl { display: block; }
214 .box.sel { box-shadow: 0 0 0 2px #fff, 0 0 0 4px var(--blue); }
215 .box.ok { border-color: var(--green); background: rgba(18, 184, 134, .13); }
216 .box.ok .box-lbl { background: var(--green); }
217 .box.warn { border-color: var(--amber); background: rgba(245, 159, 0, .13); }
218 .box.warn .box-lbl { background: var(--amber); }
219 .box.bad { border-color: var(--red); background: rgba(250, 82, 82, .13); }
220 .box.bad .box-lbl { background: var(--red); }
221 .box.gray { border-color: #8592a6; background: rgba(133, 146, 166, .15); }
222 .box.gray .box-lbl { background: #8592a6; }
223 .pill.gray { background: var(--line); color: var(--sub); }
224 table.batch .fc { font-weight: 600; font-variant-numeric: tabular-nums; }
225
226 /* Non-digital emitter inline info */
227 .survinfo {
228   background: var(--hover); border-radius: 14px; padding: 14px 16px; margin: 12px 0;
229   border-left: 4px solid #8592a6; animation: rise .3s ease both;
230 }
231 .survinfo h3 { margin: 0 0 10px; font-size: 14px; }
232 .si-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 10px 16px; }
233 .si-k { font-size: 11px; color: var(--sub); }

```

```
233 .si-v { font-size: 17px; font-weight: 700; font-variant-numeric: tabular-nums; }
234
235 @media (max-width: 420px) {
236   .form-row { grid-template-columns: 1fr; }
237   .param-grid { grid-template-columns: 1fr; }
238   .summary-body { gap: 18px; }
239 }
240
241 @media (prefers-reduced-motion: reduce) {
242   * { animation: none !important; transition: none !important; }
243 }
```

```

1  "use strict";
2  const $ = (id) => document.getElementById(id);
3  const API = "/api/analyze";
4  const SURVEY_API = "/api/survey";
5  const FS_RE = /(?:^[_.])fs(\d+(?:\.\d+)?(?:e[+-]?\d+)?(?:[_.]|$))/i;
6  const RF_RE = /(?:^[_.])rf(\d+(?:\.\d+)?(?:e[+-]?\d+)?(?:[_.]|$))/i;
7  const REDUCED = matchMedia("(prefers-reduced-motion: reduce)").matches;
8  const state = { file: null, sidecar: null, meta: null, resp: null, batch: [],
9    ..... burst: null, survey: null, drilled: false, overview: null };
10
11  /* --- filename parsing (mirror of sigio.parse_name; read-necessities only) --- */
12  const DTYPES = ["i8", "u8", "i16", "u16", "f32", "f64"];
13  // baudline-style aliases: <bits>t = two's complement (signed), <bits>o = offset (unsigned)
14  const DTYPE_ALIAS = { "8t": "i8", "8o": "u8", "16t": "i16", "16o": "u16", "32f": "f32", "64f": "f64",
15    ..... };
16  const FMTS = { cplx: "iq", real: "real", iq: "iq" }; // iq = 옛 밀줄 형식의 이름
17  /* <이름>.cplx|real.<샘플레이트>.<16t>.pcm - 샘플레이트는 접두사가 없으므로 포맷 바로
18     다음 자리로 찾는다. 옛 밀줄 형식(<이름>_fs..._iq_i16.iq)도 그대로 읽는다. sigio.py 와 규칙이
19     같아야 한다: 여기서 갈리면 웹은 되고 CLI 는 안 되는(또는 그 반대) 조용한 불일치가 된다. */
20  function parseName(name) {
21    const base = name.replace(/^\.*\/$/, "").toLowerCase();
22    const mt = FS_RE.exec(base), rt = RF_RE.exec(base), toks = base.split(/[_.]/);
23    // 진짜 포맷 토막 = "숫자 + 아는 제원 토막"을 데리고 다니는 마지막 후보. 라벨의 real/cplx/
24    // u8/be 오염과 점 켜 샘플레이트(2.4e6 -> 2 Hz)를 같은 관문으로 막는다 (sigio.py 와 동일 규칙).
25    const spec = (t) => DTYPES.includes(t) || t in DTYPE_ALIAS || t === "be" || t === "bitrev"
26      ..... || t === "pcm" || t.startsWith("rf");
27    const hit = (k) => /^d+$/ .test(toks[k + 1] || "") && spec(toks[k + 2] || "");
28    const cand = toks.map((t, k) => (t in FMTS ? k : -1)).filter((k) => k >= 0);
29    const hits = cand.filter(hit);
30    const i = hits.length ? hits[hits.length - 1] : (cand.length ? cand[0] : -1);
31    const tail = i >= 0 ? toks.slice(i + 1) : toks; // 제원은 포맷 토막 뒤에만 산다
32    const dt = tail.find((t) => DTYPES.includes(t) || t in DTYPE_ALIAS);
33    return {
34      fs: hits.includes(i) ? parseFloat(toks[i + 1]) : (mt ? parseFloat(mt[1]) : null),
35      fmt: i >= 0 ? FMTS[toks[i]] : null,
36      dtype: dt ? (DTYPE_ALIAS[dt] || dt) : "i16",
37      endian: tail.includes("be") ? "be" : "le",
38      bitrev: tail.includes("bitrev"),
39      rf: rt ? parseFloat(rt[1]) : null,
40    };
41  }
42  /* SigMF: core:datatype -> our fmt/dtype/endian */
43  const SIGMF = { cf32: ["iq", "f32"], ci16: ["iq", "i16"], ci8: ["iq", "i8"],
44    ..... cu8: ["iq", "u8"], rf32: ["real", "f32"], ri16: ["real", "i16"] };
45  function metaFromSigmf(doc) {
46    const g = doc && doc.global;
47    if (!g || !g["core:datatype"]) return null;
48    const dt = g["core:datatype"], base = dt.replace(/_[lbe]$/, "");
49    if (!(base in SIGMF)) return null;
50    const [fmt, dtype] = SIGMF[base];
51    const cap = (doc.captures || [])[0] || {};
52    return { fs: Number(g["core:sample_rate"]), fmt, dtype,
53      ..... endian: dt.endsWith("_be") ? "be" : "le", bitrev: false,
54      ..... rf: cap["core:frequency"] != null ? Number(cap["core:frequency"]) : null };
55  }
56  /* --- number formatting --- */
57  function sig(x) {
58    const a = Math.abs(x);
59    return parseFloat(a >= 100 ? x.toFixed(0) : a >= 10 ? x.toFixed(1) : x.toFixed(2)).toString();

```

```

60 }
61 function fmtHz(v) {
62   if (v == null) return "-";
63   const a = Math.abs(v);
64   if (a >= 1e6) return sig(v / 1e6) + " MHz";
65   if (a >= 1e3) return sig(v / 1e3) + " kHz";
66   return sig(v) + " Hz";
67 }
68 function fmtRf(v) { // absolute RF: keep kHz resolution even in the 100s-of-MHz range
69   return (v / 1e6).toFixed(3) + " MHz";
70 }
71
72 /* --- ideal constellation points (mirror of constellations.ideal_points) --- */
73 function idealPoints(mod) {
74   if (mod === "bpsk") return [{ i: 1, q: 0 }, { i: -1, q: 0 }];
75   if (mod === "qpsk" || mod === "8psk") {
76     const m = mod === "qpsk" ? 4 : 8, off = mod === "qpsk" ? Math.PI / 4 : 0, p = [];
77     for (let k = 0; k < m; k++) p.push({ i: Math.cos(off + 2 * Math.PI * k / m), q: Math.sin(of
78       f + 2 * Math.PI * k / m) });
79     return p;
80   }
81   const n = mod === "16qam" ? 4 : mod === "32qam" ? 6 : 8, lv = [], p = [];
82   for (let k = 0; k < n; k++) lv.push(k * 2 - (n - 1));
83   for (const qi of lv) for (const ii of lv) {
84     if (mod === "32qam" && Math.abs(ii * qi) === 25) continue; // cross: corners removed
85     p.push({ i: ii, q: qi });
86   }
87   let e = 0;
88   for (const pt of p) e += pt.i * pt.i + pt.q * pt.q;
89   const norm = Math.sqrt(e / p.length);
90   return p.map((pt) => ({ i: pt.i / norm, q: pt.q / norm }));
91 }
92
93 /* --- file intake: one data file (+sidecar) or many (batch) --- */
94 async function acceptFiles(list) {
95   const files = [...list];
96   const metas = files.filter((f) => /\. (json|sigmf-meta)$/i.test(f.name));
97   const data = files.filter((f) => !metas.includes(f));
98   if (!data.length) return showError("데이터 파일을 찾을 수 없어요.");
99   if (data.length > 1) return runBatch(data, metas);
100
101   state.file = data[0];
102   state.resp = null;
103   state.batch = [];
104   state.burst = null;
105   state.survey = null;
106   state.drilled = false;
107   state.overview = null;
108   // 사이드카는 이름이 같은 캡처의 것만 쓴다 -- 확장자만 보고 집으면 다른 신호의 fs/fmt 로
109   // 조용히 읽는다 (일괄 모드는 이미 stem 대조를 한다: 두 모드가 다르면 그게 또 함정이다)
110   const mine = (m, ext) => {
111     const stem = m.name.toLowerCase().slice(0, -ext.length);
112     const dn = state.file.name.toLowerCase();
113     return dn.startsWith(stem) && (dn.length === stem.length || dn[stem.length] === ".");
114   };
115   const sm = metas.find((m) => m.name.toLowerCase().endsWith(".sigmf-meta") && mine(m, ".sigmf-meta"
116     ));
117   const truth = metas.find((m) => m.name.toLowerCase().endsWith(".json") && mine(m, ".json"));
118   state.sidecar = truth ? await readJson(truth) : null; // client-side only, never sent
119   state.meta = (sm && metaFromSigmf(await readJson(sm))) || parseName(state.file.name);

```

```

118   if (state.meta && state.meta.fs && state.meta.fmt) runSurvey();
119   else showMetaForm();
120 }
121 const readJson = (f) => f.text().then(JSON.parse).catch(() => null);
122
123 function showMetaForm() {
124   hideAll();
125   state.meta = state.meta || {};
126   if (state.meta.fs) $("mFs").value = state.meta.fs;
127   $("mFmt").value = state.meta.fmt || "";
128   $("mDtype").value = state.meta.dtype || "i16";
129   show($("metaForm"));
130 }
131 $("metaGo").onclick = () => {
132   const fs = parseFloat($("mFs").value), fmt = $("mFmt").value;
133   if (!(fs > 0) || !fmt) return showError("샘플레이트와 포맷을 입력해주세요.");
134   state.meta = { fs, fmt, dtype: $("mDtype").value, endian: "le", bitrev: false };
135   runSurvey();
136 };
137
138 /* --- server call (contract-frozen) --- */
139 function query(file, m, burst) {
140   const q = new URLSearchParams({ name: file.name });
141   if (m.fs) q.set("fs", m.fs);
142   if (m.fmt) q.set("fmt", m.fmt);
143   if (m.dtype) q.set("dtype", m.dtype);
144   if (m.endian) q.set("endian", m.endian);
145   if (m.bitrev) q.set("bitrev", "1");
146   if (m.rf != null) q.set("rf", m.rf);
147   if (burst != null) q.set("burst", burst);
148   return q;
149 }
150 async function postTo(api, file, m, burst) {
151   const res = await fetch(api + "?" + query(file, m, burst), { method: "POST", body: file });
152   const data = await res.json();
153   if (!res.ok || data.error) throw new Error(data.error || `서버 오류 (${res.status})`);
154   return data;
155 }
156 async function runSurvey() { // entry: one capture -> single detail OR survey
157   hideAll();
158   state.drilled = false;
159   $("loadMsg").textContent = "신호를 조사하고 있어요...";
160   show($("loading"));
161   try {
162     const resp = await postTo(SURVEY_API, state.file, state.meta);
163     if (resp.mode === "single") {
164       state.survey = null; state.overview = resp.overview || null; state.resp = resp.result;
165       render(resp.result);
166       if (state.overview) renderBurstMap(state.overview, resp.result);
167     } else renderSurvey(resp);
168   } catch (e) {
169     showError(e.message);
170   }
171 }
172 async function analyze() { // burst re-selection within a single signal
173   hideAll();
174   $("loadMsg").textContent = "신호를 분석하고 있어요...";
175   show($("loading"));
176   try {
177     state.resp = await postTo(API, state.file, state.meta, state.burst);

```

```

178     render(state.resp);
179     if (state.overview) renderBurstMap(state.overview, state.resp);
180     $("backBatch").classList.toggle("hidden", !state.batch.length);
181   } catch (e) {
182     showError(e.message);
183   }
184 }
185
186 /* --- batch mode --- */
187 async function runBatch(data, metas) {
188   hideAll();
189   show($("loading"));
190   const truths = new Map(metas.filter((m) => /\.json$/i.test(m.name))
191     .map((m) => [m.name.replace(/\.json$/i, ""), m]));
192   const sigmfs = new Map(metas.filter((m) => /\.sigmf-meta$/i.test(m.name))
193     .map((m) => [m.name.replace(/\.sigmf-meta$/i, ""), m]));
194   const rows = [];
195   for (let i = 0; i < data.length; i++) {
196     const f = data[i];
197     $("loadMsg").textContent = `일괄 분석 중... (${i + 1}/${data.length}) ${f.name}`;
198     // same meta precedence as single-file mode: a .sigmf-meta sidecar (matched by stem)
199     // beats filename tokens
200     const sm = sigmfs.get(f.name.replace(/\.([^.]+\$/i, ""));
201     const meta = (sm && metaFromSigmf(await readJson(sm))) || parseName(f.name);
202     let resp = null, err = null;
203     try {
204       if (!meta.fs || !meta.fmt) throw new Error("파일명에 fs/포맷 정보가 없어요");
205       resp = await postTo(API, f, meta);
206     } catch (e) { err = e.message; }
207     const tf = truths.get(f.name);
208     // meta is stored per row: a later burst-chip re-analyze posts with THIS file's meta --
209     // it used to read state.meta, which a fresh batch never set (crash) and a previous
210     // single-file run left stale (silent wrong-fs decode)
211     rows.push({ file: f, meta, resp, err, sidecar: tf ? await readJson(tf) : null });
212   }
213   state.batch = rows;
214   hideAll();
215   renderBatch(rows);
216 }
217
218 function renderBatch(rows) {
219   $("batchBody").innerHTML = rows.map((r, i) => {
220     if (r.err) return `<tr data-i="${i}"><td class="fn">${r.file.name}</td>` +
221       `<td colspan="4" style="color:var(--red)">${r.err}</td></tr>`;
222     const d = r.resp.detected, lock = Math.round(r.resp.quality.lock);
223     const cls = lock >= 60 ? "ok" : lock >= 40 ? "warn" : "bad";
224     return `<tr data-i="${i}"><td class="fn">${r.file.name}</td>` +
225       `<td>${d.mod.toUpperCase()}</td><td>${fmtHz(d.fc)}</td><td>${fmtHz(d.baud)}</td>` +
226       `<td><span class="pill ${cls}">${lock}</span></td></tr>`;
227   }).join("");
228   $("batchBody").querySelectorAll("tr").forEach((tr) => {
229     tr.onclick = () => {
230       const r = state.batch[+tr.dataset.i];
231       if (!r.resp) return;
232       state.file = r.file; state.meta = r.meta; state.resp = r.resp; state.sidecar = r.sidecar;
233       state.burst = null; state.drilled = false; state.survey = null; state.overview = null;
234       hideAll();
235       render(r.resp);
236       const b = $("backBatch");
237       b.textContent = "← 목록으로";

```

```

238     b.onclick = () => { hideAll(); renderBatch(state.batch); };
239     b.classList.remove("hidden");
240   };
241 });
242 show($("#batchCard"));
243 }
244 $("#backBatch").onclick = () => { hideAll(); renderBatch(state.batch); };
245 $("#dlCsv").onclick = () => {
246   const head = "file,mod,fc_hz,baud_hz,rolloff,lock,mer_db,snr_est_db\n";
247   const body = state.batch.filter((r) => r.resp).map((r) => {
248     const d = r.resp.detected, q = r.resp.quality;
249     return [r.file.name, d.mod, d.fc, d.baud, d.rolloff ?? "", q.lock, q.mer_db,
250       r.resp.snr_est_db ?? ""].join(",");
251   }).join("\n");
252   saveBlob("signus_batch.csv", head + body, "text/csv");
253 };
254
255 /* --- wideband survey overview (waterfall + detected-emitter boxes) --- */
256 function renderSurvey(resp) {
257   hideAll();
258   stopPlay();
259   state.survey = resp;
260   const dig = resp.emitters.filter((e) => e.result).length;
261   $("#survCount").textContent = `${resp.emitters.length}개 신호 · 디지털 ${dig}`;
262   $("#survInfo").classList.add("hidden");
263   renderSurveyList(resp);
264   show($("#surveyCard")); // show first so the canvas has a real width
265   paintWaterfall($("#survFall"), resp.overview.waterfall, 230);
266   drawBoxes(resp);
267 }
268
269 function boxStyle(e) { // -> [css class, short label]
270   if (e.result) {
271     const lock = Math.round(e.result.quality.lock);
272     return [lock >= 60 ? "ok" : lock >= 40 ? "warn" : "bad",
273       `${e.result.detected.mod.toUpperCase()} · ${lock}`];
274   }
275   if (e.kind === "chirp") {
276     return ["gray", e.info && e.info.sf ? `LoRa SF${e.info.sf}` : "처프"];
277   }
278   return ["gray", { analog: "아날로그", tone: "톤/CW", tooshort: "짧음",
279     error: "오류" }[e.kind] || e.kind];
280 }
281
282 function drawBoxes(resp) {
283   const host = $("#survBoxes"), fs = resp.overview.fs, n = resp.overview.n || 1;
284   host.innerHTML = "";
285   resp.emitters.forEach((e) => {
286     const d = e.det, [cls, lbl] = boxStyle(e);
287     const wpct = Math.max(0.8, d.bw / fs * 100); // width = bandwidth / fs
288     const left = (d.fc + fs / 2) / fs * 100 - wpct / 2; // x = carrier on -fs/2..fs/2
289     const top = Math.max(0, d.t0 / n * 100); // y = burst start over capture
290     const b = document.createElement("button");
291     b.className = "box " + cls;
292     b.style.left = Math.max(0, Math.min(100 - wpct, left)) + "%";
293     b.style.width = wpct + "%";
294     b.style.top = top + "%";
295     b.style.height = Math.min(100 - top, Math.max(7, (d.t1 - d.t0) / n * 100)) + "%";
296     b.title = lbl;
297     b.innerHTML = `${lbl}</span>`;

```



```

298     b.onclick = () => drillEmitter(e);
299     host.appendChild(b);
300   });
301 }
302
303 function renderSurveyList(resp) {
304   const KIND = { linear: "디지털", fsk: "FSK", chirp: "처프/LoRa", analog: "아날로그",
305     ..... tone: "톤/CW", tooshort: "짧음", error: "오류" };
306   const rf0 = resp.rf_center; // real RF = capture centre + abs_fc
307   $("survBody").innerHTML = resp.emitters.map((e, i) => {
308     const [cls] = boxStyle(e), r = e.result;
309     const lock = r ? `<span class="pill ${cls}">${Math.round(r.quality.lock)}</span>`
310       ..... : `<span class="pill gray"></span>`;
311     return `<tr data-i="${i}"><td class="fc">${rf0 != null ? fmtRf(rf0 + e.abs_fc) : fmtHz(e.abs_fc)}
312     ↳ `</td>` +
313     ..... `<td>${KIND[e.kind] || e.kind}</td><td>${r ? r.detected.mod.toUpperCase() : "-"}</td>` +
314     ..... `<td>${r ? fmtHz(r.detected.baud) : "-"}</td><td>${lock}</td></tr>`;
315   }).join("");
316   $("survBody").querySelectorAll("tr").forEach((tr) => {
317     tr.onclick = () => drillEmitter(resp.emitters[+tr.dataset.i]);
318   });
319 }
320
321 function drillEmitter(e) {
322   if (e.result) { // digital -> reuse the single-signal detail view
323     state.drilled = true;
324     state.overview = null; // a cut channel has no whole-capture burst map
325     state.resp = e.result;
326     render(e.result);
327     const rf0 = state.survey && state.survey.rf_center;
328     $("fileName").textContent =
329     ..... `에미터 @ ${rf0 != null ? fmtRf(rf0 + e.abs_fc) : fmtHz(e.abs_fc)} · ${state.file.name}`;
330     const b = $("backBatch");
331     b.textContent = "← Survey로";
332     b.onclick = backToSurvey;
333     b.classList.remove("hidden");
334   } else { // non-digital -> inline info, stay on the overview
335     showSurveyInfo(e);
336   }
337 }
338
339 function backToSurvey() { state.drilled = false; renderSurvey(state.survey); }
340
341 function showSurveyInfo(e) {
342   const d = e.det;
343   let title, rows;
344   if (e.kind === "chirp" && e.info) { // linear chirp / CSS (LoRa): characterized only
345     const c = e.info;
346     title = (c.sf ? `LoRa 계열 (SF${c.sf})` : "선형 처프 / CSS") + " - 복조 대상 아님";
347     // rate + direction are robust (coherent from the beat tone); bw/SF only when the
348     // channel is cleanly isolated enough to snap to a LoRa hypothesis.
349     rows = [
350       ["중심주파수", fmtHz(e.abs_fc)],
351       ..... ["처프 방향", c.up ? "▲ 상승" : "▼ 하강"],
352       ..... ["처프율", (c.mu / 1e9).toFixed(3) + " MHz/ms"];
353     if (c.sf) rows.push(["대역폭", fmtHz(c.bw)], ["심볼레이트", fmtHz(c.rs)],
354       ..... ["심볼시간", (c.tsym * 1e3).toFixed(2) + " ms"]);
355   } else {
356     const KIND = { analog: "아날로그 (FM/음성)", tone: "톤 / CW",
357       ..... tooshort: "너무 짧은 버스트", error: "분석 오류" };
358     title = (KIND[e.kind] || e.kind) + " - 복조 대상 아님";
359     rows = [
360       ["중심주파수", fmtHz(e.abs_fc)], ["대역폭", fmtHz(d.bw)],

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357     ["SNR", d.snr_db == null ? "-" : d.snr_db.toFixed(1) + " dB"],
358     ["심볼레이트(추정)", fmtHz(d.baud_hint)]];
359 }
360 $("survInfo").innerHTML = `<h3>${title}</h3>` +
361   '<div class="si-grid">' + rows.map(([k, v]) =>
362     `<div><div class="si-k">${k}</div><div class="si-v">${v}</div></div>`).join("") + "</div>";
363 $("survInfo").classList.remove("hidden");
364 $("survInfo").scrollIntoView({ behavior: REDUCED ? "auto" : "smooth", block: "nearest" });
365 }
366
367 /* --- burst map: whole-record waterfall with clickable time-burst boxes (single signal) --- */
368 function renderBurstMap(ov, result) {
369   show($("#burstMap"));
370   paintWaterfall($("#burstFall"), ov.waterfall, 150);
371   const host = $("#burstBoxes"), n = ov.n || 1;
372   host.innerHTML = "";
373   (result.bursts || []).forEach((b, i) => {
374     const top = Math.max(0, b.start / n * 100);
375     const dur = (b.end - b.start) / ov.fs;
376     const lbl = `버스트 ${i + 1} · ${dur >= 1 ? dur.toFixed(2) + " s" : (dur * 1e3).toFixed(0) + " m
377       s"}`;
378     const box = document.createElement("button"); // full width (one signal), spans t0..t1 in time
379     box.className = "box" + (i === result.burst_idx ? " sel" : "");
380     box.style.left = "0%"; box.style.width = "100%"; box.style.top = top + "%";
381     box.style.height = Math.min(100 - top, Math.max(5, (b.end - b.start) / n * 100)) + "%";
382     box.title = lbl;
383     box.innerHTML = `<span class="box-lbl">${lbl}</span>`;
384     box.onclick = () => { state.burst = i; analyze(); }; // re-analyse that burst; map persists
385     host.appendChild(box);
386   });
387
388 /* --- rendering --- */
389 function statusOf(lock) {
390   if (lock >= 60) return ["복조 성공", "ok"];
391   if (lock >= 40) return ["부분 복조", "warn"];
392   return ["복조 실패", "bad"];
393 }
394 function render(d) {
395   hideAll();
396   show($("#results"));
397   $("#burstMap").classList.add("hidden"); // re-shown by renderBurstMap only in multi-burst singl
398   e mode
399   const lock = d.quality.lock, [txt, cls] = statusOf(lock);
400   const pill = $("#statusPill");
401   pill.textContent = txt;
402   pill.className = "pill " + cls;
403   $("#fileName").textContent =
404     `${state.file.name} · ${fmtHz(d.fs)} · ${d.fmt === "real" ? "Real PCM" : "IQ"}`;
405   $("#merVal").textContent = d.quality.mer_db == null ? "-" : d.quality.mer_db.toFixed(1) + " dB";
406   $("#evmVal").textContent = d.quality.evm == null ? "-" : (d.quality.evm * 100).toFixed(1) + " %";
407   $("#snrVal").textContent = snrText(d);
408
409   const flags = [];
410   if (d.family === "fsk") flags.push('<span class="chip warn">FSK 계열 · 주파수 판별</span>');
411   if (d.eq && d.eq.applied) flags.push('<span class="chip hit">등화기 적용 · ${
412     d.eq.mode === "fse" ? "T/2 분수간격" : "심볼간격"></span>');
413   if (d.detected.alias_resolved) flags.push('<span class="chip warn">반송파 앨리어스 보정</span>');
414   if (d.detected.baud_fallback) flags.push('<span class="chip warn">심볼레이트 대역폭 폴백</span>');
415   if (d.detected.carrier_ambiguous) flags.push('<span class="chip warn">반송파 모호 · 앨리어싱 가능<

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414     /span>');
415     $("flagRow").innerHTML = flags.join("");
416
417     renderBursts(d);
418     animateGauge(lock, cls);
419     renderParams(d);
420     drawSpectrum(d);
421     drawWaterfall(d);
422     startPlay(d);
423 }
424 function snrText(d) {
425     const s = d.snr_est_db;
426     if (s == null || d.quality.lock < 30) return "-";
427     return (s > 28 ? "≥28" : s.toFixed(1)) + " dB";
428 }
429
430 function renderBursts(d) {
431     const row = $("burstRow"), bs = d.bursts || [];
432     // a survey-drilled emitter is already a cut channel: burst re-selection would
433     // re-post the whole capture, so it is disabled while drilled. When the burst MAP
434     // is shown (multi-burst single signal) the map replaces the chips.
435     if (state.drilled || state.overview || bs.length < 2) { row.innerHTML = ""; return; }
436     const dur = (b) => {
437         const sec = (b.end - b.start) / d.fs;
438         return sec >= 1 ? sec.toFixed(2) + " s" : (sec * 1e3).toFixed(1) + " ms";
439     };
440     row.innerHTML = bs.map((b, i) =>
441         `<button class="chip-btn${i === d.burst_idx ? " on" : ""}" data-i="${i}">` +
442         `버스트 ${i + 1} · ${dur(b)}</button>`).join("");
443     row.querySelectorAll("button").forEach((el) => {
444         el.onclick = () => { state.burst = +el.dataset.i; analyze(); };
445     });
446 }
447
448 function chip(hit, truthTxt) {
449     return `<span class="chip ${hit ? "hit" : "miss"}">정답 ${truthTxt} · ${hit ? "일치" : "불일치"}</span>`;
450 }
451 function renderParams(d) {
452     const det = d.detected, t = state.sidecar && state.sidecar.truth;
453     const near = (a, b, tol) => Math.abs(a - b) <= tol;
454     const rows = [
455         ["중심주파수", det.rf_hz != null ? fmtRf(det.rf_hz) : fmtHz(det.fc),
456         t && t.fc != null && chip(near(det.fc, t.fc, Math.max(300, 0.02 * t.baud)), fmtHz(t.fc)),
457         det.rf_hz != null ? `기저대역 ${fmtHz(det.fc)}` : ""],
458         ["심볼레이트", fmtHz(det.baud), t && t.baud != null && chip(near(det.baud, t.baud, 0.03 * t.baud),
459         fmtHz(t.baud)),
460         det.baud_conf ? `스펙트럼 라인 강도 ×${Math.round(det.baud_conf)}` : ""],
461         ["변조방식", det.mod.toUpperCase(), t && t.mod != null && chip(det.mod === t.mod, t.mod.toUpperCase(),
462         det.symmetry ? `대칭 ${det.symmetry}` : "주파수 레벨"],
463     ];
464     if (d.family === "fsk") {
465         rows.push(["변조지수 h", det.h.toFixed(2),
466         t && t.h != null && chip(near(det.h, t.h, 0.15), t.h.toFixed(2)), ""]);
467     } else {
468         rows.push(["롤오프", det.rolloff.toFixed(2),
469         t && t.rolloff != null && chip(near(det.rolloff, t.rolloff, 0.11), t.rolloff.toFixed(2)), ""]);
470     }
471 }

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```

470   $("paramGrid").innerHTML = rows.map(([label, val, hit, note]) => `
471     <div class="card param">
472       <div class="param-label">${label}</div>
473       <div class="param-val">${val}</div>
474       ${note ? `<div class="param-note">${note}</div>` : ""}
475       ${hit ? `<div class="chips">${hit}</div>` : ""}
476     </div>`).join("");
477 }
478
479 function animateGauge(lock, cls) {
480   const arc = $("donutArc"), num = $("lockNum"), C = 2 * Math.PI * 52;
481   arc.style.stroke = { ok: "#12b886", warn: "#f59f00", bad: "#fa5252" }[cls];
482   arc.style.strokeDasharray = C;
483   const target = C * (1 - Math.max(0, Math.min(100, lock)) / 100);
484   if (REDUCED) { arc.style.strokeDashoffset = target; num.textContent = Math.round(lock); return; }
485   arc.style.strokeDashoffset = C;
486   requestAnimationFrame(() => { arc.style.strokeDashoffset = target; });
487   const t0 = performance.now();
488   const step = (t) => {
489     const k = Math.min(1, (t - t0) / 900);
490     num.textContent = Math.round(lock * (1 - Math.pow(1 - k, 3)));
491     if (k < 1) requestAnimationFrame(step);
492   };
493   requestAnimationFrame(step);
494 }
495
496 /* --- canvas helpers --- */
497 function fit(cv, hCss) {
498   const dpr = window.devicePixelRatio || 1, w = cv.clientWidth || 320;
499   const h = hCss || w;
500   cv.width = w * dpr; cv.height = h * dpr;
501   const g = cv.getContext("2d");
502   g.setTransform(dpr, 0, 0, dpr, 0, 0);
503   return [g, w, h];
504 }
505 function pct(a, p) {
506   const s = [...a].sort((x, y) => x - y);
507   return s[Math.min(s.length - 1, Math.max(0, Math.round(p * (s.length - 1))))];
508 }
509
510 /* --- spectrum --- */
511 function drawSpectrum(d) {
512   const [g, w, h] = fit($("specCanvas"), 120);
513   g.fillStyle = "#0d1320"; g.fillRect(0, 0, w, h);
514   if (!d.spectrum) return;
515   const f = d.spectrum.f, db = d.spectrum.db;
516   const lo = pct(db, 0.02), hi = pct(db, 1) + 2;
517   const fmin = f[0], fmax = f[f.length - 1];
518   const X = (v) => (v - fmin) / (fmax - fmin) * w;
519   const Y = (v) => h - (Math.max(lo, Math.min(hi, v)) - lo) / (hi - lo) * (h - 8) - 4;
520   const det = d.detected, half = det.baud / 2e3;
521   g.strokeStyle = "rgba(120,180,255,.35)"; g.setLineDash([4, 4]);
522   for (const gf of [det.fc / 1e3 - half, det.fc / 1e3 + half]) {
523     g.beginPath(); g.moveTo(X(gf), 0); g.lineTo(X(gf), h); g.stroke();
524   }
525   g.setLineDash([]);
526   g.strokeStyle = "rgba(255,255,255,.45)";
527   g.beginPath(); g.moveTo(X(det.fc / 1e3), 0); g.lineTo(X(det.fc / 1e3), h); g.stroke();
528   g.strokeStyle = "#5aa9ff"; g.lineWidth = 1.4; g.beginPath();
529   f.forEach((v, k) => (k ? g.lineTo(X(v), Y(db[k])) : g.moveTo(X(v), Y(db[k]))));

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530 g.stroke();
531 g.fillStyle = "rgba(255,255,255,.5)"; g.font = "10px sans-serif";
532 g.fillText(sig(fmin) + " kHz", 4, h - 4);
533 const tmax = sig(fmax) + " kHz";
534 g.fillText(tmax, w - g.measureText(tmax).width - 4, h - 4);
535 }
536
537 /* --- waterfall --- */
538 function drawWaterfall(d) { paintWaterfall($("#fallCanvas"), d.waterfall, 150); }
539 function paintWaterfall(canvas, wf, hCss) {
540   const [g, w, h] = fit(canvas, hCss);
541   g.fillStyle = "#0d1320"; g.fillRect(0, 0, w, h);
542   if (!wf || !wf.rows) return;
543   const lo = pct(wf.db, 0.05), hi = pct(wf.db, 0.95);
544   const img = g.createImageData(wf.bins, wf.rows);
545   for (let k = 0; k < wf.db.length; k++) {
546     const t = Math.max(0, Math.min(1, (wf.db[k] - lo) / (hi - lo + 1e-9)));
547     const o = k * 4; // dark -> blue -> white ramp
548     img.data[o] = t < 0.5 ? 13 + t * 120 : 73 + (t - 0.5) * 364;
549     img.data[o + 1] = t < 0.5 ? 19 + t * 260 : 149 + (t - 0.5) * 212;
550     img.data[o + 2] = t < 0.5 ? 32 + t * 446 : 255;
551     img.data[o + 3] = 255;
552   }
553   createImageBitmap(img).then((bmp) => { g.imageSmoothingEnabled = true; g.drawImage(bmp, 0, 0, w, h
554     ); });
555 }
556
557 /* --- constellation playback (the headline) --- */
558 const play = { raf: 0, i: 0, on: false, data: null };
559 let CG = null; // cached constellation geometry
560
561 function stopPlay() { cancelAnimationFrame(play.raf); play.raf = 0; play.on = false; }
562 function setPlayBtn(on) { $("#playBtn").textContent = on ? "⏏ 일시정지" : "▶ 재생"; }
563
564 function startPlay(d) {
565   stopPlay();
566   play.data = d;
567   play.i = 0;
568   const n = d.constellation.i.length;
569   $("#scrub").max = String(Math.max(1, n));
570   $("#playInfo").textContent = `${n.toLocaleString()} 심볼 · 시간 순서로 재생 (잔광 효과)`;
571   drawConstBase(d);
572   if (REDUCED) { // static full view, no motion
573     drawPoints(d, 0, n);
574     $("#scrub").value = String(n);
575     setPlayBtn(false);
576     return;
577   }
578   play.on = true;
579   setPlayBtn(true);
580   play.raf = requestAnimationFrame(loop);
581 }
582
583 $("#playBtn").onclick = () => {
584   if (!play.data) return;
585   if (play.on) { stopPlay(); setPlayBtn(false); }
586   else { play.on = true; setPlayBtn(true); play.raf = requestAnimationFrame(loop); }
587 };
588
589 $("#scrub").oninput = () => {
590   if (!play.data) return;
591   stopPlay(); setPlayBtn(false);

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589 play.i = +$("scrub").value;
590 drawConstBase(play.data);
591 drawPoints(play.data, 0, play.i);    // redraw history up to the scrub point
592 };
593
594 function loop() {
595     const d = play.data, n = d.constellation.i.length;
596     const step = Math.max(1, Math.round(n / 600) * (+$("speedSel").value));
597     const to = Math.min(n, play.i + step);
598     fade();                                // phosphor persistence
599     drawPoints(d, play.i, to);
600     play.i = to >= n ? 0 : to;
601     if (play.i === 0) drawConstBase(d); // loop: clear the trail
602     $("scrub").value = String(play.i);
603     if (play.on) play.raf = requestAnimationFrame(loop);
604 }
605
606 function drawConstBase(d) {
607     const [g, w] = fit($("constCanvas"));
608     const fsk = d.family === "fsk";
609     const I = d.constellation.i, Q = d.constellation.q;
610     const ideal = fsk ? [] : idealPoints(d.detected.mod);
611     let lim = 0;
612     for (let k = 0; k < I.length; k++) lim = Math.max(lim, Math.abs(I[k]), Math.abs(Q[k]));
613     for (const p of ideal) lim = Math.max(lim, Math.abs(p.i), Math.abs(p.q));
614     lim = (lim || 1) * 1.12;
615     const R = w / 2 * 0.9;
616     CG = { g, w, lim, R, map: (re, im) => [w / 2 + re / lim * R, w / 2 - im / lim * R] };
617
618     g.fillStyle = "#0d1320"; g.fillRect(0, 0, w, w);
619     g.strokeStyle = "rgba(255,255,255,.06)"; g.lineWidth = 1;
620     for (let f = -1; f <= 1; f += 0.5) {
621         const [gx] = CG.map(f, 0), [, gy] = CG.map(0, f);
622         g.beginPath(); g.moveTo(gx, 0); g.lineTo(gx, w); g.stroke();
623         g.beginPath(); g.moveTo(0, gy); g.lineTo(w, gy); g.stroke();
624     }
625     if (fsk) {                                // frequency levels as dashed bands
626         g.strokeStyle = "rgba(255,255,255,.5)"; g.setLineDash([6, 5]);
627         for (const lv of [...new Set(I.map((v) => Math.round(v)))] .filter((v) => Math.abs(v) <= 8)) {
628             const [, yy] = CG.map(0, lv);
629             g.beginPath(); g.moveTo(0, yy); g.lineTo(w, yy); g.stroke();
630         }
631         g.setLineDash([]);
632     } else {
633         g.strokeStyle = "rgba(255,255,255,.75)"; g.lineWidth = 1.4;
634         for (const p of ideal) {
635             const [x, y] = CG.map(p.i, p.q);
636             g.beginPath(); g.arc(x, y, 6, 0, 7); g.stroke();
637         }
638     }
639 }
640
641 function fade() {
642     if (!CG) return;
643     CG.g.fillStyle = "rgba(13,19,32,.14)"; // translucent wash = phosphor decay
644     CG.g.fillRect(0, 0, CG.w, CG.w);
645 }
646
647 function drawPoints(d, from, to) {
648     if (!CG || to <= from) return;
649     const g = CG.g, I = d.constellation.i, Q = d.constellation.q;
650     const fsk = d.family === "fsk";

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649   const a = Math.max(0.08, Math.min(0.55, 150 / (to - from)));
650   g.globalCompositeOperation = "lighter";
651   g.fillStyle = `rgba(90,170,255,${a})`;
652   for (let k = from; k < to; k++) {
653     // FSK has no I/Q plane: spread levels vertically, jitter horizontally for density
654     const [x, y] = fsk ? CG.map((k % 89) / 89 - 0.5, I[k]) : CG.map(I[k], Q[k]);
655     g.beginPath(); g.arc(x, y, 1.8, 0, 7); g.fill();
656   }
657   g.globalCompositeOperation = "source-over";
658 }
659
660 /* --- downloads --- */
661 function saveBlob(name, text, type) {
662   const url = URL.createObjectURL(new Blob([text], { type }));
663   const a = document.createElement("a");
664   a.href = url; a.download = name; a.click();
665   URL.revokeObjectURL(url);
666 }
667 const stem = () => state.file.name.replace(/\.([^.]*)$/, "");
668 $("copyBits").onclick = async (e) => { // decoded bitstream -> clipboard, with brief feedback
669   const btn = e.currentTarget, was = btn.textContent;
670   try { await navigator.clipboard.writeText(state.resp.bits); btn.textContent = "복사됨 ✓"; }
671   catch { btn.textContent = "복사 실패"; }
672   setTimeout(() => { btn.textContent = was; }, 1400);
673 };
674 $("dlBits").onclick = () => saveBlob(stem() + ".bits.txt", state.resp.bits, "text/plain");
675 $("dlSyms").onclick = () => {
676   const c = state.resp.constellation;
677   saveBlob(stem() + ".symbols.csv", "i,q\n" + c.i.map((v, k) => `${v},${c.q[k]}`).join("\n"), "text/
678   csv");
679 };
680 $("dlReport").onclick = () =>
681   saveBlob(stem() + ".report.json", JSON.stringify(state.resp, null, 2), "application/json");
682
683 /* --- view helpers --- */
684 function show(el) { el.classList.remove("hidden"); }
685 function hideAll() {
686   stopPlay();
687   ["metaForm", "loading", "errorCard", "results", "batchCard", "surveyCard"]
688     .forEach((id) => $(id).classList.add("hidden"));
689   $("backBatch").classList.add("hidden");
690 }
691 function showError(msg) { hideAll(); $("errMsg").textContent = msg; show($("errorCard")); }
692
693 /* --- dropzone wiring --- */
694 const drop = $("dropCard"), input = $("fileInput");
695 // reset value first: re-picking the SAME file fires no change event otherwise
696 const pick = () => { input.value = ""; input.click(); };
697 drop.onclick = pick;
698 drop.onkeydown = (e) => { if (e.key === "Enter" || e.key === " ") { e.preventDefault(); pick(); } };
699 input.onChange = () => { if (input.files.length) acceptFiles(input.files); };
700 drop.addEventListener("dragover", (e) => { e.preventDefault(); drop.classList.add("drag"); });
701 drop.addEventListener("dragleave", () => drop.classList.remove("drag"));
702 drop.addEventListener("drop", (e) => {
703   e.preventDefault(); drop.classList.remove("drag");
704   if (e.dataTransfer.files.length) acceptFiles(e.dataTransfer.files);
705 });

```

```

1  """sigc.py - 실신호 특징 4줄 + find_bursts 단계 관찰 프로브 (장비 필사용, signus/ 옆에 저장).
2      python3 sigc.py kat                # 자기검증 - 표지의 기대 4줄과 비교
3      python3 sigc.py <캡처> [1-6]        # 요약 4줄(받아쳐 회신) / 단계 관찰(그림 또는 ASCII)
4  """
5  import sys
6
7  import numpy as np
8  from scipy.ndimage import uniform_filter1d
9  from scipy.signal import get_window, stft
10
11 from signus import dsp, sigio
12 from signus.cli import check_code
13
14
15 def otsu(v, bins=128):
16     h, ed = np.histogram(v, bins=bins)
17     c = (ed[:-1] + ed[1:]) / 2
18     w = np.cumsum(h).astype(float)
19     m = np.cumsum(h * c)
20     mb = m / np.where(w > 0, w, 1)
21     mf = (m[-1] - m) / np.where(w[-1] - w > 0, w[-1] - w, 1)
22     return float(c[int(np.argmax(w * (w[-1] - w) * (mb - mf) ** 2))])
23
24
25 def runs_of(mask):
26     d = np.diff(mask.astype(np.int8))
27     s = list(np.where(d == 1)[0] + 1)
28     e = list(np.where(d == -1)[0] + 1)
29     if mask[0]:
30         s.insert(0, 0)
31     if mask[-1]:
32         e.append(mask.size)
33     return list(zip(s, e, strict=True))
34
35
36 if len(sys.argv) < 2:
37     raise SystemExit("사용법: python3 sigc.py kat | python3 sigc.py <캡처파일> [단계 1-6]")
38 arg, step = sys.argv[1], int(sys.argv[2]) if len(sys.argv) > 2 else 0
39 if arg == "kat":
40     fs, t = 1e6, np.arange(66000.0)      # 66000: 마지막 버스트가 파일 끝까지 이어진다
41     rng = np.random.default_rng(7)        # 시드 고정 스트림 - numpy 가 재현을 보장한다
42     x0 = 0.01 * (rng.standard_normal(66000) + 1j * rng.standard_normal(66000))
43     x0 += 0.2 * np.exp(2j * np.pi * 0.29 * t)
44     x0 += 0.5 * np.exp(-2j * np.pi * 0.11 * t) * ((t // 6000).astype(int) % 2 == 0)
45     x0 += 0.0028 * np.exp(2j * np.pi * 0.41 * t)    # 문턱 경계에 걸친 성분들: 필사 오타가
46     x0[31000:35000] += 0.0088 * np.exp(2j * np.pi * 0.35 * t[:4000])    # 상수 하나만
47     x0[42400:47600] += 0.0075 * np.exp(2j * np.pi * -0.37 * t[:5200])    # 스쳐도 아래
48     x0[200:260] = 0.95                # 요약 4줄 어딘가가 바뀌게 만든다
49     x0[21000] += 8.0
50     x0[15000] += 2.0
51 else:
52     x0, meta = sigio.read(arg)
53     fs = meta.fs
54 cx = np.iscomplexobj(x0)
55 raw = np.abs(np.asarray([x0.real, x0.imag])).max(0) if cx else np.abs(x0)
56 cp = float((raw > 0.98 * np.percentile(raw, 99.9)).mean())    # 만점(±1)이 아니라 캡처 자신의
57 # 99.9퍼센타일 기준: float 캡처는 만점 정규화가 안 돼 ±1 기준이면 통짜 오경보다(실측 cp972)
58 x = dsp.analytic(x0)
59 n, pw = x.size, np.abs(x) ** 2
60 rms = float(np.sqrt(pw.mean()) + 1e-30)

```



```

61 dc = abs(complex(x.mean())) / rms # dc 는 블록 전에 재고, 그 뒤는 pipeline 과 똑같이
62 x = x - x.mean() # DC 블록한 신호로 켜다 - 안 그러면 dc 큰 캡처에서
63 pw = np.abs(x) ** 2 # 프로브와 analyze 가 정반대 진단을 낸다
64 # 러닝합 잔차 때문에 정확히 0 인 구간(스켈치/ 뮤트)에서 음수가 나와 log10 이 NaN 이 되고,
65 # otsu 의 histogram 이 장비에서 생 트레이스백으로 죽었다 - 0 으로 눌러 막는다.
66 lp = np.log10(np.maximum(uniform_filter1d(pw, max(64, n // 1000)), 0) + 1e-20)
67 hot = lp >= (et := otsu(lp))
68 ev = float(lp[hot].mean() - lp[~hot].mean()) if 0 < hot.sum() < n else 0.0
69 ed = float(hot.mean())
70 sp = 10 * np.log10(pw.max() / (pw.mean() + 1e-30) + 1e-30)
71 nper = int(min(128, max(64, 1 << int(np.log2(max(n // 2, 64)))))
72 hop = nper // 2
73 _, _, z = stft(x, fs=fs, window=get_window("blackmanharris", nper), nperseg=nper,
74             ..., ..., noverlap=nper - hop, return_onesided=False, boundary=None, padded=False)
75 P = np.abs(z) ** 2
76 nb, nc = P.shape
77 fl = np.percentile(P, 10, axis=1)
78 r = P / (fl[:, None] + 1e-30)
79 k = max(3, nb // 16)
80 sc = uniform_filter1d(np.log10(np.sort(r, axis=0)[-k:].mean(axis=0) + 1e-30), 3)
81 thr = otsu(sc, bins=64)
82 quiet = sc[sc < thr]
83 base = float(np.median(quiet)) if quiet.size else 9.99
84 cn = float(np.log10((np.log(nb / k) + 1) / 0.105))
85 dlo, dhi = 0.25, 0.45 # find_bursts 의 두 문턱 여유. 상수는 한 곳에만 쓰고
86 lov, hiv = base + dlo, base + dhi # b 줄에 이 숫자를 그대로 되찍는다 - 오타면 그게 드러난다
87 above, hi_m = sc >= lov, sc >= hiv
88 rr = [(s, e) for s, e in runs_of(above) if hi_m[s:e].any()]
89 spans = [(max(0, s * hop) + (hop if e - s >= 6 else 0), # 6열 이상은 양끝 한 hop
90          min(n, (e - 1) * hop + nper) - (hop if e - s >= 6 else 0)) # 씹 자른다 - dsp 의
91          for s, e in rr] # 스파이크 게이트와 같은
92 segs = [pw[s:e] for s, e in spans] # 구간에서 재려고
93 sk = sum(float(sg.max() / (sg.mean() + 1e-30)) > 60.0 for sg in segs)
94 gp_ = np.array([rr[i + 1][0] - rr[i][1] for i in range(len(rr) - 1)])
95 sb = round(10 * float(np.median([sc[s:e].max() for s, e in rr]) - base)) if rr else 0
96 g = float(np.median(fl if cx else fl[:nb // 2]) + 1e-30) # real 은 해석신호라 음수 반쪽이
97 # 비어 있다: 전 빈 중앙값을 쓰면 바닥이 0 으로 내려가 밴드 절반이 '점유'로 읽힌다(실측)
98 Ps, fls = np.fft.fftshift(P, 0), np.fft.fftshift(fl)
99 du = (Ps > 40 * g).mean(1)
100 occ = du > 0.1
101 kb, kcont = int(occ.sum()), int((fls > 5 * g).sum())
102 grp = runs_of(occ) if occ.any() else []
103 wd = max((e - s for s, e in grp), default=0)
104 p95 = np.percentile(Ps, 95, axis=1) / g
105 pk = int(np.argmax(p95))
106 pgrp = next(((s, e) for s, e in grp if s <= pk < e), (pk, pk + 1))
107 m2 = np.where(occ, p95, 0.0)
108 m2[pgrp[0]:pgrp[1]] = 0.0
109 q2 = int(np.argmax(m2))
110 qo, qd, qs = (q2 - nb // 2, round(100 * float(du[q2])),
111             ..., ..., round(10 * np.log10(m2[q2] + 1e-30))) if m2[q2] > 0 else (999, 0, 0)
112 la = (f"sigc a n{n} f{fs:.0f} ev{round(100 * ev)} ed{round(100 * ed)}"
113      f" sp{round(float(sp))} dc{round(100 * dc)} cp{round(1000 * cp)}"
114      f" cx{int(cx)}")
115 lb = (f"sigc b c{nc} g{nb} b{round(100 * base)} cn{round(100 * cn)}"
116      f" t{round(100 * thr)} m{round(100 * float(sc.max()))}"
117      f" dlo{round(100 * dlo)} dhi{round(100 * dhi)}") # 뽕샘으로 되계산하면 안 된다 --
118 # (base+0.45)-base 가 44.99999999999999 라, int 로 잘못 옮기면 44 로 찍혔다 (2026-08-14 실측)
119 lc = (f"sigc c r{len(rr)} dn{round(float(np.median([e - s for s, e in rr]))} if rr else 0}"
120      f" gp{round(float(np.median(gp_))} if gp_.size else 0} sb{sb}")

```

```

121     f" av{round(100 * float(above.mean()))} ah{round(100 * float(hi_m.mean()))}"
122     f" sk{sk}")
123 ld = (f"sigc d kb{kb} kc{kcont} w{wd} p{pk - nb // 2} pd{round(100 * float(du[pk]))}"
124     f" ps{round(10 * np.log10(p95[pk] + 1e-30))} q{qo} qd{qd} qs{qs}")
125 if step == 0:                                # 요약 4줄 - 이것만 받아쳐서 회신한다
126     for s in (la, lb, lc, ld):
127         print(f"{s} #{check_code(s)}")
128     sys.exit()
129
130 # — 여기부터는 그림 단계(1~6) 전용이다. 요약 4줄만 쓸 거면 여기서 멈춰도 된다 —
131
132 try:
133     import matplotlib.pyplot as plt
134     plt = None if plt.get_backend().lower() == "agg" else plt    # 무화면(ssh)이면 ASCII 로
135 except ImportError:
136     plt = None
137
138
139 def pool(a, m):
140     a = np.asarray(a, float)
141     k = max(1, a.shape[-1] // m)
142     return a[..., :(a.shape[-1] // k) * k].reshape(*a.shape[:-1], -1, k).max(-1)
143
144
145 def curve(y, marks):
146     if plt:
147         plt.plot(y)
148         for la, v in marks:
149             plt.axhline(v, ls="--", label=f"{la} {v:.2f}")
150         plt.legend()
151         return plt.show()
152     # 얼마나 최소·최대를 같이 그린다: max 만 그리면 버스트 사이의 골이 지워져 계단이 통짜
153     # 블록으로 보이고, 관측자가 "평평하다(=베토)" 라고 정반대로 회신하게 된다.
154     kk = max(1, y.size // 100)
155     y2 = y[:y.size // kk * kk].reshape(-1, kk)
156     ylo, yhi = y2.min(1), y2.max(1)
157     bot = float(min(ylo.min(), *(v for _, v in marks)))
158     st = (float(max(yhi.max(), *(v for _, v in marks))) - bot) / 18 or 1.0
159     rl, rh = np.round((ylo - bot) / st).astype(int), np.round((yhi - bot) / st).astype(int)
160     rm = {}
161     for la, v in marks:
162         rm.setdefault(round((v - bot) / st), []).append(f"{la} {v:.2f}")
163     for r in range(18, -1, -1):
164         bg = "-" if r in rm else " "                # 문턱은 가로 전체 선으로 (겹치면 라벨 합침)
165         print("".join("#" if rl[i] >= r else ("+" if rh[i] >= r else bg)
166             for i in range(rl.size)) + "|" + " / ".join(rm.get(r, [])))
167     print("#=이 구간 내내 그 위 · +=봉우리만 그 위 · -=문턱선")
168
169
170 def image(img):
171     if plt:
172         plt.imshow(img, aspect="auto", origin="lower", interpolation="nearest")
173         return plt.show()
174     sm = pool(pool(img, 100).T, 32).T
175     hi = sm.max() + 1e-30
176     lo = max(float(np.percentile(sm, 5)), hi - 4)    # 표시폭 4 decades 고정 - real 캡처는 빈
177     # 음수 반쪽이 1e-16 이라 5퍼센타일을 쓰면 눈금이 14 decades 로 늘어나 통짜 벽이 된다
178     for row in np.clip((sm - lo) / (hi - lo) * 9.99, 0, 9).astype(int)[::-1]:
179         print("".join(" .:~+*#%@"[i] for i in row))
180

```

```

181
182 if step == 1:
183     print(f"n {n} fs {fs:.0f} 길이 {n / fs:.3f}s 형식 {'iq' if np.iscomplexobj(x0) else 'real'}")
184     print(f"rms {rms:.4f} dc {100 * dc:.1f}% 클리핑 {1000 * cp:.1f}% 피크비 {float(sp):.0f}dB")
185 elif step == 2:
186     curve(lp, [("otsu", et)])
187     print(f"포락선 분리도 {ev:.3f} decades - 0.12 미만이면 find_bursts 는 통짜 [(0,n)] (베토)")
188     print(f"포락선 듀티 {100 * ed:.0f}% (버스트가 계단으로 보여야 정상)")
189 elif step == 3:
190     image(np.log10(Ps + 1e-20))
191     print("위터폴 절대전력 (아래=-fs/2, 위=+fs/2) - 가로로 끊기지 않는 띠 = 연속 방사체")
192 elif step == 4:
193     image(np.log10(np.fft.fftshift(r, 0) + 1e-30))
194     print("빈별 바닥 대비 비율 - find_bursts 가 보는 그림. 세기가 일정한 연속 방사체는 여기서"
195           " 사라지지만, 세기가 흔들리면 가짜 버스트로 남는다 (d 줄 pd 로 확인)")
196 elif step == 5:
197     curve(sc, [("base", base), ("hi", hiv), ("c_noise", cn)])
198     print(f"열 점수 - hi 위 봉우리가 버스트 후보. 런 {len(rr)}개, base {base:.2f}, cn {cn:.2f}")
199 elif step == 6:
200     for (s, e), (a0, a1), sg in zip(rr, spans, segs, strict=True):
201         print(f"열 {s}-{e} 샘플 {a0}-{a1} 높이 {10 * (float(sc[s:e].max()) - base):.0f}dB"
202               f" 피크/평균 {float(sg.max() / (sg.mean() + 1e-30)):.0f} (60 초과=스파이크 탈락)")
203     print(f"위는 병합/가드 전 원시 후보 {len(rr)}개 - 아래가 실제로 분석에 쓰이는 답이다:")
204     fb = dsp.find_bursts(x, fs)
205     print(f"find_bursts → {fb[:6]}{'...' if len(fb) > 6 else ''}"
206           + (" ※ 통짜 = 버스트 미검출" if fb == [(0, n)] else f" 버스트 {len(fb)}개"))

```

```

1  """sa.py - 종합 관찰 프로브 (장비 필사용, signus/ 옆에 저장). 한 번에:
2  디버그 출력(포락선·버스트 표) + <캡처>.sa.png (스펙트로그램 + 검출 띠 + 버스트 번호 +
3  상위 버스트별 [PSD |  $x^2$  |  $x^4$  |  $x^8$  | AM검파] 판독 행 - 바늘이 처음 서는 판이 변조).
4  python3 sa.py kat # 자기검증: 표지 기대 줄과 비교, sa-kat.png 생성
5  python3 sa.py <캡처파일> [K] # 판독할 버스트 수 K (기본 4, 에너지 상위부터)
6   $x^m$  판은 접힌 축( $0 \sim fs/2$ , 음수 주파수를 겹쳐 최대값) - 바늘 위치가  $m \cdot fc \pmod{fs}$  다.
7  """
8  import struct
9  import sys
10 import zlib
11
12 import numpy as np
13 from scipy.signal import stft, welch
14
15 from signus import dsp, sigio
16 from signus.chirp import is_chirp, sweeps_band
17 from signus.cli import check_code
18 from signus.fsk import fsk_gate
19
20 GLYPH = {"0": (14, 17, 19, 21, 25, 17, 14), "1": (4, 12, 4, 4, 4, 4, 14),
21          "2": (14, 17, 1, 2, 4, 8, 31), "3": (31, 2, 4, 2, 1, 17, 14),
22          "4": (2, 6, 10, 18, 31, 2, 2), "5": (31, 16, 30, 1, 1, 17, 14),
23          "6": (6, 8, 16, 30, 17, 17, 14), "7": (31, 1, 2, 4, 8, 8, 8),
24          "8": (14, 17, 17, 14, 17, 17, 14), "9": (14, 17, 17, 15, 1, 2, 12)}
25
26
27 def draw_text(img, row, col, text, s=2):
28     for ch in text:
29         for r, bits in enumerate(GLYPH[ch]):
30             for c in range(5):
31                 if bits >> (4 - c) & 1:
32                     img[row + r * s:row + r * s + s, col + c * s:col + c * s + s] = 1.0
33             col += 6 * s
34
35
36 def png_gray(img, path):
37     img = np.clip(img * 255, 0, 255).astype(np.uint8)
38     h, w = img.shape
39
40     def chunk(tag, data):
41         c = tag + data
42         return struct.pack(">I", len(data)) + c + struct.pack(">I", zlib.crc32(c))
43
44     raw = b"".join(b"\x00" + img[r].tobytes() for r in range(h))
45     with open(path, "wb") as fh:
46         fh.write(b"\x89PNG\r\n\x1a\n"
47                + chunk(b"IHDR", struct.pack(">IIBBBBB", w, h, 8, 0, 0, 0, 0))
48                + chunk(b>IDAT", zlib.compress(raw, 6)) + chunk(b"IEND", b""))
49
50
51 def panel(fvals, dbvals, wp=500, hp=120, grid=()):
52     """스펙트럼 곡선 한 판 - 바늘 보존을 위해 표시 칸마다 최대값 풀림."""
53     cols = np.clip((fvals / fvals.max() * (wp - 1)).astype(int), 0, wp - 1)
54     cur = np.full(wp, -1e9)
55     np.maximum.at(cur, cols, dbvals)
56     bad = cur < -1e8
57     cur[bad] = np.interp(np.flatnonzero(bad), np.flatnonzero(~bad), cur[~bad])
58     g_lo, g_hi = np.percentile(cur, 5) - 2, cur.max() + 2
59     img = np.zeros((hp, wp))
60     for gf in grid:

```

```

61         img[:,6, int(gf / fvals.max() * (wp - 1))] = 0.35
62     rows = ((hp - 1) * (1 - np.clip((cur - g_lo) / (g_hi - g_lo), 0, 1))).astype(int)
63     for cx in range(wp):
64         r0, r1 = (rows[cx], rows[cx - 1]) if cx else (rows[0], rows[0])
65         img[min(r0, r1):max(r0, r1) + 1, cx] = 1.0
66     return img
67
68
69 def peak(fv, dbv, fmin=20.0):
70     """판 하나의 최대 바늘: (주파수 Hz, 중앙값 대비 돌출 dB) -- 숫자로 받아칠 수 있게."""
71     ok = fv >= fmin
72     i = int(np.argmax(dbv[ok]))
73     return round(float(fv[ok][i]), round(float(dbv[ok][i] - np.median(dbv[ok]))))
74
75
76 def verdict(z, fs, d2, d4, d8, dam):
77     """바늘 돌출로 버스트의 변조 부류를 판정 -- 문턱은 합성 배터리로 캘리브레이션
78     (bpsk d2 54 / qpsk d4 38 / 16qam d4 30 / 8psk d8 29 / fsk am 17, 광대역 잡음꼴 무바늘)."""
79     if is_chirp(z, fs) and sweeps_band(z, fs):
80         return "chirp" # 처프/LoRa 류 -- 성상도 복조 대상 아님
81     if fsk_gate(z, fs):
82         return "fsk"
83     if d2 >= 35 and d2 >= d4 + 5:
84         return "bpsk"
85     if d4 >= 28:
86         return "qpsk" # qpsk 또는 사각 QAM (4차에서 무너짐)
87     if d8 >= 24:
88         return "8psk"
89     return "lin" if dam >= 25 else "flat" # lin=선형(차수불명) / flat=바늘 없음(잡음꼴)
90
91
92 def burst_row(z, fs, label):
93     """한 버스트의 판독 행: [PSD | x^2 | x^4 | x^8 | AM]. x^m 은 음수 주파수를 접는다."""
94     grid = tuple(fs / 2 * k / 5 for k in (1, 2, 3, 4))
95     fr, praw = welch(z.real, fs=fs, nperseg=min(2048, z.size))
96     ps = [panel(fr, 10 * np.log10(praw + 1e-18), grid=grid)]
97     nfft, nums = 1 << 15, []
98     f = np.fft.fftfreq(nfft, 1 / fs)
99     half = nfft // 2
100    for m in (2, 4, 8):
101        mag = np.abs(np.fft.fft((z ** m) * np.hanning(z.size), nfft))
102        fold = np.maximum(mag[:half], np.r_[mag[0], mag[1:half][::-1] * 0
103        + mag[half + 1][::-1]])
104        ps.append(panel(f[:half], 20 * np.log10(fold + 1e-12), grid=grid))
105        pf, pd = peak(f[:half], 20 * np.log10(fold + 1e-12))
106        nums.append(f"p{m} f{pf} d{pd}")
107    u = np.abs(z) ** 2
108    u = u - u.mean()
109    su = np.abs(np.fft.rfft(u * np.hanning(u.size), nfft))
110    ps.append(panel(np.fft.rfftfreq(nfft, 1 / fs), 20 * np.log10(su + 1e-12), grid=grid))
111    af, ad = peak(np.fft.rfftfreq(nfft, 1 / fs), 20 * np.log10(su + 1e-12))
112    nums.append(f"am f{af} d{ad}")
113    lab = verdict(z, fs, int(nums[0].split("d")[1]), int(nums[1].split("d")[1]),
114    + int(nums[2].split("d")[1]), ad)
115    nums.insert(0, lab)
116    div = np.full((120, 4), 0.5)
117    row = np.hstack(sum([p, div] for p in ps[:-1]), []) + [ps[-1]]
118    tag = np.zeros((120, 26))
119    draw_text(tag, 4, 4, label)
120    return np.hstack([tag, row]), " ".join(nums)

```

```

121
122
123 if len(sys.argv) < 2:
124     raise SystemExit("사용법: python3 sa.py kat | python3 sa.py <캡처파일> [버스트수 K]")
125 arg = sys.argv[1]
126 kmax = int(sys.argv[2]) if len(sys.argv) > 2 else 4
127 if arg == "kat":
128     fs, n = 10000.0, 80000
129     rng = np.random.default_rng(21) # 시드 고정 -- numpy 가 재현을 보장한다
130     x0 = 0.18 * rng.standard_normal(n)
131     t = np.arange(15000) / fs
132     for s0, fc, ph in ((12000, 550.0, 2), (50000, 550.0, 4)): # 앞=BPSK, 뒤=QPSK
133         up = np.zeros(15000, complex)
134         up[:,34] = np.exp(2j * np.pi * rng.integers(0, ph, 442) / ph)
135         sym = np.convolve(up, np.hanning(68), "same") # 간이 펄스 성형
136         sym /= np.sqrt(np.mean(np.abs(sym) ** 2))
137         x0[s0:s0 + 15000] += (sym * np.exp(2j * np.pi * fc * t)).real * np.sqrt(2)
138     name = "sa-kat"
139 else:
140     x0, meta = sigio.read(arg)
141     fs = meta.fs
142     name = arg + ".sa"
143 xa = dsp.analytic(x0)
144 xa = xa - xa.mean()
145 n = xa.size
146 fb = dsp.find_bursts(xa, fs)
147 full = fb == [(0, n)]
148 pw = np.abs(xa) ** 2
149 lp = np.log10(np.maximum(np.convolve(pw, np.ones(64) / 64, "same"), 0) + 1e-20)
150 et = np.percentile(lp, 50)
151 ev = float(lp[lp >= et].mean() - lp[lp < et].mean())
152 print(f"\n {n} fs {fs:.0f} 길이 {n / fs:.3f}s 포락선 분리(중앙 기준) {ev:.2f} decades")
153 print(f"find_bursts → {'통짜 [(0,n)] = 미검출' if full else str(len(fb)) + '개'}")
154 order = sorted(range(len(fb)), key=lambda i: -float(pw[fb[i][0]:fb[i][1]].sum()))
155 pick = sorted(order[:kmax]) if not full else []
156 for i, (s0, e0) in enumerate(fb if not full else []):
157     star = " ★판독" if i in pick else ""
158     print(f" {i + 1:2} 샘플 [{s0:>8}, {e0:>8}] {1000 * (e0 - s0) / fs:8.1f} ms"
159           f" 전력 {10 * np.log10(pw[s0:e0].mean() / (pw.mean() + 1e-30) + 1e-30):+5.1f} dB{star}")
160
161 x_disp = x0.real if np.iscomplexobj(x0) else x0
162 _, _, z = stft(x_disp, fs=fs, window="hamming", nperseg=256, noverlap=128,
163               return_onesided=True, boundary=None, padded=False)
164 db = 10 * np.log10(np.abs(z) ** 2 + 1e-12)
165 lo = float(np.percentile(db, 25))
166 rg = float(max(10.0, min(35.0, np.percentile(db, 99.5) - lo)))
167 spec = np.clip((db - lo) / rg, 0, 1)[::-1]
168 kp = max(1, spec.shape[1] // 4000)
169 spec = spec[:, :spec.shape[1] // kp * kp].reshape(spec.shape[0], -1, kp).max(2)
170 pw_row = 26 + 5 * 500 + 4 * 4 # 판독 행 폭 -- 스트립을 여기에 맞춰 늘린다
171 fx = max(1, pw_row // spec.shape[1])
172 spec = np.repeat(spec, fx, axis=1)
173 ncol = spec.shape[1]
174 mark = np.zeros((18, ncol))
175 lane = np.zeros(ncol)
176 for i, (s0, e0) in enumerate(fb if not full else []):
177     c0 = (s0 // 128) // kp * fx
178     c1 = min(ncol - 1, (e0 // 128) // kp * fx)
179     lane[c0:c1 + 1] = 1.0
180     draw_text(mark, 2, min(c0 + 2, ncol - 14 * len(str(i + 1))), str(i + 1))

```

```

181 rows = [mark, spec, np.full((2, ncol), 0.35), np.tile(lane, (10, 1)),
182         np.full((6, ncol), 0.0)]
183 width = max(ncol, pw_row)
184 rows = [np.pad(r, ((0, 0), (0, width - r.shape[1]))) for r in rows]
185 for i in pick:
186     r, num = burst_row(xa[fb[i][0]:fb[i][1]], fs, str(i + 1))
187     ln = f"sa b{i + 1} {num}" # 바늘 주파수·돌출 -- 받아칠 판독 숫자줄
188     print(f"{ln} #{check_code(ln)}")
189     rows += [np.pad(r, ((0, 0), (0, width - r.shape[1]))), np.full((6, width), 0.0)]
190 png_gray(np.vstack(rows), name + ".png")
191 line = (f"sa {'kat' if arg == 'kat' else 'cap'} n{n} f{fs:.0f} s{n / fs:.1f}"
192        f" fb{0 if full else len(fb)} ev{round(100 * ev)}")
193 print(f"{line} #{check_code(line)}")
194 print(f"{name}.png 저장 - 위: 스펙트로그램+검출 띠+버스트 번호, 아래: ★버스트별"
195        " [PSD|x2|x4|x8|AM] (x2 바늘=BPSK, x4=QPSK, x8=8PSK, AM 봉우리=심볼레이트)")

```